Qualcomm Technologies, Inc.

QCS8275 data sheet

80-73475-1 Rev. AK
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Device description

QCS8275 (also known as IQ-8275) is the next generation Qualcomm® SoC designed for superior performance and power efficiency.

The QCS8275 device includes the following key components:

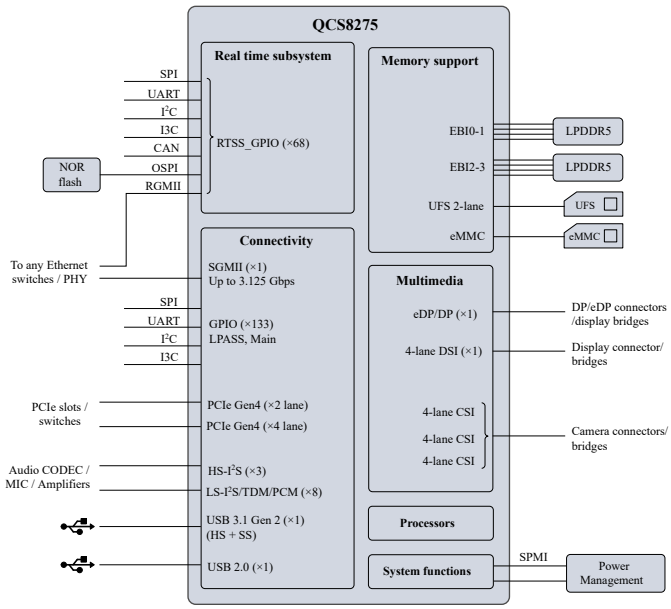
- Qualcomm® Kryo™ Gen 6 CPU built on Arm v8.2 Cortex technology
- 256 kB of L2 cache per core for Prime and Gold cores
- 64 kB of L2 cache per core and 512 kB shared L3 cache for Silver cores
- 1 MB shared L3 cache per cluster for Prime and Gold clusters
- Qualcomm® Adreno™ 623 GPU for the highest in graphics performance and power efficiency
- Qualcomm® Hexagon™ Tensor Processor integrated with Qualcomm Hexagon DSP, quad Qualcomm Hexagon Vector eXtensions (HVX) processor, and dual Qualcomm Hexagon Matrix eXtensions (HMX) coprocessors for high performance machine learning use cases.
- Four-channel (independent) high-speed LPDDR5 SDRAM – up to 3200 MHz
- 256 kB IMEM, 1.5 MB GMEM, 8 MB Vector-TCM for HTP
- AI capability up to 40 TOPS (-AA) and 20 TOPS (-AC)

Functional block diagram
Device physical dimensions
QCS8275 features

Key features

- Qualcomm Spectra 692 ISP image processing engine
- Adreno 623 VPU for high-quality, Ultra HD video encode and decode
- Qualcomm Adreno 1199 DPU for Ultra HD multi-display support
- Dedicated real time subsystem (RTSS) equipped with quad Arm Cortex-R52 CPU
- 1 × OSPI/QSPI interface (166 MHz) in RTSS domain for boot-up
- 2 × PCIe interfaces:
  - 1 × 2-lane PCIe Gen 4 (PCIe0)
  - 1 × 4-lane PCIe Gen 4 (PCIe1)
- 1 × SGMII (SGMII0) supporting up to 3.125 Gbps
- 1 × RGMII v1.3 and v2.0 up to 1 Gbps over RTSS domain, multiplexed with RTSS I/Os
- Two USB interfaces:
  - USB0 – USB 3.1 Gen 2 (HS and SS, supporting device and host modes)
  - USB1 – USB 2.0 (HS, supporting device and host modes)
- 1 × UFS 3.1 up to Gear 4 (UFS0), 2 lanes, Rate B, 11.67 Gbps (bootable)
- 1 × SDIO interface (SDC1) for eMMC 5.1 (bootable)/SD Card (storage only)
- 3 × 4-lane CSI C-PHY/D-PHY interfaces (CSI0, CSI1, CSI2)
  - 6 dedicated CCI_I2C interfaces multiplexed with GPIOs
  - 3 camera MCLKs, multiplexed with GPIOs
  - Each 4-lane interface configurable as 2 × PHY with 2 or 1 lanes each
- 1 × 4-lane DSI C-PHY/D-PHY interface (DSI0)
- 1 × DP/eDP interface up to 4 lanes; DP is compliant with VESA DisplayPort v1.4
- 133 GPIOs in main domain, 68 GPIOs in RTSS domain, multiplexed with SPI/I²C/I³C/UART/CAN
- Audio interfaces: LS-I2S, PCM/TDM, HS-I2S

QCS8275 high-level block diagram



- RTSS_IO (×68): Only the RTSS_IO_52 pin is reserved for internal use.
- HS-I²S (×3): Two out of the three HS-I²S interfaces are muxed behind the I²S interfaces.
- LS-I²S/PCM/TDM (×8): Among the eight LS-I²S interfaces, five are dedicated LS-I²S, two are muxed with HS-I²S, and one dedicated HS-I²S interface reconfigured as an LS-I²S interface.
- eMMC and UFS: Cannot co-exist, either eMMC or UFS can be used.
- EBI – External Bus Interface

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1 Introduction

This topic provides the functional block diagram and the device features.

NOTE Unless otherwise specified, QCS8275 refers to the QCS8275-AA and QCS8275-AC throughout this document. Wherever appropriate, exceptions are specified.

1.1 Functional block diagram

This functional block diagram shows the overall representation of the subsystems on this chipset.

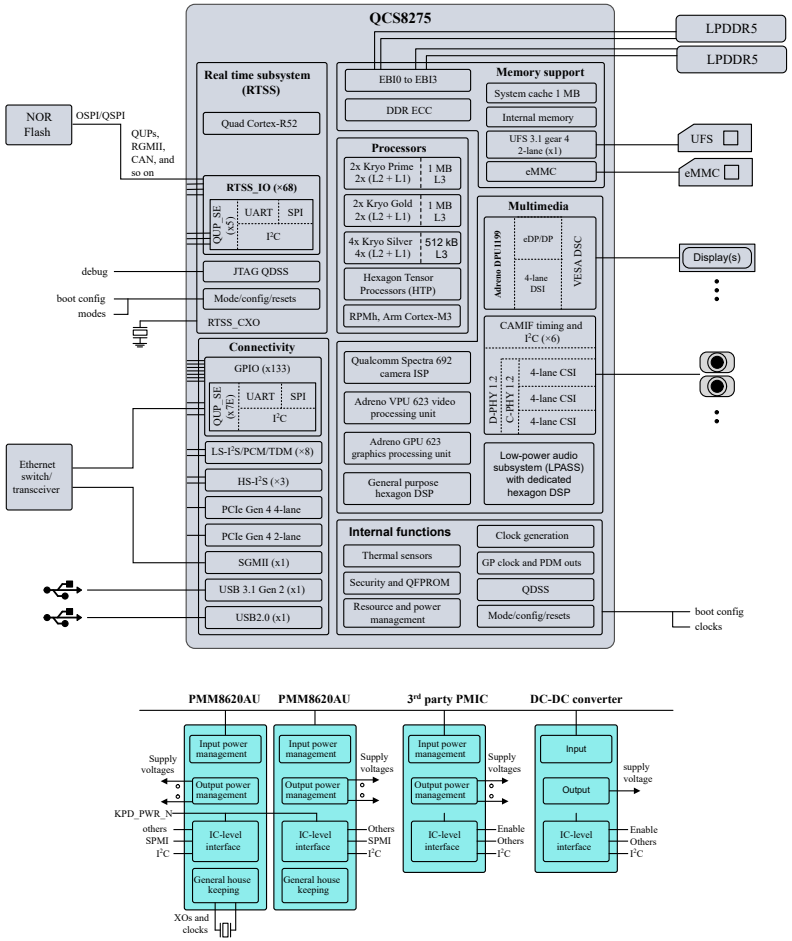


Figure 1-1 QCS8275 functional block diagram

NOTE Only 2 × PMIC (2 × PMM8620AU) are supported, 4 × PMIC support will be available later.

NOTE

- RTSS_IO (×68): Only the RTSS_IO_52 pin is reserved for internal use.
- HS-I²S (×3): Two out of the three HS-I²S interfaces are muxed behind the I²S interfaces.
- LS-I²S/PCM/TDM (×8): Among the eight LS-I²S interfaces, five are dedicated LS-I²S, two are muxed with HS-I²S, and one dedicated HS-I²S interface reconfigured as an LS-I²S interface.
- eMMC and UFS: Cannot co-exist, either eMMC or UFS can be used.

1.2 QCS8275 features

This table lists the comprehensive features of the QCS8275 chipset and its capabilities.

NOTE Some of the hardware features integrated within the QCS8275 must be enabled by software. See the latest revision of the applicable software release notes to identify the enabled QCS8275 features.

Table 1-1 QCS8275 variant feature set comparison

Feature	QCS8275-AA	QCS8275-AC
Kryo Prime	2 × 2.35 GHz	2 × 2.05 GHz
Kryo Gold	2 × 2.1 GHz	2 × 1.9 GHz
Kryo Silver	4 × 1.95 GHz	4 × 1.8 GHz
GPU	A623 at 877 MHz	A623 at 599 MHz
HMX/HVX	1.2 GHz/1.5 GHz [a]	672 MHz/960 MHz
Dense TOPS	40	20

[a] To support HMX/HVX maximum specification, 4 × PMIC configuration is required.

Processors

Function	Description
CPU	Kryo Gen 6 CPU subsystem ■ Dual Kryo Prime cores with 256 kB L2 cache per core targeting up to 2.35 GHz ■ Dual Kryo Gold cores with 256 kB L2 cache per core targeting up to 2.1 GHz ■ 1 MB shared L3 cache per cluster for Prime cluster and Gold cluster ■ Quad Kryo Silver cores with 64 kB L2 cache per core targeting up to 1.95 GHz ■ 512 kB shared L3 cache for Silver cluster
Digital signal processing	Hexagon Tensor Processor is integrated with Qualcomm Hexagon DSP, quad Hexagon vector eXtensions (quad-HVX), and dual Hexagon matrix eXtensions (HMX) co-processors ■ Common AI processing architecture for machine learning use cases ■ Matrix co-processors (one HMX-Integer and one HMX-float per Hexagon Tensor Processor) for deep neural network acceleration Audio Hexagon DSP dedicated to audio subsystem, targeting up to 1.344 GHz and with a 2 MB L2/TCM One general-purpose Hexagon DSP (GPDSP) targeting up to 1.708 GHz and with a 1 MB L2 cache for advanced audio processing and other use cases All Hexagon DSPs are cache-based processors with full access to DDR memory
Real time subsystem	■ A subsystem integrated with quad Cortex-R52 CPU ■ Separated clock and reset input ■ Dedicated logic/memory voltage domain ■ Independent booting capability through OSPI interface
Always-on system	■ Always-on subsystem with always-on processor ■ Hardware-based resource and power management (RPMh) with hardware accelerators for voltage control and regulation, clock management, and resource communication

Memory

Function	Description
System memory via EBI	■ Four-channel high-speed memory – 3200 MHz LPDDR5 SDRAM (4 × 16-bit) and supports up to 32 GB (8 GB per channel) **NOTE**: Tested only for 12 GB (2 × 6 GB) ■ 1 MB system cache
Other internal memory	■ 256 kB IMEM ■ 1.5 MB GMEM for graphics ■ 1 MB L2 cache and 8 MB Vector-TCM (vTCM) for each Hexagon Tensor Processor

External memory

Function	Description
Via UFS	One UFS 3.1 gear 4 – two lanes for on-board memory (Main domain boot up device)
Via eMMC	eMMC interface (Main domain boot up device) **NOTE**: Either UFS or eMMC can be the primary (Main domain) boot interface
Via OSPI/QSPI	NOR flash memory running up to 166 MHz as RTSS domain boot up device
Via PCIe	NVMe (non-bootable)

Multimedia

Function	Description
Display	Adreno display processing unit (DPU) 1199
Display interface/ performance	One 4-lane MIPI DSI with VESA DSC v1.2 ■ D-PHY v1.2: 2.5 Gbps/lane on four lanes per port, 10 Gbps total ■ C-PHY v1.1: 5.7 Gbps/trio on three trios per port, 17 Gbps total One DisplayPort (DP) v1.4 at 8.1 Gbps/lane, 32.4 Gbps/port, MST, and VESA DSC v1.2a and forward error correction (FEC)
Image processing	Destination scaler, fetch exclusion rectangle, inline rotation, 17 × 17 × 17 3D LUT (ViG/DSPP), HDR10 improvements WCG, rounded-corner, CCCS to fixed-point
Compression	UBWC 4.0, DSC v1.2
Camera	Qualcomm Spectra 692 ISP
Performance	■ Pixel processing: 2 × IFE + 5 × IFE_L ■ Data format inputs: 24-bit HDR RGGB/RCCB/RYYCy/RGBIR, RCCG, YUV Data format outputs: 12/10/8 bit Y, 10/8 bit UV; RGB8 in planar or interleaved order
Camera interface	Processing features: 24b HDR bayer processing, lens distortion correction, advanced tone mapping, offset correction, lens roll off correction, bad pixel correction, directional scalers, color LUTs, color space transform, noise reduction Three MIPI CSI interfaces are configurable as D-PHY or C-PHY mode ■ D-PHY v1.2: 2.5 Gbps/lane on four lanes per port, 10 Gbps/port, up to 30 Gbps total. Each 4-lane port can also be configured as 1-lane + 2-lane ports or 1-lane +1-lane ports ■ C-PHY v1.2: 10.2 Gbps/trio on three trios per port, 31 Gbps/port, up to 93 Gbps total. Each 3-trio port can also be configured as 1-trio + 2-trio ports, or three 1-trio ports
Video	■ Adreno video processing unit (VPU) 623 – fifth-generation UHD video processing unit ■ Video decode up to 4k120 ■ Video encode up to 4k60 ■ Concurrent UHD60D + UHD30E or 2x (UHD30 decode + UHD30 encode)

Function	Description
	■ Native decode support for AV1, HEVC, H.264, H.265, VP9, MPEG-2 codecs ■ Native encode support for HEVC, H.264 and H.265 ■ Embedded video analytics (EVA) for optical flow processing and stereo disparity
Graphics	■ Adreno graphic processing unit (GPU) 623 targeting up to 877 MHz ■ Graphics APIs: Vulkan 1.2, OpenGL ES 3.2 ■ Compute APIs: Vulkan 1.2, OpenCL 2.0 FP, Adreno NN direct
Low-power audio subsystem (LPASS)	■ Dedicated audio LPASS Hexagon DSP with a 2 MB L2/TCM ■ AI processor (eNPU) to accelerate neural networking use cases ■ Hardware linear echo cancellation accelerator ■ One general purpose DSP (GPDSP) to offload audio high-performance use cases
Audio interfaces	
LS-I^{2}S (muxed with PCM/TDM pins)	■ Up to 8 (5 + 2 + 1) [b] interfaces (pin multiplexed with PCM/TDM interfaces): seven (4 + 2 + 1) [b] interfaces with two data lanes each; one interface with four data lanes for a total of 18 (12 + 4 + 2) [b] data lanes ■ Three MCLKs up to 512 × 48 kHz (24.576 MHz); clock master or slave independent of the source of frame sync/word select ■ Supports two channels of 32-bit sample up to 384 kHz sample rate, for each data line
PCM/TDM (muxed with LS-I^{2}S pins)	■ Multi-lane TDM (master and slave capable) ■ Up to 10 (7 + 3*) interfaces (pin multiplexed with I^{2}S): nine (6 + 3*) interfaces with two data lanes each; one interface with four data lanes ■ Short, long, and one-slot sync mode ■ Maximum clock frequency of 24.576 MHz □ Up to 512 bits/frame, 32 bits/slot, 16 slots/interface with 48 kHz sample rate ■ Support TDM interfaces grouping for data synchronization
HS-I^{2}S	■ Three high-speed (up to 73.728 MHz) receive interfaces; two of them are muxed behind LS-I^{2}S interfaces; all of them can be configured as LS-I^{2}S interfaces if needed ■ Two receive-only data lanes per interface; clock and word select in slave mode **NOTE**: The additional LS-I^{2}S interfaces (muxed with PCM/TDM pins) are configured from the three dedicated HS-I^{2}S interfaces.

[b] Five dedicated LS-I^{2}S, two muxed with HS-I^{2}S, and one dedicated HS-I^{2}S reconfigured as LS-I^{2}S interface

Physical interfaces

Interface	Description
Qualcomm universal peripheral (QUP) serial engines	■ 17 main domain (MD) GPIO-based QUP SEs: multiplexed serial interface functions ■ 5 bits for QUP1_SE_2 and 4 bits for each of the other 15 QUP SEs; multiplexed serial interface functions ■ Five RTSS domain GPIO-based QUP SEs: 5 bits for QUP0_SE_4 and 4 bits for each of the other four QUP SEs; multiplexed serial interface functions
UART (up to 4 MHz)	UART (64 B FIFO) interface available on all MD and RTSS domains QUP SEs, except MD QUP0_SE_2/3, QUP1_SE_2/3 and RTSS QUP0_SE_2/3, which also support HS-UART (128 B FIFO)
I²C master (up to 1 MHz)	I²C interface available on QUP SEs of MD and RTSS domains, dedicated controller for each port
SPI master (up to 50 MHz)	SPI master interfaces are available on all QUP SEs of MD and RTSS domains except for MD QUP0_SE_7 and MD QUP1_SE_3
SPI slave (up to 50 MHz)	SPI slave interface available on MD QUP0_SE_4/5, QUP1_SE_4/5, and RTSS QUP0_SE_0/1
CCI I²C	Six dedicated I²C interfaces for devices using CSI ports

Interface	Description
USB	Two total USB interfaces ■ USB0 – USB 3.1 Gen 2 (HS + SS, supports device and host modes, DRD) ■ USB1 – USB 2.0 (HS, support device and host modes, DRD)
PCIe	■ One 2-lane PCIe Gen4: PCIe0 (RC + EP) ■ One 4-lane PCIe Gen4: PCIe1 (RC + EP)
RGMII	One RTSS domain RGMII interface with MDIO for Ethernet
CAN-FD	Four CAN-FD interfaces located in the RTSS domain, each of them supports up to 12 Mbps
SGMII	One SGMII interface supporting up to 2.5GbE (up to 3.125 Gbps serial line rate)

Configurable GPIOs

Function	Description
Number of main domain GPIO ports	133 – GPIO_0 to GPIO_132
Number of RTSS domain GPIO ports	68 – RTSS_IO_0 to RTSS_IO_67
Input configurations	Pull-up, pull-down, keeper, or no pull
Output configurations	Programmable drive current up to 16 mA
Top-level mode multiplexer	Provides a convenient way to program groups of GPIOs

Internal functions

Function	Description
Security	Crypto: Hardware ECC and RSA (Elliptic-Curve Cryptography), ICE, Crypto engine v5 (CE5), FIPS/CAVP certifiable, RNG (random number generation) QFPROM: Fuse bits available for OEM use Access control: Programmable security domain protection and sandboxing Secure boot and tools: Secure boot with Sectools 5.4; easy to use tool set Storage security: Secure file system (SFS); fast trusted storage TrustZone: Qualcomm® Trusted Execution Environment (QTEE v5.3) QTEE services: SHE emulation in TEE ICEMEM: An inline encryption/decryption engine used to protect the confidentiality of external RAM Debug: JTAG Others: Thermal sensors; modes and resets; peripheral subsystem
PLLs and clocks	■ Multiple clock regimes; watchdog and sleep timers ■ Dedicated 19.2 MHz RTSS_CXO for real time subsystem ■ 19.2 MHz CXO_0 and CXO_1 as clock sources for main domain of the chip ■ General-purpose outputs: M/N counter and PDM
Operating temperature	Ambient temperature -40°C to +105°C; maximum junction temperature 125°C

RF support

Function	Description
WLAN/Bluetooth	QCA6698 802.11ax Wi-Fi 6E Bluetooth (BT) v5.3
WLAN baseband data	PCIe 0 interface
Bluetooth	UART interface

2 Pin definitions

This topic explains the pin assignments of the QCS8275 chipsets. The figure and table within this topic provide details on pinouts, pad attributes, and their corresponding definitions.

2.1 I/O parameter definitions

Table 2-1 I/O parameter definitions

Symbol	Description
Pad type	
AI	Analog input (does not include pad circuitry)
AO	Analog output (does not include pad circuitry)
B	Bidirectional digital with CMOS input
DI	Digital input (CMOS)
DO	Digital output (CMOS)
H	High-voltage tolerant
S	Schmitt trigger input
Z	High impedance (Hi-Z) output
Padpull details for digital I/Os	
nppdpukp	Programmable pull resistor. The default pull direction is indicated using capital letters and is a prefix to other programmable options: NP: pdpukp = default no-pull with programmable options following the colon (:) PD: nppukp = default pull-down with programmable options following the colon (:) PU: nppdkp = default pull-up with programmable options following the colon (:) KP: nppdpu = default keeper with programmable options following the colon (:)
KP	Contains an internal weak keeper device (keepers cannot drive external buses)
NP	Contains no internal pull
PU	Contains an internal pull-up device
PD	Contains an internal pull-down device
Pad voltage groupings for baseband circuits	
PX_0	Pad group 0 for main domain (control signals); 1.8 V
PX_1	Pad group 1 (EBI I/O); 1.1 V
RTSS_PX_0	Pad group 0 for RTSS domain (control signals); 1.8 V
PX_3	Pad group 3 (most peripherals); 1.8 V
PX_7	Pad group 7 (SDC1- eMMC); 1.8 V
RTSS_PX_3	Pad group 3 for RTSS domain (most peripherals); 1.8 V
RTSS_PX_RGMII	Pad group 3 for RTSS domain (RGMII_0); 1.8 V
PX_9	Pad group 9 (UFS0_REF_CLK and UFS0_RESET); 1.2 V
PX_10	Pad group 10 (RESERVED); 1.2 V

Table 2-1 I/O parameter definitions (cont.)

Symbol	Description
PX_11	Pad group 11 (CXO); 1.2 V
RTSS_PX_11	Pad group 11 for RTSS domain (CXO); 1.8 V

2.2 Pin map

The QCS8275 chipset is available in the FCBGA1326+HS package. A high-level view of the pin assignments is shown in the following figure. For more information about package details and to create printed circuit board (PCB) footprint, see Mechanical information.

The *QCS8275 Pin Assignment and GPIO Configuration Spreadsheet* (80-73475-1A) lists all QCS8275 pad numbers (in alphanumeric order), pad names, pad voltages, pad types, and functional descriptions.

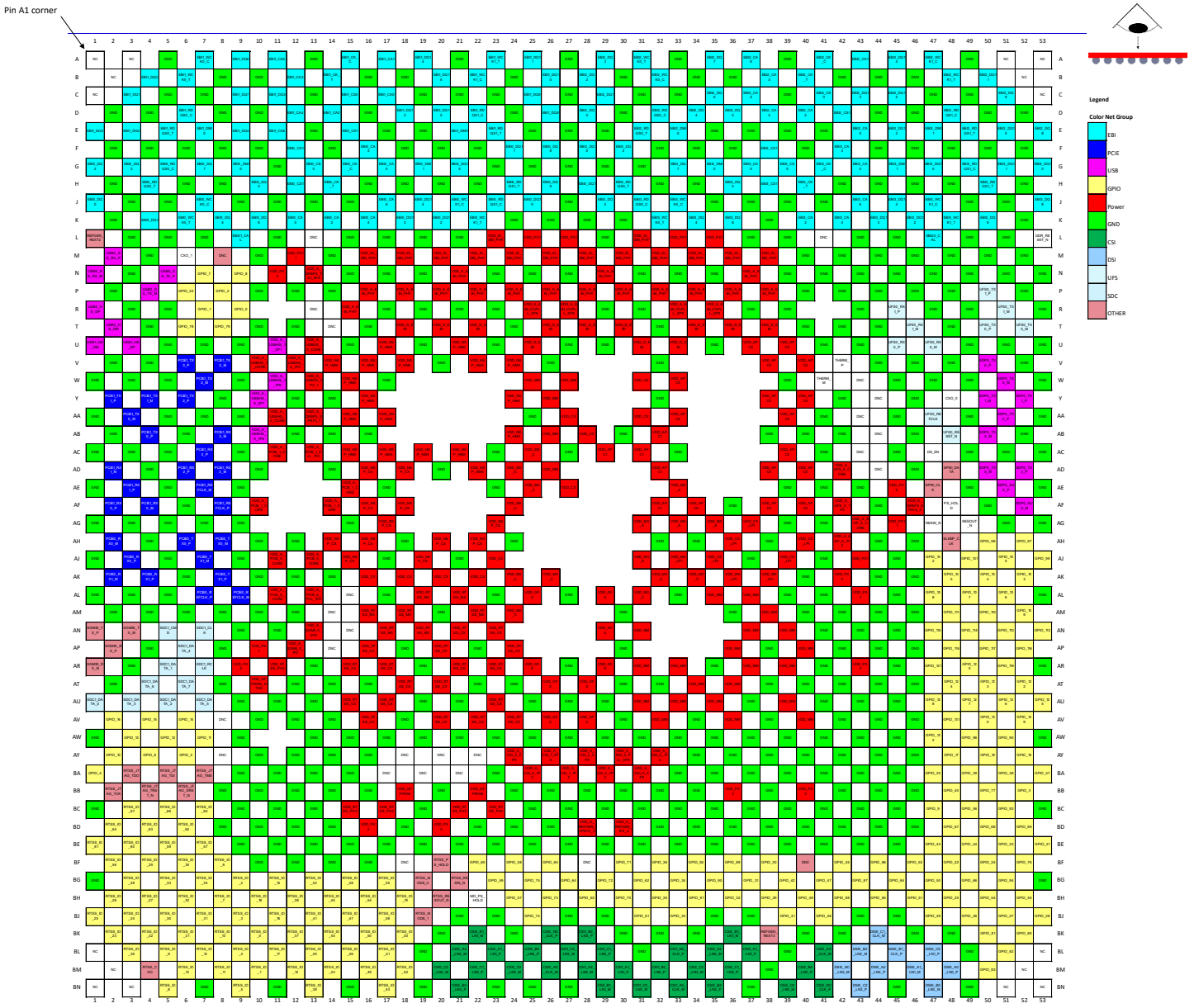


Figure 2-1 QCS8275 pin assignments

2.3 Pin descriptions

The following table describes the pin functions, pin pull types, and pin voltages. It also provides a mapping of pin numbers and pin names for all the pins on the device. These include general pins, GPIO pins, and power pins.

Table 2-2 Pin descriptions – primary pins

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
BM26	CSI0_A0_CLK_M	CSI	AI	MIPI CSI0 (D-PHY), differential clock – minus MIPI CSI0 (C-PHY), trio lane 0 – A
BL23	CSI0_A1_LN1_P	CSI	AI	MIPI CSI0 (D-PHY), differential lane 1 – plus MIPI CSI0 (C-PHY), trio lane 1 – A
BL21	CSI0_A2_LN2_M	CSI	AI	MIPI CSI0 (D-PHY), differential lane 2 – minus MIPI CSI0 (C-PHY), trio lane 2 – A
BL25	CSI0_B0_LN0_P	CSI	AI	MIPI CSI0 (D-PHY), differential lane 0 – plus MIPI CSI0 (C-PHY), trio lane 0 – B
BK22	CSI0_B1_LN1_M	CSI	AI	MIPI CSI0 (D-PHY), differential lane 1 – minus MIPI CSI0 (C-PHY), trio lane 1 – B
BN21	CSI0_B2_LN3_P	CSI	AI	MIPI CSI0 (D-PHY), differential lane 3 – plus MIPI CSI0 (C-PHY), trio lane 2 – B
BM24	CSI0_C0_LN0_M	CSI	AI	MIPI CSI0 (D-PHY), differential lane 0 – minus MIPI CSI0 (C-PHY), trio lane 0 – C
BM22	CSI0_C1_LN2_P	CSI	AI	MIPI CSI0 (D-PHY), differential lane 2 – plus MIPI CSI0 (C-PHY), trio lane 1 – C
BM20	CSI0_C2_LN3_M	CSI	AI	MIPI CSI0 (D-PHY), differential lane 3 – minus MIPI CSI0 (C-PHY), trio lane 2 – C
BK26	CSI0_NC_CLK_P	CSI	AI	MIPI CSI0 (D-PHY), differential clock – plus MIPI CSI0 (C-PHY), no connect
BN33	CSI1_A0_CLK_M	CSI	AI	MIPI CSI1 (D-PHY), differential clock – minus MIPI CSI0 (C-PHY), no connect
BM30	CSI1_A1_LN1_P	CSI	AI	MIPI CSI1 (D-PHY), differential lane 1 – plus MIPI CSI1 (C-PHY), trio lane 1 – A
BM28	CSI1_A2_LN2_M	CSI	AI	MIPI CSI1 (D-PHY), differential lane 2 – minus MIPI CSI1 (C-PHY), trio lane 2 – A
BM32	CSI1_B0_LN0_P	CSI	AI	MIPI CSI1 (D-PHY), differential lane 0 – plus MIPI CSI1 (C-PHY), trio lane 0 – B
BN29	CSI1_B1_LN1_M	CSI	AI	MIPI CSI1 (D-PHY), differential lane 1 – minus MIPI CSI1 (C-PHY), trio lane 1 – B
BK28	CSI1_B2_LN3_P	CSI	AI	MIPI CSI1 (D-PHY), differential lane 3 – plus MIPI CSI1 (C-PHY), trio lane 2 – B
BN31	CSI1_C0_LN0_M	CSI	AI	MIPI CSI1 (D-PHY), differential lane 0 – minus

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
				MIPI CSI1 (C-PHY), trio lane 0 – C
BL29	CSI1_C1_LN2_P	CSI	AI	MIPI CSI1 (D-PHY), differential lane 2 – plus MIPI CSI1 (C-PHY), trio lane 1 – C
BL27	CSI1_C2_LN3_M	CSI	AI	MIPI CSI1 (D-PHY), differential lane 3 – minus MIPI CSI1 (C-PHY), trio lane 2 – C
BL33	CSI1_NC_CLK_P	CSI	AI	MIPI CSI1 (D-PHY), differential clock – plus MIPI CSI1 (C-PHY), no connect
BL41	CSI2_A0_CLK_M	CSI	AI	MIPI CSI2 (D-PHY), differential clock – minus MIPI CSI2 (C-PHY), trio lane 0 – A
BL37	CSI2_A1_LN1_P	CSI	AI	MIPI CSI2 (D-PHY), differential lane 1 – plus MIPI CSI2 (C-PHY), trio lane 1 – A
BL35	CSI2_A2_LN2_M	CSI	AI	MIPI CSI2 (D-PHY), differential lane 2 – minus MIPI CSI2 (C-PHY), trio lane 2 – A
BM40	CSI2_B0_LN0_P	CSI	AI	MIPI CSI2 (D-PHY), differential lane 0 – plus MIPI CSI2 (C-PHY), trio lane 0 – B
BK36	CSI2_B1_LN1_M	CSI	AI	MIPI CSI2 (D-PHY), differential lane 1 – minus MIPI CSI2 (C-PHY), trio lane 1 – B
BN35	CSI2_B2_LN3_P	CSI	AI	MIPI CSI2 (D-PHY), differential lane 3 – plus MIPI CSI2 (C-PHY), trio lane 2 – B
BN39	CSI2_C0_LN0_M	CSI	AI	MIPI CSI2 (D-PHY), differential lane 0 – minus MIPI CSI2 (C-PHY), trio lane 0 – C
BM36	CSI2_C1_LN2_P	CSI	AI	MIPI CSI2 (D-PHY), differential lane 2 – plus MIPI CSI2 (C-PHY), trio lane 1 – C
BM34	CSI2_C2_LN3_M	CSI	AI	MIPI CSI2 (D-PHY), differential lane 3 – minus MIPI CSI2 (C-PHY), trio lane 2 – C
BN41	CSI2_NC_CLK_P	CSI	AI	MIPI CSI2 (D-PHY), differential clock – plus MIPI CSI2 (C-PHY), no connect
Y48	CXO_0	PX_11	DI	38.4 MHz clock input to SoC from PMIC LNBBCLK1 output.
M6	CXO_1	PX_11	DI	38.4 MHz clock input to SoC from PMIC LNBBCLK2 output.
L53	DDR_RESET_N	PX_1	DO	LPDDR5 reset shared by all DDR EBI (External Bus Interface). Active-low reset signal for the external DDR memory interface or memory subsystem.
BM48	DSI0_A0_LN0_P	DSI	AO	MIPI DSI0 (D-PHY), differential lane 0 – plus MIPI DSI0 (C-PHY), trio lane 0 – A
BM46	DSI0_A1_LN1_M	DSI	AO	MIPI DSI0 (D-PHY), differential lane 1 – minus MIPI DSI0 (C-PHY), trio lane 1 – A
BM44	DSI0_A2_LN2_P	DSI	AO	MIPI DSI0 (D-PHY), differential lane 2 – plus MIPI DSI0 (C-PHY), trio lane 2 – A
BN47	DSI0_B0_LN0_M	DSI	AO	MIPI DSI0 (D-PHY), differential lane 0 – minus

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
				MIPI DSI0 (C-PHY), trio lane 0 – B
BL45	DSI0_B1_CLK_P	DSI	AO	MIPI DSI0 (D-PHY), differential clock – plus MIPI DSI0 (C-PHY), trio lane 1 – B
BL43	DSI0_B2_LN2_M	DSI	AO	MIPI DSI0 (D-PHY), differential lane 2 – minus MIPI DSI0 (C-PHY), trio lane 2 – B
BL47	DSI0_C0_LN1_P	DSI	AO	MIPI DSI0 (D-PHY), differential lane 1 – plus MIPI DSI0 (C-PHY), trio lane 0 – C
BK44	DSI0_C1_CLK_M	DSI	AO	MIPI DSI0 (D-PHY), differential clock – minus MIPI DSI0 (C-PHY), trio lane 1 – C
BN43	DSI0_C2_LN3_P	DSI	AO	MIPI DSI0 (D-PHY), differential lane 3 – plus MIPI DSI0 (C-PHY), trio lane 2 – C
BM42	DSI0_NC_LN3_M	DSI	AO	MIPI DSI0 (D-PHY), differential lane 3 – minus MIPI DSI0 (C-PHY), no connect
AC47	DS_EN	PX_0	DI	Deep sleep handshake between PMIC and SoC. Not supported on 4-PMIC design. Future support only.
L9	EBI01_CAL	EBI	DO	External calibration resistor connection for EBI0 and EBI1. Pull up through 240 Ω ±1% resistor.
K12	EBI0_CA0	EBI	DO	EBI0 LPDDR5 command/address bit 0
F12	EBI0_CA1	EBI	DO	EBI0 LPDDR5 command/address bit 1
K14	EBI0_CA2	EBI	DO	EBI0 LPDDR5 command/address bit 2
F16	EBI0_CA3	EBI	DO	EBI0 LPDDR5 command/address bit 3
K16	EBI0_CA4	EBI	DO	EBI0 LPDDR5 command/address bit 4
G17	EBI0_CA5	EBI	DO	EBI0 LPDDR5 command/address bit 5
J17	EBI0_CA6	EBI	DO	EBI0 LPDDR5 command/address bit 6
G15	EBI0_CK_C	EBI	DO	EBI0 LPDDR5 differential clock (C)
H14	EBI0_CK_T	EBI	DO	EBI0 LPDDR5 differential clock (T)
G13	EBI0_CS0	EBI	DO	EBI0 LPDDR5 chip-select 0
H12	EBI0_CS1	EBI	DO	EBI0 LPDDR5 chip-select 1
G9	EBI0_DMI0	EBI	B	EBI0 LPDDR5 data mask for byte 0
G19	EBI0_DMI1	EBI	B	EBI0 LPDDR5 data mask for byte 1
J1	EBI0_DQ0	EBI	B	EBI0 LPDDR5 data bit 0
K4	EBI0_DQ1	EBI	B	EBI0 LPDDR5 data bit 1
J25	EBI0_DQ10	EBI	B	EBI0 LPDDR5 data bit 10
F24	EBI0_DQ11	EBI	B	EBI0 LPDDR5 data bit 11
K20	EBI0_DQ12	EBI	B	EBI0 LPDDR5 data bit 12
K18	EBI0_DQ13	EBI	B	EBI0 LPDDR5 data bit 13
J19	EBI0_DQ14	EBI	B	EBI0 LPDDR5 data bit 14
G21	EBI0_DQ15	EBI	B	EBI0 LPDDR5 data bit 15

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
G1	EBI0_DQ2	EBI	B	EBI0 LPDDR5 data bit 2
G3	EBI0_DQ3	EBI	B	EBI0 LPDDR5 data bit 3
K8	EBI0_DQ4	EBI	B	EBI0 LPDDR5 data bit 4
H10	EBI0_DQ5	EBI	B	EBI0 LPDDR5 data bit 5
K10	EBI0_DQ6	EBI	B	EBI0 LPDDR5 data bit 6
G7	EBI0_DQ7	EBI	B	EBI0 LPDDR5 data bit 7
F26	EBI0_DQ8	EBI	B	EBI0 LPDDR5 data bit 8
H26	EBI0_DQ9	EBI	B	EBI0 LPDDR5 data bit 9
G5	EBI0_RDQS0_C	EBI	B	EBI0 LPDDR5 differential read data strobe for byte 0 (C)
H4	EBI0_RDQS0_T	EBI	B	EBI0 LPDDR5 differential read data strobe for byte 0 (T)
J23	EBI0_RDQS1_C	EBI	B	EBI0 LPDDR5 differential read data strobe for byte 1 (C)
H24	EBI0_RDQS1_T	EBI	B	EBI0 LPDDR5 differential read data strobe for byte 1 (T)
J7	EBI0_WCK0_C	EBI	DO	EBI0 LPDDR5 differential data clock for byte 0 (C)
K6	EBI0_WCK0_T	EBI	DO	EBI0 LPDDR5 differential data clock for byte 0 (T)
J21	EBI0_WCK1_C	EBI	DO	EBI0 LPDDR5 differential data clock for byte 1 (C)
K22	EBI0_WCK1_T	EBI	DO	EBI0 LPDDR5 differential data clock for byte 1 (T)
C17	EBI1_CA0	EBI	DO	EBI1 LPDDR5 command/address bit 0
A17	EBI1_CA1	EBI	DO	EBI1 LPDDR5 command/address bit 1
D14	EBI1_CA2	EBI	DO	EBI1 LPDDR5 command/address bit 2
B12	EBI1_CA3	EBI	DO	EBI1 LPDDR5 command/address bit 3
D12	EBI1_CA4	EBI	DO	EBI1 LPDDR5 command/address bit 4
A11	EBI1_CA5	EBI	DO	EBI1 LPDDR5 command/address bit 5
E11	EBI1_CA6	EBI	DO	EBI1 LPDDR5 command/address bit 6
A15	EBI1_CK_C	EBI	DO	EBI1 LPDDR5 differential clock (C)
B14	EBI1_CK_T	EBI	DO	EBI1 LPDDR5 differential clock (T)
C15	EBI1_CS0	EBI	DO	EBI1 LPDDR5 chip-select 0
E15	EBI1_CS1	EBI	DO	EBI1 LPDDR5 chip-select 1
E7	EBI1_DMI0	EBI	B	EBI1 LPDDR5 data mask for byte 0
E21	EBI1_DMI1	EBI	B	EBI1 LPDDR5 data mask for byte 1
E1	EBI1_DQ0	EBI	B	EBI1 LPDDR5 data bit 0
C3	EBI1_DQ1	EBI	B	EBI1 LPDDR5 data bit 1
B26	EBI1_DQ10	EBI	B	EBI1 LPDDR5 data bit 10
A25	EBI1_DQ11	EBI	B	EBI1 LPDDR5 data bit 11
D20	EBI1_DQ12	EBI	B	EBI1 LPDDR5 data bit 12
D18	EBI1_DQ13	EBI	B	EBI1 LPDDR5 data bit 13
A19	EBI1_DQ14	EBI	B	EBI1 LPDDR5 data bit 14
B20	EBI1_DQ15	EBI	B	EBI1 LPDDR5 data bit 15

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
E3	EBI1_DQ2	EBI	B	EBI1 LPDDR5 data bit 2
B4	EBI1_DQ3	EBI	B	EBI1 LPDDR5 data bit 3
E9	EBI1_DQ4	EBI	B	EBI1 LPDDR5 data bit 4
C11	EBI1_DQ5	EBI	B	EBI1 LPDDR5 data bit 5
A9	EBI1_DQ6	EBI	B	EBI1 LPDDR5 data bit 6
C9	EBI1_DQ7	EBI	B	EBI1 LPDDR5 data bit 7
D26	EBI1_DQ8	EBI	B	EBI1 LPDDR5 data bit 8
C25	EBI1_DQ9	EBI	B	EBI1 LPDDR5 data bit 9
D6	EBI1_RDQS0_C	EBI	B	EBI1 LPDDR5 differential read data strobe for byte 0 (C)
E5	EBI1_RDQS0_T	EBI	B	EBI1 LPDDR5 differential read data strobe for byte 0 (T)
D22	EBI1_RDQS1_C	EBI	B	EBI1 LPDDR5 differential read data strobe for byte 1 (C)
E23	EBI1_RDQS1_T	EBI	B	EBI1 LPDDR5 differential read data strobe for byte 1 (T)
A7	EBI1_WCK0_C	EBI	DO	EBI1 LPDDR5 differential data clock for byte 0 (C)
B6	EBI1_WCK0_T	EBI	DO	EBI1 LPDDR5 differential data clock for byte 0 (T)
B22	EBI1_WCK1_C	EBI	DO	EBI1 LPDDR5 differential data clock for byte 1 (C)
A23	EBI1_WCK1_T	EBI	DO	EBI1 LPDDR5 differential data clock for byte 1 (T)
L47	EBI23_CAL	EBI	DO	External calibration resistor connection for EBI2 and EBI3. Pull up through 240 Ω ±1% resistor.
E43	EBI2_CA0	EBI	DO	EBI2 LPDDR5 command/address bit 0
A43	EBI2_CA1	EBI	DO	EBI2 LPDDR5 command/address bit 1
D40	EBI2_CA2	EBI	DO	EBI2 LPDDR5 command/address bit 2
B38	EBI2_CA3	EBI	DO	EBI2 LPDDR5 command/address bit 3
D38	EBI2_CA4	EBI	DO	EBI2 LPDDR5 command/address bit 4
C37	EBI2_CA5	EBI	DO	EBI2 LPDDR5 command/address bit 5
A37	EBI2_CA6	EBI	DO	EBI2 LPDDR5 command/address bit 6
A41	EBI2_CK_C	EBI	DO	EBI2 LPDDR5 differential clock (C)
B40	EBI2_CK_T	EBI	DO	EBI2 LPDDR5 differential clock (T)
C41	EBI2_CS0	EBI	DO	EBI2 LPDDR5 chip-select 0
D42	EBI2_CS1	EBI	DO	EBI2 LPDDR5 chip-select 1
E33	EBI2_DMI0	EBI	B	EBI2 LPDDR5 data mask for byte 0
E47	EBI2_DMI1	EBI	B	EBI2 LPDDR5 data mask for byte 1
D28	EBI2_DQ0	EBI	B	EBI2 LPDDR5 data bit 0
C29	EBI2_DQ1	EBI	B	EBI2 LPDDR5 data bit 1
E51	EBI2_DQ10	EBI	B	EBI2 LPDDR5 data bit 10
B50	EBI2_DQ11	EBI	B	EBI2 LPDDR5 data bit 11
E45	EBI2_DQ12	EBI	B	EBI2 LPDDR5 data bit 12
C43	EBI2_DQ13	EBI	B	EBI2 LPDDR5 data bit 13
A45	EBI2_DQ14	EBI	B	EBI2 LPDDR5 data bit 14

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
C45	EBI2_DQ15	EBI	B	EBI2 LPDDR5 data bit 15
B28	EBI2_DQ2	EBI	B	EBI2 LPDDR5 data bit 2
A29	EBI2_DQ3	EBI	B	EBI2 LPDDR5 data bit 3
D34	EBI2_DQ4	EBI	B	EBI2 LPDDR5 data bit 4
D36	EBI2_DQ5	EBI	B	EBI2 LPDDR5 data bit 5
C35	EBI2_DQ6	EBI	B	EBI2 LPDDR5 data bit 6
A35	EBI2_DQ7	EBI	B	EBI2 LPDDR5 data bit 7
E53	EBI2_DQ8	EBI	B	EBI2 LPDDR5 data bit 8
C51	EBI2_DQ9	EBI	B	EBI2 LPDDR5 data bit 9
D32	EBI2_RDQS0_C	EBI	B	EBI2 LPDDR5 differential read data strobe for byte 0 (C)
E31	EBI2_RDQS0_T	EBI	B	EBI2 LPDDR5 differential read data strobe for byte 0 (T)
D48	EBI2_RDQS1_C	EBI	B	EBI2 LPDDR5 differential read data strobe for byte 1 (C)
E49	EBI2_RDQS1_T	EBI	B	EBI2 LPDDR5 differential read data strobe for byte 1 (T)
B32	EBI2_WCK0_C	EBI	DO	EBI2 LPDDR5 differential data clock for byte 0 (C)
A31	EBI2_WCK0_T	EBI	DO	EBI2 LPDDR5 differential data clock for byte 0 (T)
A47	EBI2_WCK1_C	EBI	DO	EBI2 LPDDR5 differential data clock for byte 1 (C)
B48	EBI2_WCK1_T	EBI	DO	EBI2 LPDDR5 differential data clock for byte 1 (T)
G37	EBI3_CA0	EBI	DO	EBI3 LPDDR5 command/address bit 0
F38	EBI3_CA1	EBI	DO	EBI3 LPDDR5 command/address bit 1
K40	EBI3_CA2	EBI	DO	EBI3 LPDDR5 command/address bit 2
F42	EBI3_CA3	EBI	DO	EBI3 LPDDR5 command/address bit 3
K42	EBI3_CA4	EBI	DO	EBI3 LPDDR5 command/address bit 4
G43	EBI3_CA5	EBI	DO	EBI3 LPDDR5 command/address bit 5
J43	EBI3_CA6	EBI	DO	EBI3 LPDDR5 command/address bit 6
G41	EBI3_CK_C	EBI	DO	EBI3 LPDDR5 differential clock (C)
H40	EBI3_CK_T	EBI	DO	EBI3 LPDDR5 differential clock (T)
G39	EBI3_CS0	EBI	DO	EBI3 LPDDR5 chip-select 0
H38	EBI3_CS1	EBI	DO	EBI3 LPDDR5 chip-select 1
G35	EBI3_DMI0	EBI	B	EBI3 LPDDR5 data mask for byte 0
G45	EBI3_DMI1	EBI	B	EBI3 LPDDR5 data mask for byte 1
F28	EBI3_DQ0	EBI	B	EBI3 LPDDR5 data bit 0
H28	EBI3_DQ1	EBI	B	EBI3 LPDDR5 data bit 1
G53	EBI3_DQ10	EBI	B	EBI3 LPDDR5 data bit 10
G51	EBI3_DQ11	EBI	B	EBI3 LPDDR5 data bit 11
K46	EBI3_DQ12	EBI	B	EBI3 LPDDR5 data bit 12
K44	EBI3_DQ13	EBI	B	EBI3 LPDDR5 data bit 13
J45	EBI3_DQ14	EBI	B	EBI3 LPDDR5 data bit 14

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
G47	EBI3_DQ15	EBI	B	EBI3 LPDDR5 data bit 15
J29	EBI3_DQ2	EBI	B	EBI3 LPDDR5 data bit 2
F30	EBI3_DQ3	EBI	B	EBI3 LPDDR5 data bit 3
K34	EBI3_DQ4	EBI	B	EBI3 LPDDR5 data bit 4
H36	EBI3_DQ5	EBI	B	EBI3 LPDDR5 data bit 5
K36	EBI3_DQ6	EBI	B	EBI3 LPDDR5 data bit 6
G33	EBI3_DQ7	EBI	B	EBI3 LPDDR5 data bit 7
J53	EBI3_DQ8	EBI	B	EBI3 LPDDR5 data bit 8
K50	EBI3_DQ9	EBI	B	EBI3 LPDDR5 data bit 9
J31	EBI3_RDQS0_C	EBI	B	EBI3 LPDDR5 differential read data strobe for byte 0 (C)
H30	EBI3_RDQS0_T	EBI	B	EBI3 LPDDR5 differential read data strobe for byte 0 (T)
G49	EBI3_RDQS1_C	EBI	B	EBI3 LPDDR5 differential read data strobe for byte 1 (C)
H50	EBI3_RDQS1_T	EBI	B	EBI3 LPDDR5 differential read data strobe for byte 1 (T)
J33	EBI3_WCK0_C	EBI	DO	EBI3 LPDDR5 differential data clock for byte 0 (C)
K32	EBI3_WCK0_T	EBI	DO	EBI3 LPDDR5 differential data clock for byte 0 (T)
J47	EBI3_WCK1_C	EBI	DO	EBI3 LPDDR5 differential data clock for byte 1 (C)
K48	EBI3_WCK1_T	EBI	DO	EBI3 LPDDR5 differential data clock for byte 1 (T)
AF52	EDP0_AUX_M	EBI	AO	eDP 0 auxiliary channel
AE51	EDP0_AUX_P	–	AO	eDP 0 auxiliary channel
W51	EDP0_TX0_M	–	AO	eDP 0 transmit channel 0 – minus
V50	EDP0_TX0_P	–	AO	eDP 0 transmit channel 0 – plus
Y50	EDP0_TX1_M	–	AO	eDP 0 transmit channel 1 – minus
Y52	EDP0_TX1_P	–	AO	eDP 0 transmit channel 1 – plus
AB50	EDP0_TX2_M	–	AO	eDP 0 transmit channel 2 – minus
AA51	EDP0_TX2_P	–	AO	eDP 0 transmit channel 2 – plus
AD50	EDP0_TX3_M	–	AO	eDP 0 transmit channel 3 – minus
AD52	EDP0_TX3_P	–	AO	eDP 0 transmit channel 3 – plus
BH22	MD_PS_HOLD	PX_3	DI	The main domain PS_HOLD is an input and must be connected externally to the SoC PS_HOLD signal.
AL9	PCIE0_REFCLK_M	–	AO	PCIe 0 reference clock – minus
AL7	PCIE0_REFCLK_P	–	AO	PCIe 0 reference clock – plus
AH2	PCIE0_RX0_M	–	AI	PCIe 0 receive 0 – minus
AJ3	PCIE0_RX0_P	–	AI	PCIe 0 receive 0 – plus
AK2	PCIE0_RX1_M	–	AI	PCIe 0 receive 1 – minus
AK4	PCIE0_RX1_P	–	AI	PCIe 0 receive 1 – plus
AH8	PCIE0_TX0_M	–	AO	PCIe 0 transmit 0 – minus
AH6	PCIE0_TX0_P	–	AO	PCIe 0 transmit 0 – plus

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
AJ7	PCIE0_TX1_M	–	AO	PCIe 0 transmit 1 – minus
AK8	PCIE0_TX1_P	–	AO	PCIe 0 transmit 1 – plus
AE7	PCIE1_REFCLK_M	–	AO	PCIe 1 reference clock – minus
AF8	PCIE1_REFCLK_P	–	AO	PCIe 1 reference clock – plus
AF4	PCIE1_RX0_M	–	AI	PCIe 1 receive 0 – minus
AF2	PCIE1_RX0_P	–	AI	PCIe 1 receive 0 – plus
AD2	PCIE1_RX1_M	–	AI	PCIe 1 receive 1 – minus
AE3	PCIE1_RX1_P	–	AI	PCIe 1 receive 1 – plus
AD8	PCIE1_RX2_M	–	AI	PCIe 1 receive 2 – minus
AD6	PCIE1_RX2_P	–	AI	PCIe 1 receive 2 – plus
AB8	PCIE1_RX3_M	–	AI	PCIe 1 receive 3 – minus
AC7	PCIE1_RX3_P	–	AI	PCIe 1 receive 3 – plus
AA3	PCIE1_TX0_M	–	AO	PCIe 1 transmit 0 – minus
AB4	PCIE1_TX0_P	–	AO	PCIe 1 transmit 0 – plus
Y4	PCIE1_TX1_M	–	AO	PCIe 1 transmit 1 – minus
Y2	PCIE1_TX1_P	–	AO	PCIe 1 transmit 1 – plus
W7	PCIE1_TX2_M	–	AO	PCIe 1 transmit 2 – minus
Y6	PCIE1_TX2_P	–	AO	PCIe 1 transmit 2 – plus
V8	PCIE1_TX3_M	–	AO	PCIe 1 transmit 3 – minus
V6	PCIE1_TX3_P	–	AO	PCIe 1 transmit 3 – plus
AF48	PS_HOLD	PX_3	DO-Z	The Power Supply HOLD (PS_HOLD) signal is driven by the SoC to the PMIC. It is active HIGH and has an internal pull-down by default. During boot, PS_HOLD is set HIGH (logic 1) to indicate a successful boot and to request that the PMIC keep the power supplies enabled. During shutdown, watchdog expiry, thermal trip, or certain error conditions, software or hardware clears PS_HOLD to LOW (logic 0) to instruct the PMIC to shut down or reset the system.
L1	REFGEN_REXT2	–	AI	External resistor connection for the internal reference generator or bias network. Pull down through 100 Ω ±1% resistor.
BK38	REFGEN_REXT3	–	AI	External resistor connection for the internal reference generator or bias network. Pull down through 100 Ω ±1% resistor.
AG47	RESIN_N	PX_0	DI	The main domain hardware reset input to the SoC is provided by the PMIC. This is an active-low reset pin. ▪ **RESIN_N = 0 (LOW):** The SoC is held in reset. ▪ **RESIN_N = 1 (HIGH):** The SoC reset is released, allowing boot to proceed and basic power rails to turn ON. If the SoC does not assert PS_HOLD HIGH within a certain period after RESIN_N goes HIGH, the PMIC will trigger a reset. RESIN_N connects to the PON_RESET_N pin on the PMIC side.

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
AG49	RESOUT_N	PX_3	DO	The main domain active-low reset output follows RESIN_N with a short delay. It can be used to reset external circuitry. This signal is asserted or released a few clock cycles after a change on RESIN_N or after a watchdog expiration.
BM4	RTSS_CXO	PX_11	DI	Core crystal oscillator for the RTSS domain from PMIC LNBBCLK3 output.
BB6	RTSS_JTAG_SRST_N	PX_3	DI	RTSS subsystem JTAG reset for debug
BB2	RTSS_JTAG_TCK	PX_3	DI	RTSS subsystem JTAG clock input
BA5	RTSS_JTAG_TDI	PX_3	DI	RTSS subsystem JTAG data input
BA3	RTSS_JTAG_TDO	PX_3	DO-Z	RTSS subsystem JTAG data output
BA7	RTSS_JTAG_TMS	PX_3	B	RTSS subsystem JTAG mode select input
BB4	RTSS_JTAG_TRST_N	PX_3	DI	RTSS subsystem JTAG reset
BG19	RTSS_MODE_0	PX_3	DI	RTSS subsystem mode control bit 0
BJ19	RTSS_MODE_1	PX_3	DI	RTSS subsystem mode control bit 1
BF20	RTSS_PS_HOLD	PX_3	DO	When the RTSS domain is active, this signal will be HIGH. For debugging purposes, a test point can be added to this signal. RTSS temperature fatal events, power sequence watchdog expiration, MD PMIC fault indications, or software faults can cause this signal to go LOW.
BG21	RTSS_RESIN_N	PX_0	DI	The RTSS domain hardware reset input to the SoC is provided by the PMIC. This input should be connected to the main domain RESIN_N signal.
BH20	RTSS_RESOUT_N	PX_3	DO	The RTSS domain active-low reset output follows RTSS_RESIN_N with a short delay. It can be used to reset external circuitry. This signal is asserted or released a few clock cycles after a change on RTSS_RESIN_N or based on watchdog expire status.
AN7	SDC1_CLK	PX_7	DO	Secure digital controller 1 clock
AN5	SDC1_CMD	PX_7	B	Secure digital controller 1 command
AU1	SDC1_DATA_0	PX_7	B	Secure digital controller 1 data bit 0
AR5	SDC1_DATA_1	PX_7	B	Secure digital controller 1 data bit 1
AU5	SDC1_DATA_2	PX_7	B	Secure digital controller 1 data bit 2
AU3	SDC1_DATA_3	PX_7	B	Secure digital controller 1 data bit 3
AP6	SDC1_DATA_4	PX_7	B	Secure digital controller 1 data bit 4
AU7	SDC1_DATA_5	PX_7	B	Secure digital controller 1 data bit 5
AT4	SDC1_DATA_6	PX_7	B	Secure digital controller 1 data bit 6
AT6	SDC1_DATA_7	PX_7	B	Secure digital controller 1 data bit 7
AR7	SDC1_RCLK	PX_7	DI	Secure digital controller 1 return clock
AR1	SGMII0_RX_M	–	AI, AO	SGMII 0 receive M – minus
AP2	SGMII0_RX_P	–	AI, AO	SGMII 0 receive P – plus

Table 2-2 Pin descriptions – primary pins (cont.)

Pin number	Pin name	Pad characteristics [a]		Functional description
		Voltage	Type	
AN3	SGMII0_TX_M	–	AI, AO	SGMII interface 0 serial data Tx – minus
AN1	SGMII0_TX_P	–	AI, AO	SGMII interface 0 serial data Tx – plus
AH48	SLEEP_CLK	PX_3	DI	Sleep clock
AE47	SPMI_CLK	PX_0	B	Slave and PBUS interface for PMICs – clock
AD48	SPMI_DATA	PX_0	B	Slave and PBUS interface for PMICs – data
AA47	UFS0_REFCLK	PX_9	DO	UFS0 reference clock
AB48	UFS0_RESET_N	PX_9	DO	UFS0 reset
U47	UFS0_RX0_M	–	AI	UFS0 receive 0 – minus
U45	UFS0_RX0_P	–	AI	UFS0 receive 0 – plus
T46	UFS0_RX1_M	–	AI	UFS0 receive 1 – minus
R45	UFS0_RX1_P	–	AI	UFS0 receive 1 – plus
T52	UFS0_TX0_M	–	AO	UFS0 transmit 0 – minus
T50	UFS0_TX0_P	–	AO	UFS0 transmit 0 – plus
R51	UFS0_TX1_M	–	AO	UFS0 transmit 1 – minus
P50	UFS0_TX1_P	–	AO	UFS0 transmit 1 – plus
T2	USB0_HS_DM	–	AI, AO	USB0 high-speed data – minus
R1	USB0_HS_DP	–	AI, AO	USB0 high-speed data – plus
N1	USB0_SS_RX_M	–	AI	USB0 SuperSpeed receive 0 – minus
M2	USB0_SS_RX_P	–	AI	USB0 SuperSpeed receive 0 – plus
P4	USB0_SS_TX_M	–	AO	USB0 SuperSpeed transmit 0 – minus
N5	USB0_SS_TX_P	–	AO	USB0 SuperSpeed transmit 0 – plus
U1	USB1_HS_DM	–	AI, AO	USB1 high-speed data – minus
U3	USB1_HS_DP	–	AI, AO	USB1 high-speed data – plus

[a] For parameter and acronym definitions, see I/O parameter definitions.

GPIO pins can support multiple functions. To assign GPIOs to particular functions (such as the options listed in the preceding table), identify all their application's requirements and map each GPIO to its function. See the following table for a list of all supported functions for each GPIO.

Some GPIOs can wake up the SoC from sleep. For more information, see Table 2-3.

To avoid excessive current leakage, examine each GPIO external connection and programmed configuration. Ensure control of the following factors:

- GPIO configuration
  - Input vs. output
  - Pull-up or pull-down

- External connections
  - Unused inputs
  - Connections to high-impedance (tri-state) outputs
  - Connections to external devices that may not be attached

For GPIO assignments, see the *QCS8275 Pin Assignment and GPIO Configuration Spreadsheet* (80-73475-1A). It lists all pad names and/or functions for QCS8275 GPIOs (in numeric order), pad numbers, pad voltages, pull states, and available configurations.

Table 2-3 Pin descriptions – general-purpose input/output ports

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
R9	GPIO_0		PX_3	PD:nppukp	Configurable I/O	Y
R7	GPIO_1		PX_3	PU:nppdkp	Configurable I/O	Y
		PCIE0_CLKREQ_N			PCIE0 clock request	
P8	GPIO_2		PX_3	PD:nppukp	Configurable I/O	Y
BB52	GPIO_3		PX_3	PD:nppukp	Configurable I/O	Y
BA1	GPIO_4		PX_3	PD:nppukp	Configurable I/O	Y
		SGMII_PHY_INTR0_N			SGMII 0 PHY interrupt	
AY6	GPIO_5		PX_3	PD:nppukp	Configurable I/O	N
		EMAC0_MDC			EMAC 0 management interface clock	
AY4	GPIO_6		PX_3	PD:nppukp	Configurable I/O	N
		EMAC0_MDIO			EMAC 0 management interface I/O	
N7	GPIO_7		PX_3	PD:nppukp	Configurable I/O	N
		USB2PHY_AC_EN0			USB AC coupling 0 control	
		BOOT_CONFIG[12]			Boot configuration bit 12	
N9	GPIO_8		PX_3	PD:nppukp	Configurable I/O	N
		USB2PHY_AC_EN2			USB AC coupling 2 control	
		BOOT_CONFIG[14]			Boot configuration bit 14	
BC47	GPIO_9		PX_3	PD:nppukp	Configurable I/O	Y
AY2	GPIO_10		PX_3	PD:nppukp	Configurable I/O	Y
		QSPI_DATA[1]			QSPI_DATA[1]	
		EMAC0_MCG_PST_TRIG_LW[0]			EMAC0 trigger input 0 for media clock generation	
AW7	GPIO_11		PX_3	PD:nppukp	Configurable I/O	Y
		QSPI_DATA[0]			QSPI_DATA[0]	
		EMAC0_MCG_PST_TRIG_LW[1]			EMAC0 trigger input 1 for media clock generation	
AW5	GPIO_12		PX_3	PD:nppukp	Configurable I/O	N

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		QSPI_CLK			QSPI_CLK	
AW3	GPIO_13		PX_3	PD:nppukp	Configurable I/O	Y
		QSPI_CS			QSPI_CS	
AV6	GPIO_14		PX_3	PD:nppukp	Configurable I/O	N
		QSPI_DATA[2]			QSPI_DATA[2]	
AV4	GPIO_15		PX_3	PD:nppukp	Configurable I/O	N
		QSPI_DATA[3]			QSPI_DATA[3]	
AV2	GPIO_16		PX_3	PD:nppukp	Configurable I/O	Y
		QSPI_INT			QSPI_INT	
		GP_PDM_MIRA[2]			General-purpose PDM output 2 A	
AY48	GPIO_17		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE0_L0			QUP 0 SE 0 lane 0: I2C_SDA/SPI_MISO/UART_CTS	
		QUP0_SE1_L2			QUP 0 SE 1 lane 2: SPI_SCLK/UART_TX	
		IBI_I3C_QUP0_SE0_SDA			QUP0 In-band interrupt I3C SE clock 0	
		GP_PDM_MIRB[1]			General-purpose PDM output 1 B	
AY50	GPIO_18		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE0_L1			QUP 0 SE 0 lane 1: I2C_SCL/SPI_MOSI/UART_RFR	
		QUP0_SE1_L3			QUP 0 SE 1 lane 3: SPI_CS_0/UART_RX	
		IBI_I3C_QUP0_SE0_SCL			QUP0 In-Band interrupt I3C SE data 0	
AY52	GPIO_19		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE1_L0			QUP 0 SE 1 lane 0: I2C_SDA/SPI_MISO/UART_CTS	
		QUP0_SE0_L2			QUP 0 SE 0 lane 2: SPI_SCLK/UART_TX	
		CCI_TIMER5			Camera control interface timer 5	
		IBI_I3C_QUP0_SE1_SDA			QUP0 In-Band interrupt I3C SE clock 0	
		GP_PDM_MIRB[2]			General-purpose PDM output 2 B	
BA47	GPIO_20		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE1_L1			QUP 0 SE 1 lane 1: I2C_SCL/SPI_MOSI/UART_RFR	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		QUP0_SE0_L3			QUP 0 SE 0 lane 3: SPI_CS_0/UART_RX	
		CCI_TIMER6			Camera control interface timer 6	
		IBI_I3C_QUP0_SE1_SCL			QUP0 In-Band interrupt I3C SE data 0	
		GP_PDM_MIRA[1]			General-purpose PDM output 1 A	
BE53	GPIO_21		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE5_L0			QUP 0 SE 5 lane 0: I2C-SDA/SPI_MISO/SPI-S_MISO/UART_CTS	
		CCI_TIMER7			Camera control interface timer 7	
BF48	GPIO_22		PX_3	PU:nppdkp	Configurable I/O	Y
		PCIE1_CLKREQ_N			PCIe1 clock request	
		QUP0_SE5_L1			QUP 0 SE 5 lane 1: I2C_SCL/SPI_MOSI/SPI-S_MOSI/UART_RFR	
		CCI_TIMER8			Camera control interface timer 8	
BE51	GPIO_23		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE5_L2			QUP 0 SE 5 lane 2: SPI_SCLK/SPI-S_SCLK/UART_TX	
		CCI_TIMER9			Camera control interface timer 9	
BF50	GPIO_24		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE5_L3			QUP 0 SE 5 lane 3: SPI_CS_0/SPI-S_CS/UART_RX	
		EMAC0_PTP_AUX_TS_I_2			EMAC 0 trigger 2 A for auxiliary snapshot	
		EMAC0_PTP_PPS_O_2			EMAC 0 configurable PPS output 2	
		EMAC0_MCG_PST_TRIG_LW[2]			EMAC0 trigger input 2 for media clock generation	
BH48	GPIO_25		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE3_L0			QUP 0 SE 3 lane 0: I2C_SDA/SPI_MISO/UART_CTS/HS-UART_CTS	
BJ49	GPIO_26		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE3_L1			QUP 0 SE 3 lane 1: I2C_SCL/SPI_MOSI/UART_RFR/HS-UART_RFR	
BJ51	GPIO_27		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE3_L2			QUP 0 SE 3 lane 2: SPI_SCLK/UART_TX/HS-UART_TX	
BJ53	GPIO_28		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE3_L3			QUP 0 SE 3 lane 3: SPI_CS_0/UART_RX/HS-UART_RX	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
BH38	GPIO_29		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE4_L0			QUP 0 SE 4 lane 0: I2C_SDA/SPI_MISO/SPI-S_MISO/UART_CTS	
		CCI_I2C_SDA1			Camera I²C data 1	
		CCI_ASYNC_IN3			Camera control interface async 3	
		EMAC0_PTP_PPS_O_0			EMAC 0 configurable PPS output 0	
BF38	GPIO_30		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE4_L1			QUP 0 SE 4 lane 1: I2C_SCL/SPI_MOSI/SPI-S_MOSI/UART_RFR	
		CCI_I2C_SCL1			Camera I²C clock 1	
		CCI_ASYNC_IN4			Camera control interface async 4	
		EMAC0_PTP_PPS_O_1			EMAC 0 configurable PPS output 1	
BG37	GPIO_31		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE4_L2			QUP 0 SE 4 lane 2: I2C_SCL/SPI_MOSI/SPI-S_MOSI	
		CCI_I2C_SDA3			Camera I²C data 3	
		CCI_ASYNC_IN5			Camera control interface async 5	
		EMAC0_PTP_AUX_TS_I_0			EMAC 0 trigger 0 A for auxiliary snapshot	
BH36	GPIO_32		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE4_L3			QUP 0 SE 4 lane 3: SPI_CS_0/SPI-S_CS/UART_RX	
		CCI_I2C_SCL3			Camera I²C clock 3	
		EMAC0_PTP_AUX_TS_I_1			EMAC 0 trigger 1 A for auxiliary snapshot	
		MDP_VSYNC_S			MDP vertical sync – secondary	
		GP_MN			General-purpose M/N:D counter output	
BJ33	GPIO_33		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE2_L0			QUP 0 SE 2 lane 0: I2C_SDA/SPI_MISO/UART_CTS/HS-UART_CTS	
BG33	GPIO_34		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE2_L1			QUP 0 SE 2 lane 1: I2C_SCL/SPI_MOSI/UART_RFR/HS-UART_RFR	
BH32	GPIO_35		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE2_L2			QUP 0 SE 2 lane 2: SPI_SCLK/UART_TX/HS-UART_TX	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		GCC_GP1_CLK_MIRB			General-purpose clock 1 B	
BF32	GPIO_36		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE2_L3			QUP 0 SE 2 lane 3: SPI_CS_0/UART_RX/HS-UART_RX	
		GCC_GP2_CLK_MIRB			General-purpose clock 2 B	
BA53	GPIO_37		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE0_L0			QUP 1 SE 0 lane 0: I2C_SDA/SPI_MISO/UART_CTS	
		IBI_I3C_QUP1_SE0_SDA			QUP1 In-Band interrupt I3C SE clock 1	
		QUP1_SE1_L2			QUP 1 SE 1 lane 2: SPI_SCLK/UART_TX	
BA51	GPIO_38		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE0_L1			QUP 1 SE 0 lane 1: I2C_SCL/SPI_MOSI/UART_RFR	
		IBI_I3C_QUP1_SE0_SCL			QUP1 In-Band interrupt I3C SE data 1	
		QUP1_SE1_L3			QUP 1 SE 1 lane 3: SPI_CS_0/UART_RX	
BA49	GPIO_39		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE1_L0			QUP 1 SE 1 lane 0: I2C_SDA/SPI_MISO/UART_CTS	
		IBI_I3C_QUP1_SE1_SDA			QUP1 In-Band interrupt I3C SE clock 1	
		QUP1_SE0_L2			QUP 1 SE 0 lane 2: SPI_SCLK/UART_TX	
BB48	GPIO_40		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE1_L1			QUP 1 SE 1 lane 1: I2C_SCL/SPI_MOSI/UART_RFR	
		IBI_I3C_QUP1_SE1_SCL			QUP1 In-Band interrupt I3C SE data 1	
		QUP1_SE0_L3			QUP 1 SE 0 lane 3: SPI_CS_0/UART_RX	
BJ39	GPIO_41		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE3_L2			QUP 1 SE 3 lane 2: UART_TX/HS-UART_TX	
		QUP1_SE3_L0			QUP 1 SE 3 lane 0: UART_CTS/HS-UART_CTS	
		BOOT_CONFIG[9]			Boot configuration bit 9	
BG39	GPIO_42		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE3_L3			QUP 1 SE 3 lane 3: UART_RX/HS-UART_RX	
		QUP1_SE3_L1			QUP 1 SE 3 lane 1: UART_RFR/HS-UART_RFR	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		MDP_VSYNC_E			MDP vertical sync – external	
BE47	GPIO_43		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE7_L2			QUP 0 SE 7 lane 2: UART_TX	
		QUP0_SE7_L0			QUP 0 SE 7 lane 0: I2C_SDA/UART_CTS	
BE49	GPIO_44		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE7_L3			QUP 0 SE 7 lane 3: UART_RX	
		QUP0_SE7_L1			QUP 0 SE 7 lane 1: I2C_SCL/UART_RFR	
BF42	GPIO_45		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE4_L0			QUP 1 SE 4 lane 0: I2C_SDA/SPI_MISO/SPI-S_MISO/UART_CTS	
BJ41	GPIO_46		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE4_L1			QUP 1 SE 4 lane 1: I2C_SCL/SPI_MOSI/SPI_MOSI/UART_RFR	
BG41	GPIO_47		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE4_L2			QUP 1 SE 4 lane 2: SPI_SCLK/SPI-S_SCLK/UART_TX	
BH40	GPIO_48		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE4_L3			QUP 1 SE 4 lane 3: SPI_CS_0/SPI_CS/UART_RX	
BJ47	GPIO_49		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE5_L0			QUP 1 SE 5 lane 0: I2C_SDA/SPI_MISO/SPI-S_MISO/UART_CTS	
		CCI_TIMER3			Camera control interface timer 3	
		BOOT_CONFIG[6]			Boot configuration bit 6	
BG47	GPIO_50		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE5_L1			QUP 1 SE 5 lane 1: I2C_SCL/SPI_MOSI/SPI-S_MOSI	
		CCI_TIMER4			Camera control interface timer 4	
BH46	GPIO_51		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE5_L2			QUP 1 SE 5 lane 2	
		BOOT_CONFIG[7]			Boot configuration bit 7	
BF46	GPIO_52		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE5_L3			QUP 1 SE 5 lane 3 SPI_CS_0/SPI-S_CS/UART_RX	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		MDP_VSYNC_P			MDP vertical sync - primary	
		FORCED_USB_BOOT			Forced USB boot	
P6	GPIO_53		PX_3	PD:nppukp	Configurable I/O	Y
		BOOT_CONFIG[11]			Boot configuration bit 11	
BH50	GPIO_54		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SDA5			Camera I²C data 5	
BH52	GPIO_55		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SCL5			Camera I²C clock 5	
BC49	GPIO_56		PX_3	PD:nppukp	Configurable I/O	Y
BH24	GPIO_57		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SDA0			Camera I²C data 0	
BF24	GPIO_58		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SCL0			Camera I²C clock 0	
BG23	GPIO_59		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SDA2			Camera I²C data 2	
BF22	GPIO_60		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SCL2			Camera I²C clock 2	
		EDL_INDICATOR			EDL indicator	
BJ31	GPIO_61		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SDA4			Camera I²C data 4	
BG31	GPIO_62		PX_3	PD:nppukp	Configurable I/O	N
		CCI_I2C_SCL4			Camera I²C clock 4	
BH28	GPIO_63		PX_3	PD:nppukp	Configurable I/O	N
		CCI_TIMER0			Camera control interface timer 0	
		CCI_ASYNC_IN0			Camera control interface async 0	
BG27	GPIO_64		PX_3	PD:nppukp	Configurable I/O	N
		CCI_TIMER1			Camera control interface timer 1	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		CCI_ASYNC_IN1			Camera control interface async 1	
BF26	GPIO_65		PX_3	PD:nppukp	Configurable I/O	N
		CCI_TIMER2			Camera control interface timer 2	
		CCI_ASYNC_IN2			Camera control interface async 2	
BG49	GPIO_66		PX_3	PD:nppukp	Configurable I/O	Y
		BOOT_CONFIG[13]			Boot configuration bit 13	
BD48	GPIO_67		PX_3	PD:nppukp	Configurable I/O	Y
		CAM_MCLK0			Camera 0 MCLK	
BD50	GPIO_68		PX_3	PD:nppukp	Configurable I/O	Y
		CAM_MCLK1			Camera 1 MCLK	
		GCC_GP4_CLK_MIRB			General-purpose clock 4 B	
BD52	GPIO_69		PX_3	PD:nppukp	Configurable I/O	Y
		CAM_MCLK2			Camera 2 MCLK	
		GCC_GP3_CLK_MIRB			General-purpose clock 3 B	
BH30	GPIO_70		PX_3	PD:nppukp	Configurable I/O	Y
BF30	GPIO_71		PX_3	PD:nppukp	Configurable I/O	Y
BG29	GPIO_72		PX_3	PD:nppukp	Configurable I/O	Y
BH26	GPIO_73		PX_3	PD:nppukp	Configurable I/O	Y
		GP_PDM_MIRA[0]			General-purpose PDM output 0 A	
BJ25	GPIO_74		PX_3	PD:nppukp	Configurable I/O	Y
BG25	GPIO_75		PX_3	PD:nppukp	Configurable I/O	Y
		GP_PDM_MIRB[0]			General-purpose PDM output 0 B	
BF52	GPIO_76		PX_3	PD:nppukp	Configurable I/O	Y
		GCC_GP5_CLK_MIRA			General-purpose clock 5 A	
		BOOT_CONFIG[1]			Boot configuration bit 1	
BB50	GPIO_77		PX_3	PD:nppukp	Configurable I/O	Y
		GCC_GP5_CLK_MIRB			General-purpose clock 5 B	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
T6	GPIO_78		PX_3	PD:nppukp	Configurable I/O	Y
T8	GPIO_79		PX_3	PD:nppukp	Configurable I/O	Y
		EMAC0_PTP_AUX_TS_I_3			EMAC 0 trigger 3 A for auxiliary snapshot	
		EMAC0_PTP_PPS_O_3			EMAC 0 configurable PPS output 3	
		EMAC0_MCG_PST_TRIG_LW[3]			EMAC0 trigger input 3 for media clock generation	
BK52	GPIO_80		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE6_L0			QUP 0 SE 6 lane 0: I2C_SDA/SPI_MISO/UART_CTS	
		MDP0_VSYNC6_OUT			MDP 0 vertical sync 6 output	
BK50	GPIO_81		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE6_L1			QUP 0 SE 6 lane 1: I2C_SCL/SPI_MOSI/UART_RFR	
		MDP0_VSYNC7_OUT			MDP 0 vertical sync 7 output	
		GCC_GP2_CLK_MIRA			General-purpose clock 2 A	
BL51	GPIO_82		PX_3	PD:nppukp	Configurable I/O	N
		QUP0_SE6_L2			QUP 0 SE 6 lane 2: SPI_SCLK/UART_TX	
		GCC_GP3_CLK_MIRA			General-purpose clock 3 A	
		BOOT_CONFIG[10]			Boot configuration bit 10	
BM50	GPIO_83		PX_3	PD:nppukp	Configurable I/O	Y
		QUP0_SE6_L3			QUP 0 SE 6 lane 3: SPI_CS_0/UART_RX	
		GCC_GP4_CLK_MIRA			General-purpose clock 4 A	
BG45	GPIO_84		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE2_L0			QUP 1 SE 2 lane 0: I2C_SDA/SPI_MISO/UART_CTS/HS-UART_CTS	
		GCC_GP1_CLK_MIRA			General-purpose clock 1 A	
BH44	GPIO_85		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE2_L1			QUP 1 SE 2 lane 1: I2C_SCL/SPI_MOSI/UART_RFR/HS-UART_RFR	
BF44	GPIO_86		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE2_L2			QUP 1 SE 2 lane 2: SPI_SCLK/UART_TX/HS-UART_TX	
BG43	GPIO_87		PX_3	PD:nppukp	Configurable I/O	Y

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		QUP1_SE2_L3			QUP 1 SE 2 lane 3: SPI_CS_0/UART_RX/HS-UART_RX	
BH42	GPIO_88		PX_3	PD:nppukp	Configurable I/O	N
		QUP1_SE2_L4			QUP 1 SE 2 lane 4: SPI_CS_1	
BF36	GPIO_89		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE6_L0			QUP 1 SE 6 lane 0: I2C_SDA/SPI_MISO/UART_CTS	
		QUP1_SE7_L2			QUP 1 SE 7 lane 2: SPI_SCLK/UART_TX	
		MDP0_VSYNC0_OUT			MDP 0 vertical sync 0 output	
BG35	GPIO_90		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE6_L1			QUP 1 SE 6 lane 1: I2C_SCL/SPI_MOSI/UART_RFR	
		QUP1_SE7_L3			QUP 1 SE 7 lane 3: SPI_CS_0/UART_RX	
		MDP0_VSYNC1_OUT			MDP 0 vertical sync 1 output	
BH34	GPIO_91		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE7_L0			QUP 1 SE 7 lane 0: I2C_SDA/SPI_MISO/UART_CTS	
		QUP1_SE6_L2			QUP 1 SE 6 lane 2: SPI_SCLK/UART_TX	
		MDP0_VSYNC3_OUT			MDP 0 vertical sync 3 output	
		BOOT_CONFIG[0]			Boot configuration bit 0	
BF34	GPIO_92		PX_3	PD:nppukp	Configurable I/O	Y
		QUP1_SE7_L1			QUP 1 SE 7 lane 1: I2C_SCL/SPI_MOSI/UART_RFR	
		QUP1_SE6_L3			QUP 1 SE 6 lane 3: SPI_CS_0/UART_RX	
BC51	GPIO_93		PX_3	PD:nppukp	Configurable I/O	Y
		TIC_SEL_MD				
BG51	GPIO_94		PX_3	PD:nppukp	Configurable I/O	Y
		EDP0_HOT_PLUG_DETECT			Enhanced DisplayPort 0 hot plug detect	
AW51	GPIO_95		PX_3	PD:nppukp	Configurable I/O	Y
AW49	GPIO_96		PX_3	PD:nppukp	Configurable I/O	N
		BOOT_CONFIG[8]			Boot configuration bit 8	
AH52	GPIO_97		PX_3	PD:nppukp	Configurable I/O	Y

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
		MI2S_MCLK0			External Master Clock 0	
		BOOT_CONFIG[2]			Boot configuration bit 2	
AH50	GPIO_98		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S1_SCK_OR_HS3_MI2S_SCK			MI^{2}S 1 clock or high-speed 3 MI^{2}S clock	
		BOOT_CONFIG[3]			Boot configuration bit 3	
AJ53	GPIO_99		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S1_WS_OR_HS3_MI2S_WS			MI^{2}S 1 word select or high-speed 3 MI^{2}S word select and synchronization	
AJ51	GPIO_100		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S1_DATA0_OR_HS3_MI2S_DATA0			MI^{2}S 1 serial data channel 0 or high speed 3 MI^{2}S serial data channel 0	
AJ49	GPIO_101		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S1_DATA1_OR_HS3_MI2S_DATA1			MI^{2}S 1 serial data channel 1 or high speed 3 MI^{2}S serial data channel 1	
AJ47	GPIO_102		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S2_SCK_OR_HS4_MI2S_SCK			MI^{2}S 2 clock or high-speed 4 MI^{2}S clock	
		BOOT_CONFIG[4]			Boot configuration bit 4	
AK52	GPIO_103		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S2_WS_OR_HS4_MI2S_WS			MI^{2}S 2 word select or high-speed 4 MI^{2}S word select and synchronization	
		BOOT_CONFIG[5]			Boot configuration bit 5	
AK50	GPIO_104		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S2_DATA0_OR_HS4_MI2S_DATA0			MI^{2}S 2 serial data channel 0 or high speed 4 MI^{2}S serial data channel 0	
AK48	GPIO_105		PX_3	PD:nppukp	Configurable I/O	Y
		MI2S2_DATA1_OR_HS4_MI2S_DATA1			MI^{2}S 2 serial data channel 1 or high speed 4 MI2S serial data channel 1	
		AUDIO_REF_CLK			Audio reference clock	
AL51	GPIO_106		PX_3	PD:nppukp	Configurable I/O	N
		HS_I2S0_SCK			High-speed 0 MI^{2}S clock	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
AL49	GPIO_107		PX_3	PD:nppukp	Configurable I/O	N
		HS_I2S0_WS			High-speed 0 MI^{2}S word select	
AL47	GPIO_108		PX_3	PD:nppukp	Configurable I/O	N
		HS_I2S0_DATA0			High-speed 0 MI^{2}S data 0	
AM52	GPIO_109		PX_3	PD:nppukp	Configurable I/O	N
		HS_I2S0_DATA1			High-speed 0 MI^{2}S data 1	
		MI2S_MCLK1			External master clock 1	
AM50	GPIO_110		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_0			See Table 2-5 for details	
AM48	GPIO_111		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_1			See Table 2-5 for details	
AN53	GPIO_112		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_2			See Table 2-5 for details	
AN51	GPIO_113		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_3			See Table 2-5 for details	
AN49	GPIO_114		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_4			See Table 2-5 for details	
AN47	GPIO_115		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_5			See Table 2-5 for details	
AP52	GPIO_116		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_6			See Table 2-5 for details	
AP50	GPIO_117		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_7			See Table 2-5 for details	
AP48	GPIO_118		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_8			See Table 2-5 for details	
AR51	GPIO_119		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_9			See Table 2-5 for details	

Table 2-3 Pin descriptions – general-purpose input/output ports (cont.)

Pin number	Pin name	Configurable function	Pad characteristics [a]		Functional description	Wakeup
			Voltage	Type		
AR49	GPIO_120		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_10			See Table 2-5 for details	
AR47	GPIO_121		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_11			See Table 2-5 for details	
AT52	GPIO_122		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_15			See Table 2-5 for details	
AT50	GPIO_123		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_16			See Table 2-5 for details	
AT48	GPIO_124		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_14			See Table 2-5 for details	
AU53	GPIO_125		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_12			See Table 2-5 for details	
AU51	GPIO_126		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_13			See Table 2-5 for details	
AU49	GPIO_127		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_17			See Table 2-5 for details	
AU47	GPIO_128		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_18			See Table 2-5 for details	
AV52	GPIO_129		PX_3	PD:nppukp	Configurable I/O	Y
		LPASS_19			See Table 2-5 for details	
AV50	GPIO_130		PX_3	PD:nppukp	Configurable I/O	Y
		LPASS_20			See Table 2-5 for details	
AV48	GPIO_131		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_21			See Table 2-5 for details	
AW47	GPIO_132		PX_3	PD:nppukp	Configurable I/O	N
		LPASS_22			See Table 2-5 for details	

[a] For parameter and acronym definitions, see I/O parameter definitions.

Table 2-4 Pin descriptions – RTSS I/O

Pin number	RTSS I/O	Alternate functions	Pad voltage	Pad type	Functional description	Wakeup function
BK10	RTSS_IO_0		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE0_L0			RTSS QUP0 SE0 lane 0	
BM10	RTSS_IO_1		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE0_L1			RTSS QUP0 SE0 lane 1	
BG9	RTSS_IO_2		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE0_L2			RTSS QUP0 SE0 lane 2	
BJ9	RTSS_IO_3		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_QUP0_SE0_L3			RTSS QUP0 SE0 lane 3	
BL9	RTSS_IO_4		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE1_L0			RTSS QUP0 SE1 lane 0	
BN9	RTSS_IO_5		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE1_L1			RTSS QUP0 SE1 lane 2	
BF8	RTSS_IO_6		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE1_L2			RTSS QUP0 SE1 lane 2	
		RTSS_BOOT_CONFIG[13]			RTSS boot configuration bit 13	
BH8	RTSS_IO_7		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_QUP0_SE1_L3			RTSS QUP0 SE1 lane 3	
		RTSS_BOOT_CONFIG[14]			RTSS boot configuration bit 14	
BN5	RTSS_IO_8		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE2_L2			RTSS QUP0 SE2 lane 2	
BL5	RTSS_IO_9		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_QUP0_SE2_L3			RTSS QUP0 SE2 lane 3	
BK8	RTSS_IO_10		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE3_L0			RTSS QUP0 SE3 lane 0	
		RTSS_BOOT_CONFIG[1]			RTSS boot configuration bit 1	
BM8	RTSS_IO_11		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE3_L1			RTSS QUP0 SE3 lane 1	
		RTSS_BOOT_CONFIG[2]			RTSS boot configuration bit 2	

Table 2-4 Pin descriptions – RTSS I/O (cont.)

Pin number	RTSS I/O	Alternate functions	Pad voltage	Pad type	Functional description	Wakeup function
BL7	RTSS_IO_12		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE3_L2			RTSS QUP0 SE3 lane 2	
		RTSS_QUP0_SE2_L0			RTSS QUP0 SE2 lane 0	
		RTSS_BOOT_CONFIG[3]			RTSS boot configuration bit 3	
BM6	RTSS_IO_13		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_QUP0_SE3_L3			RTSS QUP0 SE3 lane 3	
		RTSS_QUP0_SE2_L1			RTSS QUP0 SE2 lane 1	
BM12	RTSS_IO_14		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE4_L0			RTSS QUP0 SE4 lane 0	
		RTSS_BOOT_CONFIG[4]			RTSS boot configuration bit 4	
BG11	RTSS_IO_15		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE4_L1			RTSS QUP0 SE4 lane 2	
		RTSS_BOOT_CONFIG[5]			RTSS boot configuration bit 5	
BJ11	RTSS_IO_16		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE4_L2			RTSS QUP0 SE4 lane 2	
		RTSS_BOOT_CONFIG[11]			RTSS boot configuration bit 11	
BL11	RTSS_IO_17		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_QUP0_SE4_L3			RTSS QUP0 SE4 lane 3	
		RTSS_BOOT_CONFIG[12]			RTSS boot configuration bit 12	
BH10	RTSS_IO_18		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QUP0_SE4_L4			RTSS QUP0 SE4 lane 4	
		RTSS_BOOT_CONFIG[6]			RTSS boot configuration bit 6	
BH18	RTSS_IO_19		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_ERR1				
BK18	RTSS_IO_20		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_ERR2				
BK6	RTSS_IO_21		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_EMAC0_PHY_INTR_N			RTSS Ethernet interrupt 0	

Table 2-4 Pin descriptions – RTSS I/O (cont.)

Pin number	RTSS I/O	Alternate functions	Pad voltage	Pad type	Functional description	Wakeup function
		RTSS_RGMII_VOL_SEL			Voltage selection pin for RTSS RGMII interface	
BK4	RTSS_IO_22		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_MDC			RTSS RGMII 0 management interface clock	
BK2	RTSS_IO_23		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_MDIO			RTSS RGMII0 management interface I/O	
BJ3	RTSS_IO_24		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_RXC			RTSS RGMII 0 receive clock signal	
BJ1	RTSS_IO_25		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_RX_CTL			RTSS RGMII 0 receive control	
BH2	RTSS_IO_26		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_RXD0			RTSS RGMII 0 receive data 0	
BH4	RTSS_IO_27		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_RXD1			RTSS RGMII 0 receive data 1	
BG3	RTSS_IO_28		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_RXD2			RTSS RGMII 0 receive data 2	
BF4	RTSS_IO_29		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_RXD3			RTSS RGMII 0 receive data 3	
BJ5	RTSS_IO_30		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_TXC			RTSS RGMII 0 transmit clock signal	
BJ7	RTSS_IO_31		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_TX_CTL			RTSS RGMII 0 transmit control	
BH6	RTSS_IO_32		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_TXD0			RTSS RGMII 0 transmit data 0	
BG5	RTSS_IO_33		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_TXD1			RTSS RGMII 0 transmit data 1	
BG7	RTSS_IO_34		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_RGMII0_TXD2			RTSS RGMII 0 transmit data 2	
BF6	RTSS_IO_35		PX_3	PD:nppukp	Configurable I/O	N

Table 2-4 Pin descriptions – RTSS I/O (cont.)

Pin number	RTSS I/O	Alternate functions	Pad voltage	Pad type	Functional description	Wakeup function
		RTSS_RGMII0_TXD3			RTSS RGMII 0 transmit data 3	
BL3	RTSS_IO_36		PX_3	PD:nppukp	Configurable I/O	N
BK12	RTSS_IO_37		PX_3	PU:nppdkp	Configurable I/O	N
		RTSS_CAN0_TX			RTSS CAN 0 transmit	
BH12	RTSS_IO_38		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN0_RX			RTSS CAN 0 receive	
		RTSS_CC_GP4_CLK_MIRB			RTSS global general-purpose clock 4 B	
BN13	RTSS_IO_39		PX_3	PU:nppdkp	Configurable I/O	N
		RTSS_CAN1_TX			RTSS CAN 1 transmit	
		RTSS_CC_GP5_CLK_MIRA			RTSS global general-purpose clock 5 A	
BL13	RTSS_IO_40		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN1_RX			RTSS CAN 1 receive	
		RTSS_CC_GP5_CLK_MIRB			RTSS global general-purpose clock 5 B	
BJ13	RTSS_IO_41		PX_3	PU:nppdkp	Configurable I/O	N
		RTSS_CAN2_TX			RTSS CAN 2 transmit	
		RTSS_EMAC0_PTP_PPS_O_1_MIRA			RTSS EMAC0 configurable PPS output 1 A	
		RTSS_EMAC0_PTP_AUX_TS_I_0_MIRA			RTSS EMAC0 trigger 0 A for auxiliary snapshot	
		RTSS_CC_GP1_CLK_MIRA			RTSS global general-purpose clock 1 A	
BG13	RTSS_IO_42		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN2_RX			RTSS CAN 2 receive	
		RTSS_EMAC0_PTP_PPS_O_0_MIRA			RTSS EMAC0 configurable PPS output 0 A	
		RTSS_CC_GP2_CLK_MIRA			RTSS global general-purpose clock 2 A	
BM14	RTSS_IO_43		PX_3	PU:nppdkp	Configurable I/O	N
		RTSS_CAN3_TX			RTSS CAN 3 transmit	
		RTSS_EMAC0_PTP_PPS_O_0_MIRB			RTSS EMAC0 configurable PPS output 0 B	

Table 2-4 Pin descriptions – RTSS I/O (cont.)

Pin number	RTSS I/O	Alternate functions	Pad voltage	Pad type	Functional description	Wakeup function
		RTSS_CC_GP2_CLK_MIRB			RTSS global general-purpose clock 2 B	
BK14	RTSS_IO_44		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN3_RX			RTSS CAN 3 receive	
		RTSS_EMAC0_PTP_PPS_O_1_MIRB			RTSS EMAC0 configurable PPS output 1 B	
		RTSS_CC_GP3_CLK_MIRA			RTSS global general-purpose clock 3 A	
BH14	RTSS_IO_45		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN0_CTL			RTSS CAN0 control	
		RTSS_EMAC0_PTP_PPS_O_2			RTSS EMAC0 configurable PPS output 2	
		RTSS_CC_GP3_CLK_MIRB			RTSS global general-purpose clock 3 B	
BL15	RTSS_IO_46		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN1_CTL			RTSS CAN1 control	
		RTSS_EMAC0_PTP_AUX_TS_I_0_MIRB			RTSS EMAC0 trigger 0 B for auxiliary snapshot	
		RTSS_EMAC0_PTP_PPS_O_3			RTSS EMAC0 configurable PPS output 3	
		RTSS_CC_GP1_CLK_MIRB			RTSS global general-purpose clock 1 B	
BJ15	RTSS_IO_47		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN2_CTL			RTSS CAN2 control	
		RTSS_EMAC0_MCG_PST_TRIG_LW[0]			RTSS EMAC0 trigger input 0 for media clock generation	
		RTSS_CC_GP4_CLK_MIRA			RTSS global general-purpose clock 4 A	
BG15	RTSS_IO_48		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN3_CTL			RTSS CAN3 control	
		RTSS_EMAC0_MCG_PST_TRIG_LW[1]			RTSS EMAC0 trigger input 1 for media clock generation	
BM16	RTSS_IO_49		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_CAN4_CTL			RTSS CAN4 control	
		RTSS_EMAC0_MCG_PST_TRIG_LW[2]			RTSS EMAC0 trigger input 2 for media clock generation	
BK16	RTSS_IO_50		PX_3	PD:nppukp	Configurable I/O	Y

Table 2-4 Pin descriptions – RTSS I/O (cont.)

Pin number	RTSS I/O	Alternate functions	Pad voltage	Pad type	Functional description	Wakeup function
		RTSS_CAN5_CTL			RTSS CAN5 control	
		RTSS_EMAC0_MCG_PST_TRIG_LW[3]			RTSS EMAC0 trigger input 3 for media clock generation	
BL17	RTSS_IO_51	RTSS_INT	PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_BOOT_CONFIG[0]			RTSS boot configuration bit 0	
BM18	RTSS_IO_52		PX_3	PD:nppukp	Configurable I/O	Y
		PRTSS_ERR_N			PMIC RTSS domain error signal active low (internally pulled up to 1.8 V)	
BN17	RTSS_IO_53	RTSS_PMIC_INT_N	PX_3	PD:nppukp	Configurable I/O	Y
BH16	RTSS_IO_54		PX_3	PD:nppukp	Configurable I/O	Y
BG17	RTSS_IO_55		PX_3	PD:nppukp	Configurable I/O	Y
		RTSS_SLP_EN			RTSS sleep enable	
BF2	RTSS_IO_56		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_CS_N_0			RTSS QSPI0 chip-select 0	
BE7	RTSS_IO_57		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DQS			RTSS QSPI0 differential data strobe	
BJ17	RTSS_IO_58		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_FORCED_USB_BOOT			Bootstrap to enter USB0 emergency download mode for RTSS domain	
BE5	RTSS_IO_59		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_CLK			RTSS QSPI0 clock	
BE3	RTSS_IO_60		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[0]			RTSS QSPI0 data 0	
BE1	RTSS_IO_61		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[1]			RTSS QSPI0 data 1	
BD6	RTSS_IO_62		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[2]			RTSS QSPI0 data 2	
BD4	RTSS_IO_63		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[3]			RTSS QSPI0 data 3	

Table 2-4 Pin descriptions – RTSS I/O (cont.)

Pin number	RTSS I/O	Alternate functions	Pad voltage	Pad type	Functional description	Wakeup function
BD2	RTSS_IO_64		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[4]			RTSS QSPI0 data 4	
BC7	RTSS_IO_65		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[5]			RTSS QSPI0 data 5	
BC5	RTSS_IO_66		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[6]			RTSS QSPI0 data 6	
BC3	RTSS_IO_67		PX_3	PD:nppukp	Configurable I/O	N
		RTSS_QSPI0_DATA[7]			RTSS QSPI0 data 7	

Table 2-5 Pin descriptions – MD_LPASS

LPASS I/O	Configurable function	Functional description
LPASS_0		(GPIO_110)
	LPI_QUA_MI2S_SCK	LPI quad I²S clock
LPASS_1		(GPIO_111)
	LPI_QUA_MI2S_WS	LPI quad I²S word select
LPASS_2		(GPIO_112)
	LPI_QUA_MI2S_DATA0	LPI quad I²S serial channel 0
LPASS_3		(GPIO_113)
	LPI_QUA_MI2S_DATA1	LPI quad I²S serial channel 1
LPASS_4		(GPIO_114)
	LPI_QUA_MI2S_DATA2	LPI quad I²S serial channel 2
LPASS_5		(GPIO_115)
	LPI_QUA_MI2S_DATA3	LPI quad I²S serial channel 3
	EXT_MCLK1_C	MI2S external MCLK1 C
LPASS_6		(GPIO_116)
	LPI_I2S1_CLK	LPI I²S 1 clock
LPASS_7		(GPIO_117)
	LPI_I2S1_WS	LPI I²S 1 word select
LPASS_8		(GPIO_118)
	LPI_I2S1_DATA0	LPI I²S 1 serial channel 0
LPASS_9		(GPIO_119)
	LPI_I2S1_DATA1	LPI I²S 1 data
LPASS_10		(GPIO_120)
	LPI_I2S2_CLK	LPI I²S 2 clock
LPASS_11		(GPIO_121)
	LPI_I2S2_WS	LPI I²S 2 word select
LPASS_12		(GPIO_125)
	LPI_I2S4_CLK	LPI I²S 4 clock
LPASS_13		(GPIO_126)
	LPI_I2S4_WS	LPI I²S 4 word select
	EXT_MCLK1_A	MI²S external MCLK1 A
LPASS_14		(GPIO_124)
	EXT_MCLK1_D	MI²S external MCLK1 D
LPASS_15		(GPIO_122)
	LPI_I2S2_DATA0	LPI I²S 2 serial channel 0
LPASS_16		(GPIO_123)
	LPI_I2S2_DATA1	LPI I²S 2 serial channel 1
LPASS_17		(GPIO_127)
	LPI_I2S4_DATA0	LPI I²S 4 serial channel 0
LPASS_18		(GPIO_128)

Table 2-5 Pin descriptions – MD_LPASS (cont.)

LPASS I/O	Configurable function	Functional description
	LPI_I2S4_DATA1	LPI I²S 4 serial channel 1
LPASS_19		(GPIO_129)
	LPI_I2S3_CLK	LPI I²S 3 clock
LPASS_20		(GPIO_130)
	LPI_I2S3_WS	LPI I²S 3 word select
LPASS_21		(GPIO_131)
	LPI_I2S3_DATA0	LPI I²S 3 serial channel 0
LPASS_22		(GPIO_132)
	LPI_I2S3_DATA1	LPI I²S 3 serial channel 1
	EXT_MCLK1_E	MI2S external MCLK1 E

Table 2-6 Pin descriptions – power supply pins

Pin number	Pin name	Functional description
AA33, AA39, AC39, AD38, AD40, AF34, AF38, AF40, U37, U39, V38, V40, W33, Y38, Y40	VDD_APC0	Power for the Kryo application processor
AB32, AC29, AC31, AD32, AF32	VDD_APC1	Power for the Kryo application processor
AY24	VDD_A_CSI_0_0P9	Power for MIPI CSI0 0.9 V analog circuits
BA25	VDD_A_CSI_0_1P2	Power for MIPI CSI0 1.2 V analog circuits
AY26	VDD_A_CSI_1_0P9	Power for MIPI CSI1 0.9 V analog circuits
BA27	VDD_A_CSI_1_1P2	Power for MIPI CSI1 1.2 V analog circuits
AY28	VDD_A_CSI_2_0P9	Power for MIPI CSI2 0.9 V analog circuits
BA29	VDD_A_CSI_2_1P2	Power for MIPI CSI2 1.2 V analog circuits
BA31	VDD_A_DSI_0_0P9	Power for MIPI DSI0 0.9 V analog circuits
AY32	VDD_A_DSI_0_1P2	Power for MIPI DSI0 1.2 V analog circuits
AY30	VDD_A_DSI_0_PLL_0P9	Power for MIPI DSI0 PLL 0.9 V
R25, R27, R33, R35	VDD_A_EBI_CCPLL_0P9	Power for EBI clock PLL 0.9 V circuits
N21, N29, N37, P16, P18, P20, P22, P24, P26, P28, P30, P32, P34, P36, P38, R15	VDD_A_EBI_PHY	Power for EBI PHY circuits
AH42	VDD_A_EDP_0_1P2	Power for eDP 0 1.2 V analog circuits
AG43	VDD_A_EDP_0_CORE	Power for eDP 0 analog circuits
AJ11, AJ13, AL11	VDD_A_PCIE_0_CORE	Power for PCIe 0 core analog circuits
AL13	VDD_A_PCIE_0_PLL_1P2	Power for PCIe 0 PLL 1.2 V analog circuits
AC11, AE15, AF10, AF14	VDD_A_PCIE_1_CORE	Power for PCIe 1 core analog circuits
AC13	VDD_A_PCIE_1_PLL_1P2	Power for PCIe 1 PLL 1.2 V analog circuits
AF46	VDD_A_QREFS_0P875_0	Reference voltage for QREFS 0 0.875 V circuits
AA13	VDD_A_QREFS_0P875_1	Reference voltage for QREFS 1 0.875 V circuits
W13	VDD_A_QREFS_1P2_1	Reference voltage for QREFS 1 1.2 V circuits
N13	VDD_A_QREFS_TX1_0P9	Reference voltage 0.88 V

Table 2-6 Pin descriptions – power supply pins (cont.)

Pin number	Pin name	Functional description
BD28	VDD_A_REFGEN_0P875_3	Reference voltage 0.875 V
BD30	VDD_A_REFGEN_1P2_3	Reference voltage 1.2 V
AN13	VDD_A_SGMII_0_0P9	Power for the first Ethernet interface
AP12	VDD_A_SGMII_0_1P2	Power for the second Ethernet interface
AF42	VDD_A_UFS_0_1P2	Power for the UFS 0 1.2 V analog circuits
AD42	VDD_A_UFS_0_CORE	Power for UFS 0 core circuits
AB10	VDD_A_USBHS_0_1P8	Power for the USB high-speed 0 1.8 V analog circuits
Y10	VDD_A_USBHS_0_3P1	Power for the USB high-speed 0 3.1 V analog circuits
AA11	VDD_A_USBHS_0_CORE	Power for the USB high-speed core circuits
W11	VDD_A_USBHS_1_1P8	Power for the USB high-speed 1 1.8 V analog circuits
U11	VDD_A_USBHS_1_3P1	Power for the USB high-speed 1 3.1 V analog circuits
V10	VDD_A_USBHS_1_CORE	Power for the USB high-speed 1 core circuits
V12	VDD_A_USBSS_0_1P2	Power for the USB SuperSpeed 0 1.2 V analog circuits
U13	VDD_A_USBSS_0_CORE	Power for the USB SuperSpeed 0 core circuits
AA27, AA31, AB28, AE27, AJ23, AK16, AK18, AK20, AK22, W31	VDD_CX	Power for digital core circuits
AG37, AH36, AH40, AJ35, AJ39	VDD_CX_LPI	Power for LPI digital core circuits
T18, T20, T22, T26, T28, T30, T34, T36, T38, U25, U31, U33	VDD_D_EBI	Power for EBI digital circuits
AL25, AL29, AN29, AR25, AR29, AT26, AT28, AV26, AV28	VDD_GFX	Power for graphics
L23, L31, M16, M18, M20, M22, M24, M26, M28, M30, M32, M34, M36, M38	VDD_IO_EBI_PHY	Power for EBI PHY circuits
AB26, AD26, AL35, AL37, AM38, AN31, AN37, AN39, AP36, AP40, AR31, AR33, AR37, AR39, AT34, AT36, AU31, AU33, AU35, AU39, AV32, AV36, AV40, W25, W27, Y26	VDD_MM	Power for multimedia subsystem circuits
AE33, AG31, AG33, AG35, AJ31	VDD_MX_A	Power for memory circuits
AC25, AD24, AE25, AF24, AK24, AK26, AK32, AK34, AL31, AM24	VDD_MX_C	Power for memory circuits
AJ33, AK36, AK38	VDD_MX_LPI	Power for LPI memory circuit
AD16, AD18, AF16, AF18, AG17, AG23, AH14, AH16, AH20, AH22, AJ15, AJ19	VDD_NSP_CX	Power for HTP current sensor
AA15, AA17, AB24, AC15, AC17, AC19, AC21, AD22, U17, U21, V14, V16, V18, V22, V24, W15, Y16, Y24	VDD_NSP_HMX	Power for HTP HMX
AK42	VDD_PX0	Power for pad group 0
AJ43, L25, L27, L33, L35	VDD_PX1	Power for pad group 1
AG45, M12	VDD_PX11	Power for pad group 11
AL43, AR43, AR9, BB36, BB40, BD16, BD20, N11	VDD_PX3	Power for pad group 3

Table 2-6 Pin descriptions – power supply pins (cont.)

Pin number	Pin name	Functional description
AP10	VDD_PX7	Power for pad group 7
AE45	VDD_PX9	Power for pad group 9
BB18, BB22	VDD_QFPROM	Power for programming the QFPROM
AT10	VDD_QFPROM_RTSS	Power for programming the QFPROM; RTSS
AN21, AN23, AP16, AP20, AP24, AR15, AR17, AR21, AR23, AT18, AT20, AU15, AU17, AU23, AV16, AV20, AV22, AV24	VDD_RTSS_CX	Power for RTSS core circuits
AL19, AL21, AM16, AM18, AM22, AN17, AN19	VDD_RTSS_MX	Power for RTSS on-chip memory circuits
BC23	VDD_RTSS_PX0	Power for RTSS pad group 0
BC15	VDD_RTSS_PX11	Power for RTSS pad group 11
AR11, BC17, BC21	VDD_RTSS_PX3	Power for RTSS pad group 3

Table 2-7 Pin descriptions – ground pins

Pin number	Pin name	Functional description
A13, A21, A27, A33, A39, A49, A5, AA1, AA25, AA41, AA45, AA49, AA5, AA53, AA7, AA9, AB12, AB14, AB16, AB2, AB30, AB38, AB40, AB42, AB46, AB52, AB6, AC1, AC23, AC27, AC3, AC33, AC41, AC45, AC49, AC5, AC51, AC53, AC9, AD10, AD14, AD20, AD4, AD46, AE1, AE17, AE23, AE39, AE41, AE43, AE49, AE5, AE53, AE9, AF36, AF44, AF50, AF6, AG1, AG15, AG3, AG39, AG41, AG5, AG51, AG53, AG7, AG9, AH10, AH12, AH18, AH24, AH32, AH34, AH38, AH4, AH44, AH46, AJ1, AJ17, AJ21, AJ37, AJ41, AJ45, AJ5, AJ9, AK10, AK12, AK14, AK40, AK44, AK46, AK6, AL1, AL17, AL23, AL27, AL3, AL33, AL39, AL41, AL45, AL5, AL53, AM10, AM12, AM2, AM20, AM30, AM36, AM4, AM40, AM42, AM44, AM46, AM6, AM8, AN11, AN41, AN43, AN45, AN9, AP18, AP22, AP30, AP38, AP4, AP42, AP44, AP46, AP8, AR13, AR19, AR27, AR3, AR35, AR41, AR45, AR53, AT12, AT14, AT16, AT2, AT22, AT24, AT30, AT32, AT38, AT40, AT42, AT44, AT46, AT8, AU11, AU13, AU19, AU21, AU25, AU27, AU29, AU37, AU41, AU43, AU45, AU9, AV10, AV12, AV14, AV18, AV30, AV34, AV38, AV42, AV44, AV46, AW1, AW13, AW15, AW17, AW19, AW21, AW23, AW25, AW27, AW29, AW31, AW33, AW35, AW37, AW39, AW41, AW43, AW45, AW53, AW9, AY10, AY12, AY14, AY16, AY34, AY36, AY38, AY40, AY42, AY44, AY46, B10, B16, B18, B24, B30, B34, B36, B42, B44, B46, B8, BA11, BA13, BA15, BA23, BA33, BA35, BA37, BA39, BA41, BA43, BA45, BA9, BB10, BB12, BB14, BB16, BB20, BB24, BB26, BB28, BB30, BB32, BB34, BB38, BB42, BB44, BB46, BB8, BC1, BC11, BC13, BC19, BC25, BC27, BC29, BC31, BC33, BC35, BC37, BC39, BC41, BC43, BC45, BC53, BC9, BD10, BD12, BD14, BD18, BD22, BD24, BD26, BD32, BD34, BD36, BD38, BD40, BD42, BD44, BD46, BD8, BE11, BE13, BE15, BE17, BE19, BE21, BE23, BE25, BE27, BE29, BE31, BE33, BE35, BE37, BE39, BE41, BE43, BE45, BE9, BF10, BF12, BF14, BF16, BG1, BG53, BJ21, BJ23, BJ27, BJ29, BJ35, BJ37, BJ43, BJ45, BK20, BK24, BK30, BK32, BK34, BK40, BK42, BK46, BK48, BL19, BL31, BL39, BL49, BM38, BN11, BN15, BN19, BN23, BN25, BN27, BN37, BN45, BN49, BN7, C13, C19, C21, C23, C27, C31, C33, C39, C47, C49, C5, C7, D10, D16, D2, D24, D30, D4, D44, D46, D50, D52, D8, E13, E17, E19, E25, E27, E29, E35, E37, E39, E41, F10, F14, F18, F2, F20, F22, F32, F34, F36, F4, F40, F44, F46, F48, F50, F52, F6, F8, G11, G23, G25, G27, G29, G31, H16, H18, H2, H20, H22, H32, H34, H42, H44, H46, H48, H52, H6, H8, J11, J13, J15, J27, J3, J35, J37, J39, J41, J49, J5, J51, J9, K2, K24, K26, K28, K30, K38, K52, L11, L15, L17, L19, L21, L29, L3, L37, L39, L43, L45, L49, L5, L51, L7, M10, M14, M4, M40, M42, M44, M46, M48, M50, M52, N15, N17, N19, N23, N25, N27, N3, N31, N33, N35, N39, N41, N43, N45, N47, N49, N51, N53, P10, P12, P14, P2, P40, P42, P44, P46, P48, P52, R17, R19, R21, R23, R29, R3, R31, R37, R39, R43, R47, R49, R5, R53, T10, T12, T16, T24, T32, T4, T40, T42, T44, T48, U15, U19, U23, U27, U29, U35, U41, U43, U49, U5, U51, U53, U7, U9, V2, V20, V26, V32, V4, V44, V46, V48, V52, W1, W17, W3, W39, W45, W47, W49, W5, W53, W9, Y12, Y14, Y32, Y42, Y46, Y8	GND	Ground

Table 2-8 Pin descriptions – NC and DNC pins

Pin number	Pin name	Functional description
A1, A3, A51, A53, B2, B52, BL1, BL53, BM2, BM52, BN1, BN3, BN51, BN53, C1, C53	NC	No connect; not connected internally.
AY8, AV8, AD44, Y44, BA21, AB44, AP14, AL15, T14, BA19, BA17, L13, L41, BF28, M8, BF40, BF18, AC43, W43, AY22, AA43, AN15, AM14, R13, AY20, AY18, W41, V42	DNC	Do not connect; connected internally, do not connect externally.

3 Electrical specifications

This topic defines the device electrical performance specifications, including absolute maximum ratings, operating conditions, and subsystem (such as memory and connectivity) parameters, which are useful in deploying the device with optimal processing capabilities and high efficiency.

3.1 Absolute maximum ratings

The absolute maximum ratings shown in the following table reflect the stress levels that, if exceeded, may cause permanent damage to the device. No functionality is guaranteed outside the operating specifications. Functionality and reliability are only guaranteed within the operating conditions described in Section 3.2.

Table 3-1 Absolute maximum ratings

Parameter		Min	Max	Unit
Power-supply voltages				
VDD_APC0	Power for the cluster 0 Kryo processors	-0.3	1.0648	V
VDD_APC1	Power for the cluster 1 Kryo processors	-0.3	1.0648	V
VDD_A_EBI_0P9	Power for EBI PHY 0.9 V circuits	-0.3	1.0472	V
VDD_A_EBI_PLL	Power for EBI PHY PLL circuits			
VDD_CX	Power for digital core circuits	-0.3	1.0648	V
VDD_D_EBI	Power for EBI digital circuits			
VDD_CX_LPI	Power for Low-power island digital core circuits	-0.3	0.9768	V
VDD_GFX	Power for graphics	-0.3	1.0648	V
VDD_MM	Power for multimedia subsystem circuits	-0.3	1.0648	V
VDD_MX_A	Power for always-ON memory circuits	-0.3	0.979	V
VDD_MX_C	Power for collapsible memory circuits	-0.3	0.979	V
VDD_MX_LPI	Power for Low-power island memory circuit	-0.3	0.9768	V
VDD_NSP_CX	Power for Hexagon Tensor Processor current sensor	-0.3	1.0648	V
VDD_NSP_HMX	Power for Hexagon Tensor Processor HMX	-0.3	1.0648	V
VDD_RTSS_CX	Power for RTSS core circuits	-0.3	0.979	V
VDD_RTSS_MX	Power for RTSS on-chip memory circuits	-0.3	0.979	V
VDD_A_CSI_0_0P9	Power for MIPI CSI0 0.9 V analog circuits	-0.3	1.012	V
VDD_A_CSI_1_0P9	Power for MIPI CSI1 0.9 V analog circuits			
VDD_A_DSI_0_0P9	Power for MIPI DSI0 0.9 V analog circuits			
VDD_A_DSI_0_PLL_0P9	Power for MIPI DSI0 PLL 0.9 V			
VDD_A_EDP_0_CORE	Power for eDP 0 analog circuits			
VDD_A_QREFS_TX1_0P9	Reference voltage 0.875V			
VDD_A_REFGEN_0P875_3	Reference voltage 0.875V			
VDD_A_SGMII_0_0P9	Power for the first Ethernet interface			
VDD_A_UFS_0_CORE	Power for UFS 0 core circuits			
VDD_A_QREFS_0P875_0	Reference voltage for QREFS 0 0.875 V circuits			

Table 3-1 Absolute maximum ratings (cont.)

Parameter		Min	Max	Unit
VDD_A_QREFS_0P875_1	Reference voltage for QREFS 1 0.875 V circuits			
VDD_A_RSVD1_0P9	Power for reserved1 0.9 V rail			
VDD_A_PCIE_0_CORE	Power for PCIE 0 core analog circuits	-0.3	1.045	V
VDD_A_PCIE_1_CORE	Power for PCIE 1 core analog circuits			
VDD_A_CSI_0_1P2	Power for MIPI CSI0 1.2 V analog circuits	-0.3	1.375	V
VDD_A_CSI_1_1P2	Power for MIPI CSI1 1.2 V analog circuits			
VDD_A_DSI_0_1P2	Power for MIPI DSI0 1.2 V analog circuits			
VDD_A_EDP_0_1P2	Power for eDP 0 1.2 V analog circuits			
VDD_A_PCIE_0_PLL_1P2	Power for PCIE 0 PLL 1.2 V analog circuits			
VDD_A_PCIE_1_PLL_1P2	Power for PCIE 1 PLL 1.2 V analog circuits			
VDD_A_QREFS_1P2_1	Reference voltage for QREFS 1 1.2 V circuits			
VDD_A_REFGEN_1P2_3	Reference voltage 1.2 V			
VDD_A_SGMII_0_1P2	Power for the second Ethernet interface			
VDD_A_UFS_0_1P2	Power for the UFS 0 1.2 V analog circuits			
VDD_A_USBSS_0_1P2	Power for the USB SuperSpeed 0 1.2 V analog circuits			
VDD_A_RSVD2_1P2	Power for reserved 2 1.2 V rail			
VDD_A_RSVD5_3P1	Power for reserved 5 3.1 V rail	-0.3	3.993	V
VDD_A_USBHS_0_3P1	Power for the USB high-speed 0 3.1 V analog circuits			
VDD_A_RSVD7_1P8	Power for reserved 7 1.8 V rail	-0.3	2.178	V
VDD_A_USBHS_0_1P8	Power for the USB high-speed 0 1.8 V analog circuits			
VDD_A_USBHS_0_CORE	Power for the USB high-speed core circuits	-0.3	1.012	V
VDD_A_USBSS_0_CORE	Power for the USB SuperSpeed 0 core circuits			
VDD_A_RSVD6_CORE	Power for reserved 6 Core rail			
VDD_IO_EBI_PHY	Power for EBI PHY circuits	-0.3	0.627	V
VDD_PX9	Power for pad group 9	-0.3	1.43	V
VDD_PX1	Power for pad group 1	-0.3	1.287	V
VDD_PX11	Power for pad group 11	-0.3	1.43	V
VDD_QFPROM	Power for QFPROM	-0.3	2.079	V
VDD_PX0	Power for pad group 0	-0.3	2.145	V
VDD_RTSS_PX11	Power for RTSS pad group 11	-0.3	2.145	V
VDD_PX3	Power for pad group 3	-0.3	2.09	V
VDD_PX7	Power for pad group 7			
VDD_QFPROM_RTSS	Power for RTSS QFPROM	-0.3	2.09	V
VDD_RTSS_PX3	Power for RTSS pad group 3			
VDD_RTSS_PX0	Power for RTSS pad group 0	-0.3	2.145	
Thermal conditions				
T_s	Storage temperature [a]	-55	150	°C

[a] The storage temperature range applies when the device is in the OFF state (the device is not assembled in any platform and is not electrically connected to any voltage or I/O signals). Damage may occur when the device is subjected to this temperature for any length of time.

3.2 Operating conditions

Operating conditions include parameters such as power supply voltage, power distribution impedances, and thermal conditions. The QCS8275 chipset meets all performance specifications listed in the following table when used within the operating conditions.

Table 3-2 Operating conditions for voltage rails with AVS Type-1

Parameter		Min	Max	Unit
Power-supply voltages				
VDD_APC0	Power for the Kryo application processor	0.312	0.968	V
VDD_APC1	Power for the Kryo application processor	0.312	0.968	V
VDD_A_EBI_CCPLL_0P9	Power for EBI clock PLL 0.9 V circuits	0.312	0.952	V
VDD_A_EBI_PHY	Power for EBI PHY circuits			
VDD_CX	Power for digital core circuits	0.312	0.968	V
VDD_D_EBI	Power for EBI digital circuits			
VDD_CX_LPI	Power for LPI digital core circuits	0.312	0.888	V
VDD_GFX	Power for graphics	0.312	0.968	V
VDD_MM	Power for multimedia subsystem circuits	0.312	0.968	V
VDD_MX_A	Power for memory circuits	0.504	0.888	V
VDD_MX_C	Power for memory circuits	0.504	0.888	V
VDD_MX_LPI	Power for LPI memory circuit	0.568	0.888	V
VDD_NSP_CX	Power for Hexagon Tensor Processor current sensor	0.312	0.968	V
VDD_NSP_HMX	Power for Hexagon Tensor Processor HMX	0.312	0.968	V
VDD_RTSS_CX	Power for RTSS core circuits	0.320	0.890	V
VDD_RTSS_MX	Power for RTSS on-chip memory circuits	0.560	0.890	V

Table 3-3 Operating conditions for voltage rails with non-AVS

Parameter [a]		Min	Typ	Max	Unit
Power-supply voltages					
VDD_A_CSI_0_0P9	Power for MIPI CSI0 0.9 V analog circuits	0.83	0.88	0.92	V
VDD_A_CSI_1_0P9	Power for MIPI CSI1 0.9 V analog circuits				
VDD_A_DSI_0_0P9	Power for MIPI DSI0 0.9 V analog circuits				
VDD_A_DSI_0_PLL_0P9	Power for MIPI DSI0 PLL 0.9 V				
VDD_A_EDP_0_CORE	Power for eDP 0 analog circuits				
VDD_A_QREFS_TX1_0P9	Reference voltage (0.875 V)				
VDD_A_REFGEN_0P875_3	Reference voltage (0.875 V)				
VDD_A_SGMII_0_0P9	Power for the first Ethernet interface				
VDD_A_UFS_0_CORE	Power for UFS 0 core circuits				
VDD_A_QREFS_0P875_0	Reference voltage for QREFS0 0.875 V circuits				
VDD_A_QREFS_0P875_1	Reference voltage for QREFS1 0.875 V circuits				
VDD_A_RSVD1_0P9	Power for reserved 1 0.9 V rail				

Table 3-3 Operating conditions for voltage rails with non-AVS (cont.)

Parameter [a]		Min	Typ	Max	Unit
VDD_A_PCIE_0_CORE	Power for PCIe 0 core analog circuits	0.88	0.910	0.96	V
VDD_A_PCIE_1_CORE	Power for PCIe 1 core analog circuits				
VDD_A_CSI_0_1P2	Power for MIPI CSI0 1.2 V analog circuits	1.15	1.2	1.25	V
VDD_A_CSI_1_1P2	Power for MIPI CSI1 1.2 V analog circuits				
VDD_A_DSI_0_1P2	Power for MIPI DSI0 1.2 V analog circuits				
VDD_A_EDP_0_1P2	Power for eDP 0 1.2 V analog circuits				
VDD_A_PCIE_0_PLL_1P2	Power for PCIe 0 PLL 1.2 V analog circuits				
VDD_A_PCIE_1_PLL_1P2	Power for PCIe 1 PLL 1.2 V analog circuits				
VDD_A_QREFS_1P2_1	Reference voltage for QREFS 1 1.2 V circuits				
VDD_A_REFGEN_1P2_3	Reference voltage (1.2 V)				
VDD_A_SGMII_0_1P2	Power for the second Ethernet interface				
VDD_A_UFS_0_1P2	Power for the UFS 0 1.2 V analog circuits				
VDD_A_USBSS_0_1P2	Power for the USB SuperSpeed 0 1.2 V analog circuits				
VDD_A_RSVD2_1P2	Power for reserved 2 1.2 V rail				
VDD_A_RSVD5_3P1	Power for reserved 5 3.1 V rail	2.97	3.070	3.63	V
VDD_A_USBHS_0_3P1	Power for the USB high-speed 0 3.1 V analog circuits				
VDD_A_RSVD7_1P8	Power for reserved 7 1.8 V rail	1.62	1.800	1.98	V
VDD_A_USBHS_0_1P8	Power for the USB high-speed 0 1.8 V analog circuits				
VDD_A_USBHS_0_CORE	Power for the USB high-speed core circuits	0.83	0.880	0.92	V
VDD_A_USBSS_0_CORE	Power for the USB SuperSpeed 0 core circuits				
VDD_A_RSVD6_CORE	Power for reserved 6 core rail				
VDD_IO_EBI_PHY	Power for EBI PHY circuits	0.27	0.500	0.57	V
VDD_PX0	Power for pad group 0	1.65	1.800	1.95	V
VDD_PX1	Power for pad group 1	1.01	1.100	TBD	V
VDD_PX11	Power for pad group 11	1.10	1.200	1.3	V
VDD_PX3	Power for pad group 3	1.7	1.800	1.9	V
VDD_PX7	Power for pad group 7				
VDD_PX9	Power for pad group 9	1.1	1.200	1.3	V
VDD_QFPROM	Power for programming the QFPROM	1.71	1.800	1.89	V
VDD_QFPROM_RTSS	Power for programming the QFPROM; RTSS	1.7	1.800	1.9	V
VDD_RTSS_PX0	Power for RTSS pad group 0	1.65	1.800	1.95	V
VDD_RTSS_PX11	Power for RTSS pad group 11	1.65	1.800	1.95	V
VDD_RTSS_PX3	Power for RTSS pad group 3	1.7	1.800	1.9	V
VDD_RTSS_PX8	Power for RTSS pad group 8	1.7	1.800	1.9	V
Thermal conditions					
T	Device operating temperature	T_a = -40	–	T_j = +125	°C

[a] Parts with voltages outside of the specified ranges are not guaranteed to operate properly.

3.3 Power distribution network

A detailed power delivery network specification is available in the *QCS8275 Chipset Power Delivery Network Specification* (80-73475-1P) document.

3.4 Average operating current

Detailed current consumption information and details about the operating modes tested are available in *QCS8275 Current Consumption Data* (to be released).

3.5 Digital logic characteristics

A digital I/O's performance specification depends on its pad type, its usage, and/or its supply voltage.

- Some are dedicated for interconnections between the QCS8275 device and other ICs within the QTI chipset; therefore, specifications are not required.
- Some are defined by existing standards, such as I^2C and SPI. QTI devices comply with those standards; therefore, additional specifications are not required.
- All other digital I/Os require performance specifications.
- Back powering protection is not supported in digital pads unless otherwise specified in the following table. All non-bpp digital pads must be kept at less than 50 mV and with a maximum of 1 µA current being injected into the pad to avoid potential back-power scenarios.

NOTE Unless otherwise specified, all device characterizations are carried out across the specified operating temperature and voltage ranges.

Table 3-4 Digital I/Os specified in this section

Pad voltage	Usage	Table
1.8 V	VDDPX_0, VDD_RTSS_PX_0, VDDPX_3, VDD_RTSS_PX_3	Table 3-5
1.8 V	VDD_RTSS_PX_RGMII	Table 3-7
1.2 V	VDDPX_9/VDDPX_10	Table 3-9

Table 3-5 DC specification of 1.8 V I/Os

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.65 × VDDPX_x	VDDPX_x [a] + 0.3	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	-0.3	0.35 × VDDPX_x [a]	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.7 × VDDPX_x	VDDPX_x [a] + 0.3	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	-0.3	0.3 × VDDPX_x [a]	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN= low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	300	–	mV
I_{IH}	Input high leakage current [b]	–	1.0	µA
I_{IL}	Input low leakage current [b]	-1.0	–	µA
R_{PD}	Pull-down resistance	20k	60k	Ω
R_{PU}	Pull-up resistance	20k	60k	Ω

Table 3-5 DC specification of 1.8 V I/Os (cont.)

Parameter	Description	Min	Max	Unit
R_{KP}	Bus keeper resistor	20k	60k	Ω
V_{OH}	High-level output voltage, CMOS	VDDPX_x [a] - 0.45	VDDPX_x [a]	V
V_{OL}	Low-level output voltage, CMOS	0.0	0.45	V

[a] VDDPX_x can be VDDPX_0, VDD_RTSS_PX_0, VDDPX_3, VDD_RTSS_PX_3 depending on the pad group.
[b] I_{IH} and I_{IL} values are based on nominal PVT (TT/25°C).

NOTE V_{OL} and V_{OH} are measured at I_{OL} and I_{OH} respectively, with 16 mA current across min/max voltages and operating temperatures. For more information regarding possible drive strength settings, see Table 3-6.

Table 3-6 Possible drive strength settings

BIT configuration	Drive strength (mA)
000	2
001	4
010	6
011	8
100	10
101	12
110	14
111	16

Table 3-7 DC specifications for RGMII 1.8 V mode (VDD_RTSS_PX_RGMII)

Parameter	Description	Min	Max	Unit
V_{OL}	Low-level output voltage [a]	–	0.45	V
V_{OH}	High-level output voltage [a]	VDDPX_x - 0.45	–	V
V_{IL}	Low-level input voltage	-0.3	0.35× VDDPX_x	V
V_{IH}	High-level input voltage	0.65× VDDPX_x	VDDPX_x+ 0.3	V
V_{HYS}	Input Hysteresis	100	–	mV
$R_{PULL-UP}$	Pull-up resistance	20	50	kΩ
$R_{PULL-DOWN}$	Pull-down resistance	20	50	kΩ
$R_{KEEPERUP}$	Keeper-up resistance	20	50	kΩ
$R_{KEEPERDOWN}$	Keeper-down resistance	20	50	kΩ
I_{IL}	Input low leakage current	–	5	µA
I_{IH}	Input high leakage current	-5	–	µA

[a] V_{OL} and V_{OH} are measured at I_{OL} and I_{OH} respectively, with 12 mA current across min and max voltages and operating temperatures. For more information regarding possible drive strength settings, see Table 3-6

Table 3-8 Possible drive strength

Drive strength	IOL (mA)	IOH (mA)
000	1.5	1.5
001	3.0	3.0
010	4.5	4.5
011	6.0	6.0
100	7.5	7.5
101	9.0	9.0
110	10.5	10.5
111	12.0	12.0

NOTE There is no industry-standard specification for a 1.8 V RGMII interface. Therefore, these specifications are based on JESD8-7A: *"1.8 V ± 0.15 V (normal range) and 1.2 V to 1.95 V (wide range) power supply voltage and interface standard for nonterminated digital integrated circuits."*

Table 3-9 Digital I/O characteristics for UFS_RESET and UFS_REF_CLK (VDDPX_9/VDDPX_10)

Parameter	Description	Min	Max	Unit
V_{OL}	Low-level output voltage	0	0.25 × VDDPX_x	V
V_{OH}	High-level output voltage	VDDPX_x × 0.75	VDDPX_x	V
$R_{PULL-UP}$	Pull-up resistance	20	–	kΩ
$R_{PULL-DOWN}$	Pull-down resistance	20	–	kΩ
I_{LEAK}	Standby leakage	-10	10	µA

NOTE VDDPX_x can be either VDDPX_9 or VDDPX_10.

V_{OL} and V_{OH} are measured at I_{OL} and I_{OH} respectively, with 4.8 mA current across min/max voltages and operating temperatures.

3.6 Memory support

QCS8275 supports four channels up to 32 GB LPDDR5 memory (JESD209-5C) with maximum speed at 3200 MHz.

3.7 Multimedia

Multimedia parameters requiring performance specifications are addressed in this section.

3.7.1 Camera interfaces

The QCS8275 device supports up to four D-PHY or C-PHY camera interfaces.

Table 3-10 Supported MIPI-CSI standards and exceptions

Applicable standard	Feature exceptions
MIPI Alliance Specification for CSI-2 v1.3	RAW7 is not supported; DPCM predictor 2 is not supported.
MIPI Alliance Specification for D-PHY v1.2	None
MIPI Alliance Specification for C-PHY v1.2	None

The following tables summarize the electrical and timing characteristics for CSI C-PHY and D-PHY interfaces at the SoC pin level.

Table 3-11 CSI C-PHY electrical characteristics

Parameter	Condition	Min	Typ	Max	Unit
Symbol rate per trio	–	–	4.5	–	Gsym/s per trio
Equivalent data rate	Per trio	–	10.26	–	Gb/s
Supported trios	–	1	–	3	trios
Single-ended signal swing	HS mode	TBD	450	535	mVpp
Common-mode voltage	HS mode	95	225	390	mV
Intra-trio skew	HS mode	–	–	5	ps
Inter-trio skew	HS mode	–	–	30	ps
HS jitter (total)	at max rate	–	–	0.55	UI
LP input low (VIL)	LP mode	–	–	550	mV
LP input high (VIH)	LP mode	740	–	–	mV

Table 3-12 CSI D-PHY electrical characteristics

Parameter	Condition	Min	Typ	Max	Unit
Data rate per lane	HS mode	–	2.5	–	Gb/s
Supported data lanes	–	1	–	4	lanes
HS differential swing	HS mode	140	200	270	mVpp
HS common-mode voltage	HS mode	70	–	330	mV
Receiver sensitivity	HS mode	40	–	–	mVpp
Intra-pair skew (P–N)	HS mode	–	–	5	ps
Lane-to-lane skew	HS mode	–	–	15	ps
HS clock frequency	Dedicated lane	–	1.25	–	GHz
HS clock jitter (total)	at max rate	–	–	0.5	UI
HS eye height	HS mode	140	–	–	mV
LP input high (VIH)	LP mode	–	–	740	mV
LP input low (VIL)	LP mode	550	–	–	mV
Hold time	–	0.3	–	0.5	UI
Setup time	–	–0.5	–	–0.3	UI

3.7.2 Display support

The QCS8275 device supports one D-PHY or C-PHY display.

Table 3-13 Supported MIPI-DSI standards and exceptions

Applicable standard	Feature exceptions
MIPI Alliance Specification for Display Serial Interface	None
MIPI Alliance Specification for D-PHY v1.2	None
MIPI Alliance Specification for C-PHY v1.1	None

The following tables summarize the electrical and timing characteristics for DSI C-PHY and D-PHY interfaces at the SoC pin level.

Table 3-14 DSI C-PHY electrical characteristics

Parameter	Condition	Min	Typ	Max	Unit
Symbol rate per trio	–	–	2.5	–	Gsym/s per trio
Equivalent data rate	Per trio	–	5.7	–	Gb/s
Supported trios	–	1	–	3	trios
Static common point voltage	HS mode	175	–	310	mV
Intra-trio skew [a]	HS mode	–	–	5	ps
Inter-trio skew [a]	HS mode	–	–	30	ps
HS jitter (total) [a]	at max rate	–	–	0.3	UI
HS eye height [a]	HS mode	80	–	–	mV
HS eye width [a]	HS mode	0.5	–	–	UI
LP output high (VOH)	LP mode	0.95	–	1.3	V
LP output low (VOL)	LP mode	-0.05	–	0.05	V

[a] Not characterized; guaranteed by design

Table 3-15 DSI D-PHY electrical characteristics

Parameter	Condition	Min	Typ	Max	Unit
Supported data rates	2.5, 2.1, 1.312				Gb/s per lane
Data rate per lane	HS mode	–	–	2.5	Gb/s
Supported data lanes	–	1	–	4	lanes
HS differential swing	HS mode	140	200	270	mVpp
HS common-mode voltage	HS mode	0.15	0.2	0.25	V
Intra-pair skew (P–N) [a]	HS mode	–	–	5	ps
Lane-to-lane skew [a]	HS mode	–	–	15	ps
HS jitter (total) [a]	at max rate	–	–	0.3	UI
HS eye height [a]	HS mode	80	–	–	mV
HS data rise time	HS mode	–	–	160	ps
HS data fall time	HS mode	–	–	160	ps
HS clock rise time	HS mode	–	–	160	ps
HS clock fall time	HS mode	–	–	160	ps
LP output high (VOH)	LP mode	0.95	–	1.3	V
LP output low (VOL)	LP mode	-0.05	–	0.05	V
LP input high (VIH)	LP mode	740	–	–	mV
LP input low (VIL)	LP mode	–	–	550	mV
LP Tx rise time	LP mode	–	–	25	ns
LP Tx fall time	LP mode	–	–	25	ns

[a] Not characterized, guaranteed by design.

3.7.3 DisplayPort (DP)

The DisplayPort (DP) interface provides high-speed digital video and audio transmission and complies with industry-standard specifications for interoperability.

Table 3-16 Supported DisplayPort standards and exceptions

Applicable standard	Feature exceptions
VESA DisplayPort v1.4	None

The following table summarizes the key electrical and link characteristics for the DisplayPort and embedded DisplayPort (eDP) interface.

Table 3-17 DP/eDP electrical and link characteristics

Parameter	Min	Typ	Max	Unit
Main link lane count	1	2	4	lanes
Main link per-lane rate (DP 1.4)	1.612	–	8.1	Gb/s per lane
Encoding/clocking	–	8b/10b	–	–
Total raw bandwidth (4 lanes)	–	–	32.4	Gb/s
AUX channel data rate	–	1	–	Mb/s
AUX differential peak-to-peak voltage (DP)	0.29	–	1.38	Vpp (diff)
AUX differential peak-to-peak voltage (eDP)	0.14	–	1.36	Vpp (diff)
AUX AC-coupling capacitance	75	100	200	nF

3.8 Connectivity

The connectivity functions supported by the QCS8275 that require electrical specifications include:

- USB host/slave support with built-in physical layer (PHY)
- Peripheral component interconnect express (PCIe) interfaces
- Universal flash storage (UFS)
- NOR flash storages (OSPI/QSPI)
- Reduced gigabit media independent interface (RGMII)
- High-speed serial gigabit media-independent interface (HSGMII)
- Through proper configuration of 20 main domain (MD) GPIO-based QUP SEs and 5 RTSS domain RTSS_IO-based QUP SEs:
  - Universal asynchronous receiver/transmitter (UART) ports
  - Inter-integrated circuit (I²C) interfaces
  - Serial peripheral interface (SPI) ports
- Dedicated I²C interfaces for camera (CCI I²C)

Pertinent specifications for these functions are detailed in the following subsections.

NOTE In addition to the following hardware specifications, see the latest software release notes for software-based performance features or limitations.

3.8.1 USB interfaces

The USB interface supports high-speed connectivity and complies with industry-standard specifications for interoperability.

Table 3-18 Supported USB standards and exceptions

Applicable standard	Feature exceptions
Universal Serial Bus Specification, Revision 3.1 (August 11, 2014 or later)	None
UTMI specification version 1.05, released on 3/29/2001	None
On-The-Go and Embedded Host Supplement to the USB 3.0 Specification (May 10, 2012, Revision 1.1 or later)	Attach detection protocol (ADP), role swap protocol (RSP), session request protocol (SRP), and host negotiation protocol (HNP)
Universal Serial Bus Serial Bus Specification, Revision 2.0 (April 27, 2000 or later)	None

The following table summarizes the high-speed USB electrical and timing characteristics at the SoC pin level.

Table 3-19 USB high-speed electrical characteristics

Parameter	Condition	Min	Typ	Max	Unit
Signaling rate	High-speed mode	479.76	480	480.24	Mb/s
Unit interval (UI)	at 480 Mb/s	–	2.083	–	ns
Differential output voltage (high)	HS driver	360	400	440	mVpp
Differential output voltage (low)	HS driver	–	–	10	mVpp
Differential interface impedance	Link (cable + termination)	76.5	90	103.5	Ω
Termination resistance (per line)	HS mode	40.5	45	49.5	Ω
Differential receiver sensitivity	Receiver	200	–	–	mV
Differential common-mode range	HS signaling	0.8	–	2.5	V
Rise time (10–90%)	Signal edge	0.5	–	–	ns
Fall time (10–90%)	Signal edge	0.5	–	–	ns
Chirp K/J width	HS detection	40	–	60	µs
Chirp J level (differential)	HS driver	700	–	1100	mV
Chirp K level (differential)	HS driver	-900	–	-500	mV
Bus pull-up (upstream port)	–	1.425	–	1.575	kΩ
Bus pull-down (downstream port)	–	14.25	–	15.75	kΩ

NOTE

- Compliant with the USB 2.0 high-speed specification (480 Mbps); values reflect SoC pin-level electrical and timing characteristics
- High-speed signaling uses bit-stuffed NRZI encoding with differential D+ and D- signaling
- High-speed operation requires chirp signaling during reset for speed negotiation with downstream devices
- No external reference clock is required; clock is recovered from the data stream
- Differential impedance target is 90 Ω ±10%; tight control of D+ and D- skew is recommended for margin

- On-die series termination is assumed unless otherwise specified in the hardware design guide
- All specifications apply across recommended operating voltage and temperature unless otherwise noted

The following table summarizes the SuperSpeed USB electrical and timing characteristics for Gen1 and Gen2 operation at the SoC pin level.

Table 3-20 USB SuperSpeed electrical characteristics

Parameter	Min	Typ	Max	Unit
Supported generation	–	Gen1, Gen2	–	–
Signaling rate (Gen1)	–	5	–	Gb/s
Signaling rate (Gen2)	–	10	–	Gb/s
Unit interval (Gen1)	199.94	200	200.065	ps
Unit interval (Gen2)	99.97	100	100.0325	ps
Differential output voltage	800	900	1200	mVpp-diff
Tx rise / fall time (Gen1)	20	–	80	ps
Tx rise / fall time (Gen2)	10	–	50	ps
Differential output impedance	70	90	110	Ω
Intra-pair skew (P–N)	–	–	5	ps
Lane-to-lane skew	–	–	70	ps
Tx total jitter (Gen1)	–	–	132	ps
Tx total jitter (Gen2)	–	–	67	ps
Rx eye height (Gen2)	0.07	–	–	mV
Rx eye height (Gen1)	0.1	–	–	mV

NOTE

- Compliant with USB-IF USB 3.1/USB 3.2 specifications for SuperSpeed operation.
- This table applies to SuperSpeed differential pairs (TX/RX); USB 2.0 D+/D− signaling is specified separately.
- SuperSpeed signaling uses embedded clocking; an external reference clock is required only for PHY PLL operation.
- Differential routing requirement: 90 Ω ±10% impedance, with strict control of intra-pair and inter-pair skew.
- AC coupling capacitors are required on Tx SuperSpeed lanes and should be placed near SoC pins.
- Supports LFPS-based power management, including U1, U2, and U3 states.
- All parameters apply across the recommended voltage and temperature range unless otherwise stated.

3.8.2 PCIe interface

The PCI Express (PCIe) interface supports high-speed serial communication and complies with industry-standard specifications for interoperability.

Table 3-21 Supported PCIe standards and exceptions

Applicable standard	Feature exceptions
PCI_Express_Base_Specification_Revision_4.0	Link configure capability
	Lane margining at receiver
	L0s link state

The following table summarizes the electrical and timing characteristics for the high-speed serial interface operating at Gen4 and Gen3 data rates. The parameters represent SoC pin-level requirements across supported operating conditions.

Table 3-22 PCIe Gen4 and Gen3 electrical and timing characteristics

Parameter	Gen4			Gen3			Unit
	Min	Typ	Max	Min	Typ	Max	
Signaling rate	–	–	16	–	–	8	GT/s
Unit interval (UI)	62.48	62.5	62.64	124.96	125	125.28	ps
Differential voltage (Tx)	600	–	1200	800	–	1200	mVpp-diff
Single-ended swing (Tx)	300	–	600	400	–	600	mVpp
Tx common-mode voltage	0.7	–	0.9	0.7	–	0.9	V
Differential output impedance	80	–	120	80	–	120	Ω
Receiver input sensitivity	30	–	–	85	–	–	mV
AC coupling capacitance	75	100	200	75	100	200	nF
Intra-pair skew (P–N)	–	–	5	–	–	20	ps
Lane-to-lane skew	–	–	20	–	–	50	ps
Tx total jitter (TJ)	–	–	15.85	–	–	33.83	ps
REFCLK frequency	99.8	100	100.2	99.8	100	100.2	MHz
REFCLK duty cycle	45	50	55	45	50	55	%
REFCLK RMS jitter (no SSC)	–	–	1	–	–	1	ps
REFCLK peak-to-peak jitter	–	–	10	–	–	10	ps
Spread spectrum clocking (down spread)	-0.5	-0.3	0	-0.5	-0.3	0	%
PERST# assertion width	100	–	–	100	–	–	ms
PERST# deassertion to LTSSM	2	–	–	2	–	–	ms
Receiver detect voltage	200	–	600	200	–	600	mV
Receiver detect resistance	40	–	60	40	–	60	Ω

3.8.3 UFS interface

Table 3-23 Supported UFS standards and exceptions

Applicable standard	Feature exceptions
Universal Flash Storage (UFS), Version 3.1	–

3.8.4 SGMII interface

The Serial Gigabit Media Independent Interface (SGMII) provides a serial link between MAC and PHY, enabling Ethernet connectivity over a differential pair with embedded clocking.

Table 3-24 Supported SGMII standards and exceptions

Applicable standard	Feature exceptions
IEEE 802.3	None

The following tables summarize the SoC pin-level electrical and timing characteristics for SGMII operation at *3.125 Gbps* and *1.25 Gbps*.

Table 3-25 SGMII characteristics at 3.125 Gbps

Parameter	Min	Typ	Max	Unit
Supported link speed	–	–	2500	Mb/s
Line coding	–	8b/10b	–	–
Serial line rate	–	3.125	–	Gb/s
Unit interval (UI)	–	320	–	ps
Tx rise/fall time	60	–	130	ps
Intra-pair skew (P–N)	–	–	15	ps
Lane-to-lane skew	–	–	100	ps
Tx total jitter (TJ)	–	–	0.35	UI
Tx random jitter (RJ, RMS)	–	–	0.18	UI
Tx deterministic jitter	–	–	0.17	UI
Tx duty cycle distortion (DCD)	–	–	2	%
Rx eye height	200	–	–	mV
Rx eye width	0.35	–	–	UI

Table 3-26 SGMII characteristics at 1.25 Gbps

Parameter	Min	Typ	Max	Unit
Supported link speed	–	–	1000	Mb/s
Line coding	–	8b/10b	–	–
Serial line rate	–	1.25	–	Gb/s
Unit interval (UI)	–	800	–	ps
Tx rise/fall time	100	–	200	ps
Intra-pair skew (P–N)	–	–	20	ps

Table 3-26 SGMII characteristics at 1.25 Gbps (cont.)

Parameter	Min	Typ	Max	Unit
Lane-to-lane skew	–	–	100	ps
Tx total jitter (TJ)	–	–	0.30	UI
Tx random jitter (RJ, RMS)	–	–	0.20	UI
Tx deterministic jitter	–	–	0.10	UI
Rx eye height	100	–	–	mV
Rx eye width	0.35	–	–	UI

NOTE

- Compliant with SGMII specification and IEEE 802.3 (1000BASE-X PCS/PMA)
- Uses 8b/10b encoding with a fixed 1.25 Gb/s serial line rate, independent of negotiated link speed
- Clock is embedded in the serial data stream; an optional 25 MHz reference clock may be used for PLL operation, depending on the SoC implementation
- Differential routing requires controlled impedance of 100 Ω ± 10% with tight control of intra-pair skew
- Auto-negotiation supports 10/100/1000 Mb/s operation over a single differential pair
- AC coupling is required on Tx differential signals unless otherwise specified in the platform design
- Specifications apply across recommended operating voltage and temperature unless otherwise noted

3.8.5 Octa-SPI/Quad-SPI interface

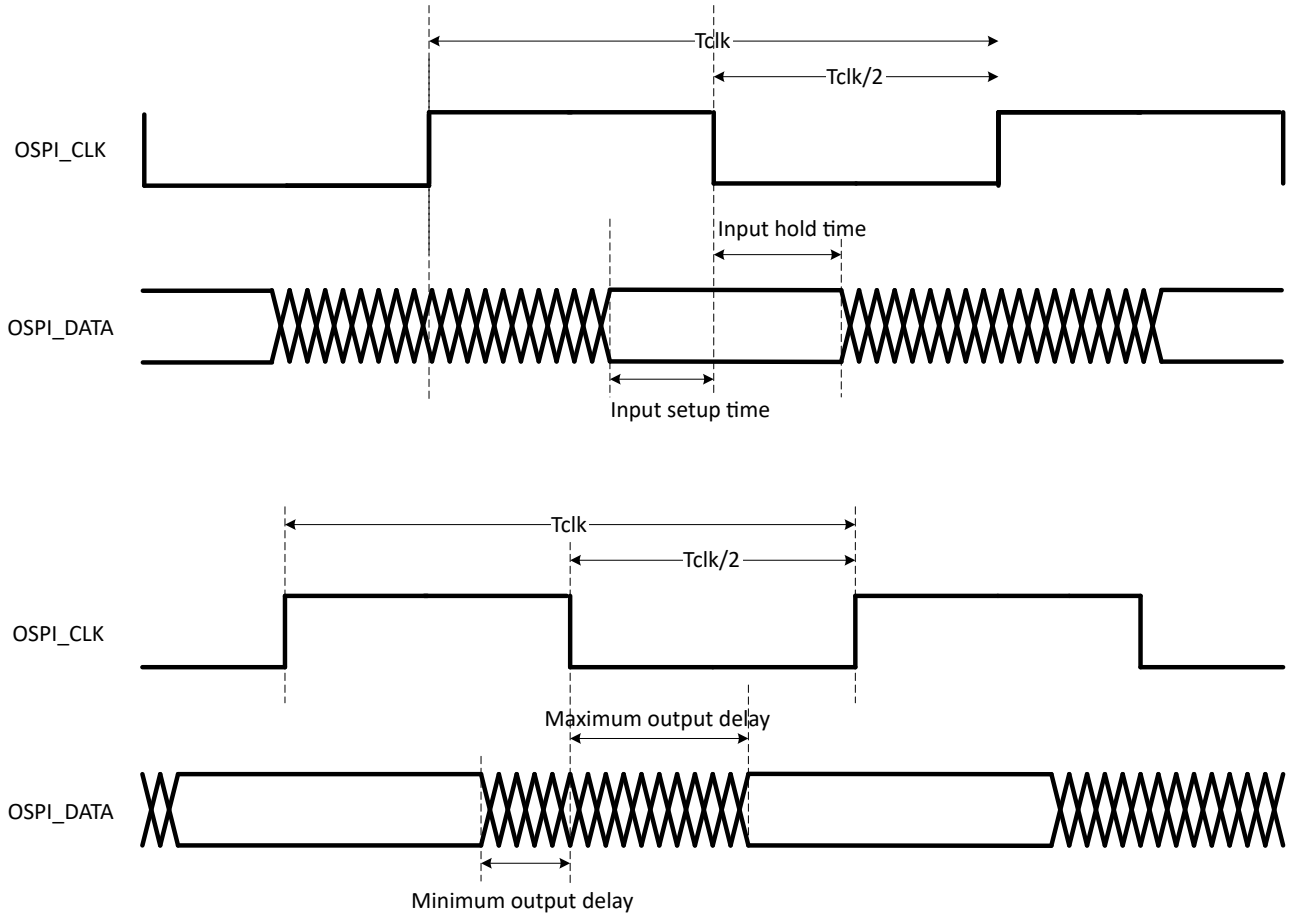


Figure 3-1 Octa-SPI/Quad-SPI SDR timing diagram

Table 3-27 Octa-SPI/Quad-SPI interface SDR timing

Parameter	Comments	Min	Max	Unit
1/T	OSPI/QSPI clock frequency	–	90	MHz
t(mis)	Master input setup time	1.7	–	ns
t(mih)	Master input hold time	1.3	–	ns
t(mod)	Master output delay	-2.00	2.00	ns

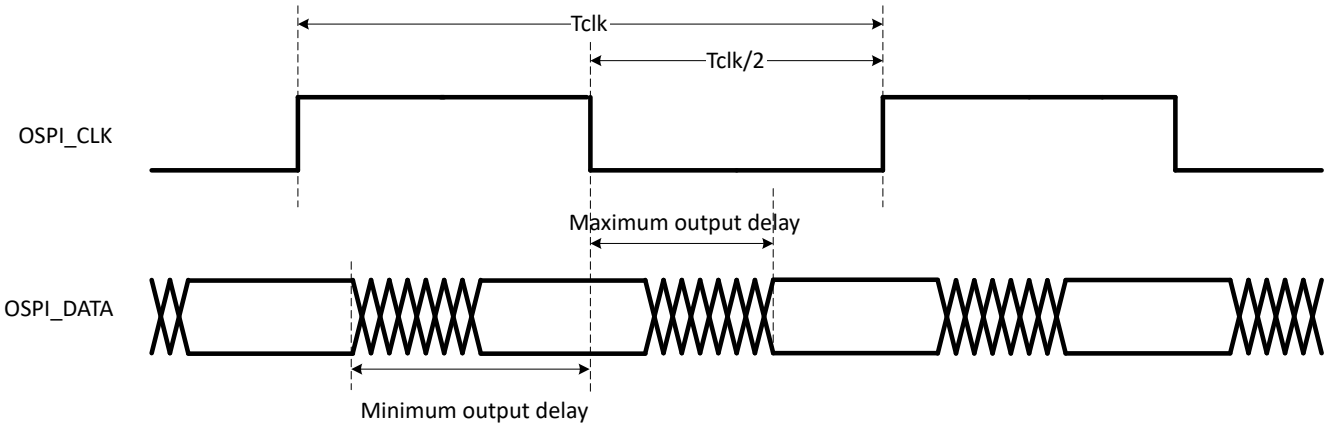


Figure 3-2 Octa-SPI DDR timing diagram

Table 3-28 Octa-SPI interface DDR timing

Parameter	Comments	Min	Max	Unit
1/T	OSPI clock frequency	–	166	MHz
t(skew)	DQ to DQS skew	–	530	ps
t(mod)	Master output delay	-2.00	2.00	ns

NOTE DDR is not supported on Quad-SPI operation.

3.8.6 RGMII interface

Table 3-29 Supported RGMII standards and exceptions

Applicable standard	Feature exceptions
Reduced Gigabit Media Independent Interface (RGMII) v1.3 and v2.0	None

Parameter	Min	Typ	Max	Unit
Supported speeds	10/100/1000			Mb/s
Reference clock (TXC/RXC)	2.5/25/125			MHz
Data transfer	DDR at 1000 Mbps only			
Signal electrical levels (CMOS)				
I/O supply voltage (VDDIO)	1.7	1.8	1.9	V
Input high voltage (VIH)	0.65 × VDD_PXx	—	VDD_PXx + 0.3	V
Input low voltage (VIL)	-0.3	—	0.35 × VDD_PXx	V
Output high voltage (VOH)	VDD_PXx - 0.45	—	—	V
Output low voltage (VOL)	—	—	0.45	V
Timing (1000BASE-T)				
TXD setup time	1.2	—	—	ns
TXD hold time	1.2	—	—	ns
RXD setup time	1	—	—	ns
RXD hold time	1	—	—	ns

Parameter	Min	Typ	Max	Unit
Clock duty cycle	45	50	55	%
Management interface (MDIO/MDC)				
MDC clock frequency	—	2.5		MHz
Input high voltage (VIH)	0.65 × VDD_PXx	—	VDD_PXx + 0.3	
Input low voltage (VIL)	-0.3	—	0.35 × VDD_PXx	
Output high voltage (VOH)	VDD_PXx - 0.45	—	—	V
Output low voltage (VOL)	—	—	0.45	V

When the MDIO signal is sourced by the PHY, it is sampled by the SoC synchronous to the rising edge of MDC. The clock-to-output delay from the PHY shall be a minimum of 0 ns and a maximum of 300 ns.

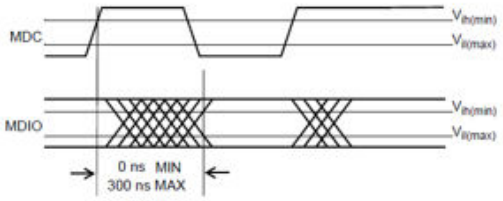


Figure 3-3 MDIO sourced by PHY

When the SoC sources the MDIO signal, it shall provide a minimum setup time of 10 ns and a minimum hold time of 10 ns, referenced to the rising edge of MDC.

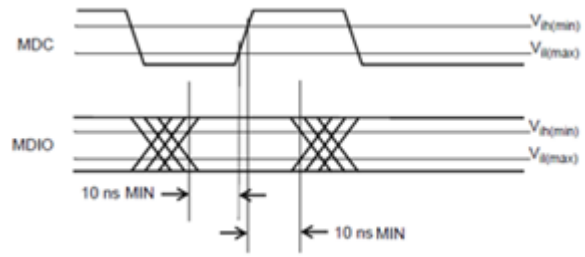


Figure 3-4 MDIO sourced by SoC

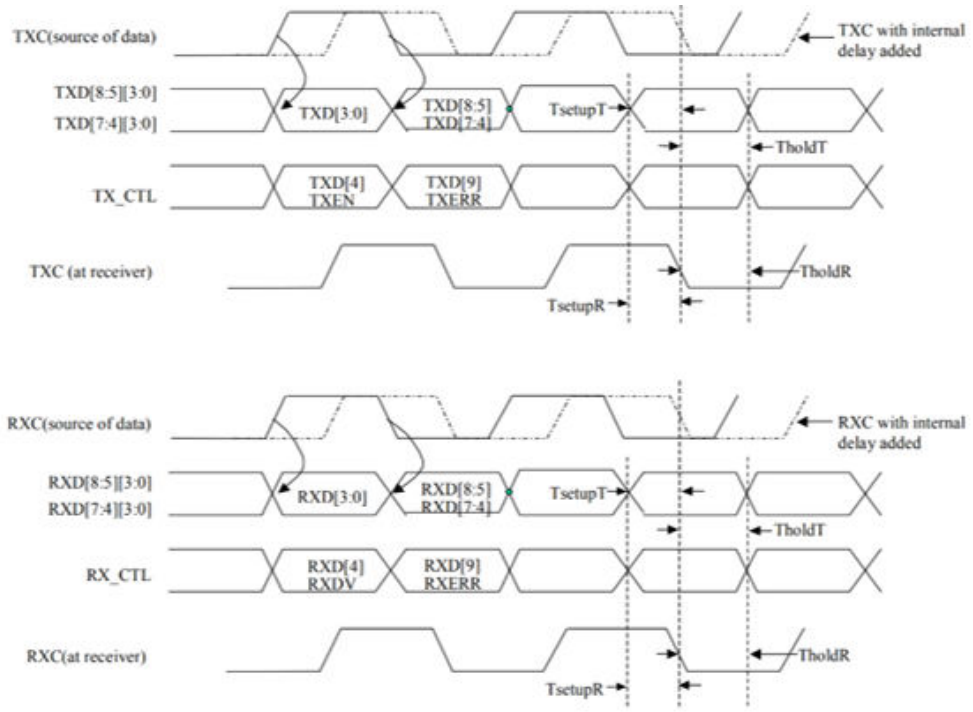


Figure 3-5 RGMII Tx and Rx timing

3.8.7 Audio support

The Audio-related interfaces supported with QCS8275 include:

- I²S: I²S interfaces
- PCM/TDM: PCM/TDM interfaces

3.8.7.1 I²S interfaces

Table 3-30 Supported I²S standards and exceptions

Applicable standards	Feature exceptions
Philips I²S Bus Specifications revised June 5, 1996	None

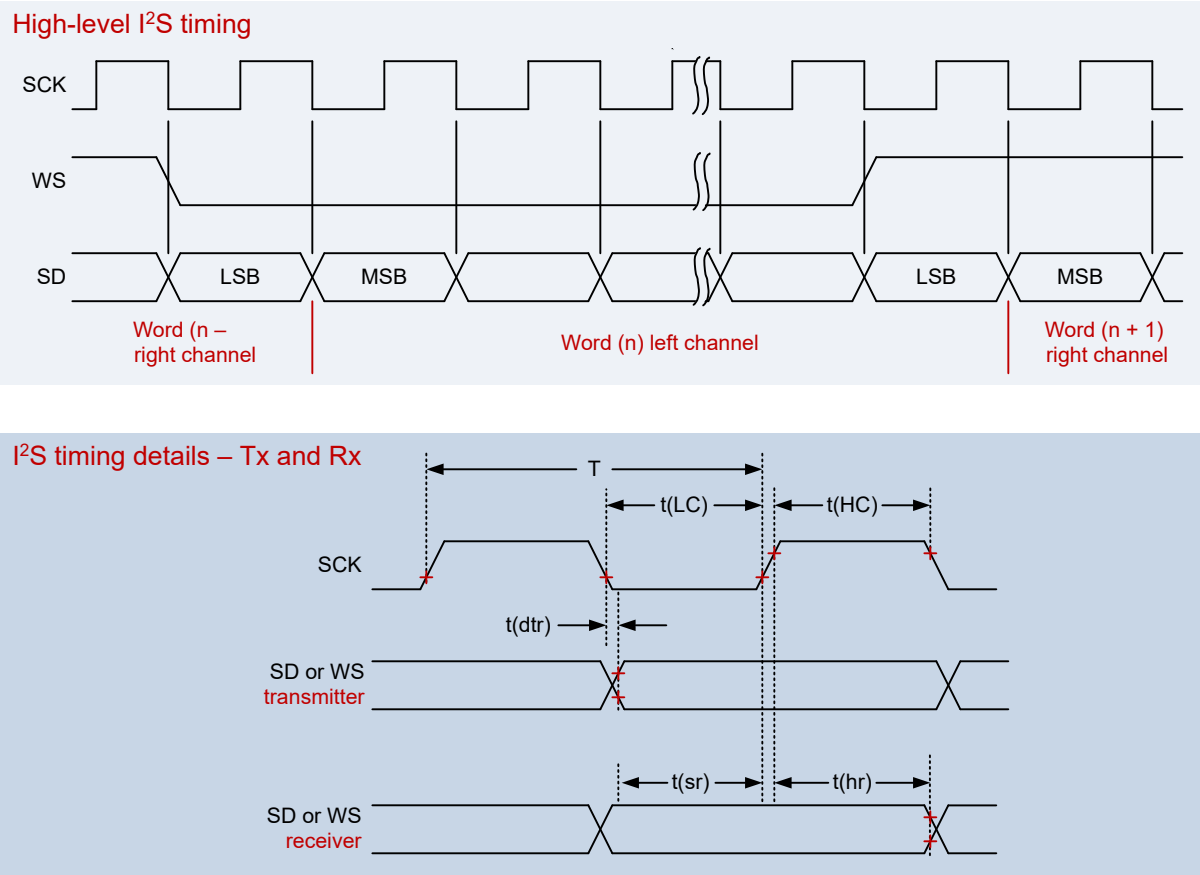


Figure 3-6 I²S timing diagram

Table 3-31 I²S interface timing

Parameter		Comments [a]	Min	Typ	Max	Unit
Using internal SCK						
Frequency			–	–	24.576	MHz
T	Clock period		40.69	–	–	ns
t(HC)	Clock high		0.45 × T	–	0.55 × T	ns
t(LC)	Clock low		0.45 × T	–	0.55 × T	ns
t(sr)	SD and WS input setup time		8.14	–	–	ns
t(hr)	SD and WS input hold time		1.5	–	–	ns
t(dtr)	SD and WS output delay		–	–	8.14	ns
Using external SCK						
Frequency			–	–	24.576	MHz
T	Clock period		40.69	–	–	ns
t(HC)	Clock high		0.45 × T	–	0.55 × T	ns
t(LC)	Clock low		0.45 × T	–	0.55 × T	ns
t(sr)	SD and WS input setup time		8.14	–	–	ns
t(hr)	SD and WS input hold time		1.5	–	–	ns
t(dtr)	SD and WS output delay		–	–	8.14	ns

[a] Load capacitance is between 10 pF and 40 pF.

Table 3-32 HS-I²S Rx interface timing

Parameter		Comments [a]	Min	Typ	Max	Unit
Using external SCK						
	Frequency			–	73.728	MHz
T	Clock period		13.56	–	–	ns
t(HC)	Clock high		0.45 × T	–	0.55 × T	ns
t(LC)	Clock low		0.45 × T	–	0.55 × T	ns
t(sr)	SD and WS input setup time		2.71	–	–	ns
t(hr)	SD and WS input hold time		1.5	–	–	ns

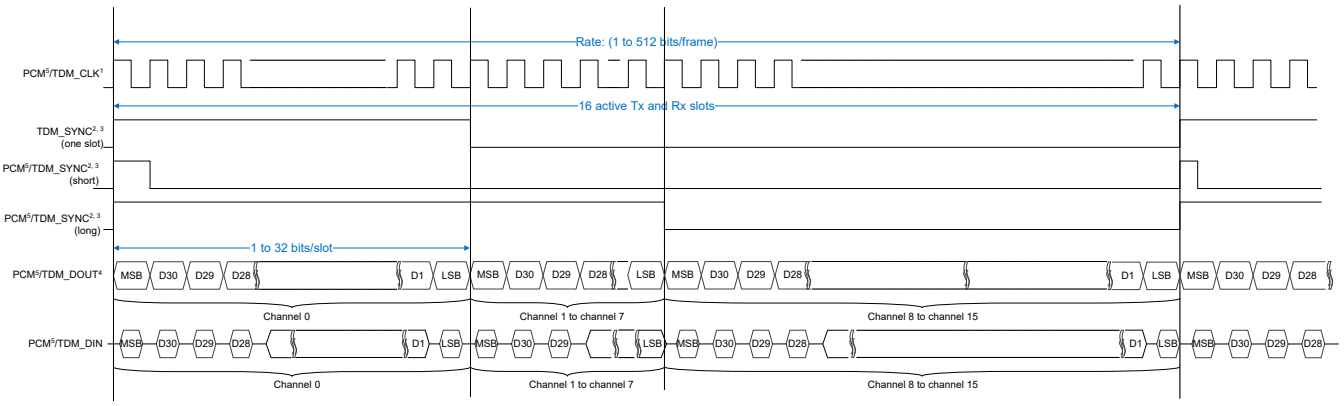
[a] Load capacitance is between 10 pF and 40 pF.

Table 3-33 HS-I²S Tx interface timing

Parameter		Comments [a]	Min	Typ	Max	Unit
Using external SCK						
	Frequency			–	24.576	MHz
T	Clock period		40.69	–	–	ns
t(HC)	Clock high		0.45 × T	–	0.55 × T	ns
t(LC)	Clock low		0.45 × T	–	0.55 × T	ns
t(dtr)	SD and WS output delay		–	–	8.14	ns

[a] Load capacitance is between 10 pF and 40 pF.

3.8.7.2 PCM/TDM interfaces



32 bits/slot; 512 bits/frame; 0 frame sync delay; 16 active Tx and Rx slots (TDM interface) or mono channel (PCM interface)

Notes:
1. Internal clock can also be inverted (180 degrees out of phase) relative to the external clock.
2. Frame sync signal can also be inverted.
3. Supports 0 to 2 cycle delays between the frame sync pulse edge and PCM_DOUT/DIN data.
4. PCM data per slot can be smaller or equal to the slot size:
   - If data size < slot size, remaining data bits are padded with zeroes.
   - If data size > slot size, extra data bits will be ignored.
5. PCM audio interface:
   - Supports only mono channel.
   - Does not support one-slot mode.
   - PCM_SYNC period is equivalent to 1 frame.

Figure 3-7 PCM/TDM audio format with different sync modes

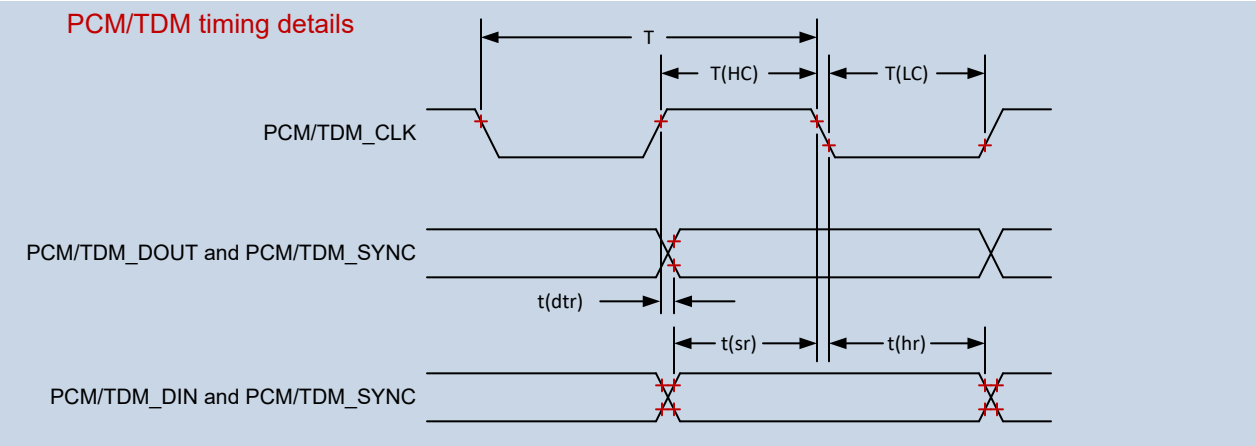


Figure 3-8 PCM/TDM timing diagram

Table 3-34 PCM/TDM interface timing parameters

Parameter		Comments	Min	Max	Unit
Master mode					
Frequency			–	24.576	MHz
T	Clock period		40.69	–	ns
t(HC)	Clock high		0.45× T	0.55× T	ns
t(LC)	Clock low		0.45× T	0.55× T	ns
t(sr)	PCM/TDM_DIN and PCM/TDM_SYNC setup time		8.14	–	ns
t(hr)	PCM/TDM_DIN and PCM/TDM_SYNC hold time		1.5	–	ns
t(dtr)	PCM/TDM_DOUT and PCM/TDM_SYNC output delay		–	8.14	ns
Slave mode					
Frequency			–	24.576	MHz
T	Clock period		40.69	–	ns
t(HC)	Clock high		0.45× T	0.55× T	ns
t(LC)	Clock low		0.45× T	0.55× T	ns
t(sr)	PCM/TDM_DIN and PCM/TDM_SYNC setup time		8.14	–	ns
t(hr)	PCM/TDM_DIN and PCM/TDM_SYNC hold time		1.5	–	ns
t(dtr)	PCM/TDM_DOUT and PCM/TDM_SYNC output delay		–	8.14	ns

[a] Load capacitance is between 10 pF to 40 pF.

Table 3-35 High speed PCM/TDM Rx interface timing

Parameter		Comments [a]	Min	Typ	Max	Unit
Using external SCK						
	Frequency			–	73.728	MHz
T	Clock period		13.56	–	–	ns
t(HC)	Clock high		0.45 × T	–	0.55 × T	ns
t(LC)	Clock low		0.45 × T	–	0.55 × T	ns
t(sr)	PCM/TDM_DIN and PCM/TDM_SYNC setup time		2.71	–	–	ns
t(hr)	PCM/TDM_DIN and PCM/TDM_SYNC hold time		1.5	–	–	ns

[a] Load capacitance is between 10 pF and 40 pF.

Table 3-36 High speed PCM/TDM Tx interface timing

Parameter		Comments [a]	Min	Typ	Max	Unit
Using external SCK						
	Frequency			–	24.576	MHz
T	Clock period		40.69	–	–	ns
t(HC)	Clock high		0.45 × T	–	0.55 × T	ns
t(LC)	Clock low		0.45 × T	–	0.55 × T	ns
t(dtr)	PCM/TDM_DOUT and PCM/TDM_SYNC delay		–	–	8.14	ns

[a] Load capacitance is between 10 pF and 40 pF.

3.8.8 I²C interface

Table 3-37 Supported I²C standards and exceptions

Applicable standard	Feature exceptions
I²C Specification, version 3.0	HS mode, slave mode, multi-master mode, and 10-bit addressing are not supported.

3.8.9 Serial peripheral interface

The QCS8275 supports SPI as a master on 24 QUP ports. SPI slave mode is supported on the eight QUP ports. See the *QCS8275 + PMM8620AU Pin Assignment and GPIO Configuration Spreadsheet* (80-73475-1A) document for more details.

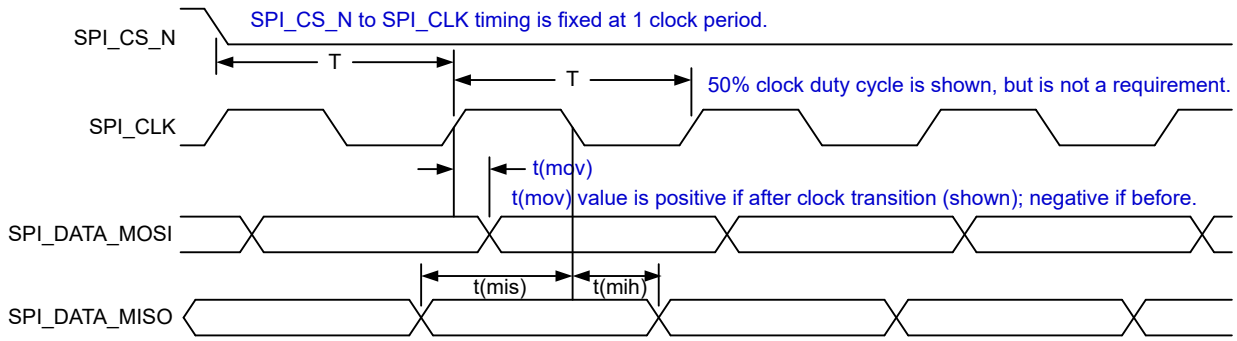


Figure 3-9 SPI master timing diagram

NOTE Depending on the mode configuration (Clock PHA and POL settings), Tx and Rx sampling edge could differ. The timing diagram above is using mode 1 as an example

Table 3-38 SPI master timing characteristics

Parameter	Comments	Min	Typ	Max	Unit
T (SPI clock period)	50 MHz maximum	20	–	–	ns
t(ch)	Clock high	8	–	–	ns
t(cl)	Clock low	8	–	–	ns

Table 3-38 SPI master timing characteristics (cont.)

Parameter	Comments	Min	Typ	Max	Unit
t(mov)	Master output valid	-5	–	5	ns
t(mis)	Master input setup	5	–	–	ns
t(mih)	Master input hold	1	–	–	ns

[a] The minimum clock period includes 1% jitter of maximum frequency.

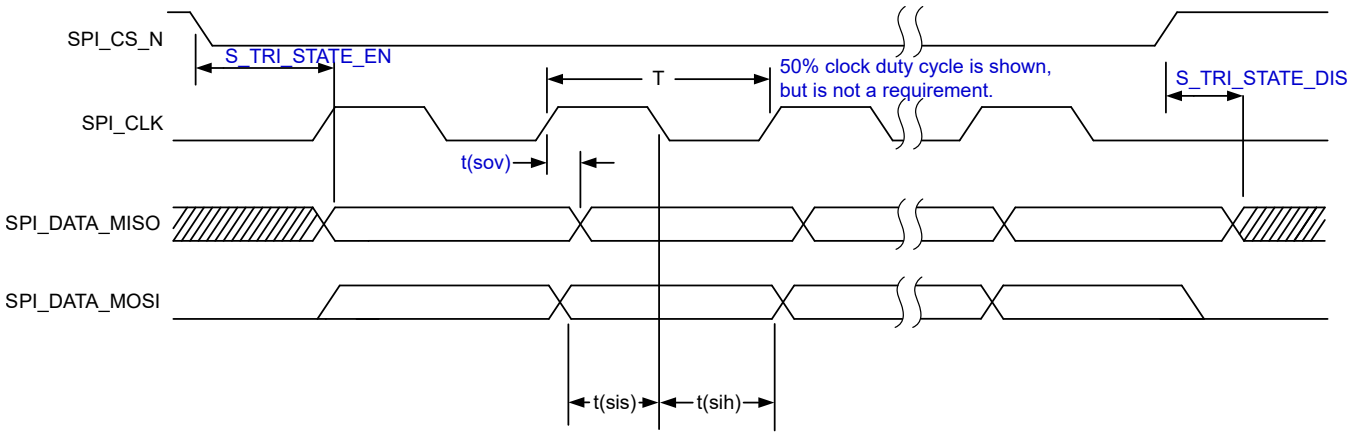


Figure 3-10 SPI slave timing diagram

Table 3-39 SPI slave timing characteristics

Parameter	Comments	Min	Max	Unit
T (SPI clock period)	50 MHz maximum	20.0	–	ns
t(ch)	Clock high	9	–	ns
t(cl)	Clock low	9	–	ns
t(sov)	Slave output valid	2	13	ns
t(sis)	Slave input setup	3	–	ns
t(sih)	Slave input hold	3	–	ns
S_TRI_STATE_EN [a]	Slave tri-state enable	–	14	ns
S_TRI_STATE_DIS [a]	Slave tri-state disable	–	14	ns

[a] The total capacitive load must not exceed 30 pF for a single slave system. For each additional slave, 10 pF should be added to the clock and data lines (not needed for chip select since it is one-to-one). For example, 40 pF for two slaves, although most systems have only a single slave. Care must be taken to verify that every slave can drive the data line fast enough to meet the timing requirement or the system must use a reduced frequency.

NOTE Only SPI slave mode 1 (PHA = 1 and POL = 0) is supported.

3.8.10 CAN-FD-interface

QCS8275 supports up to four CAN-FD interfaces multiplexed through RTSS_IOs in the RTSS domain.

For voltage characteristics, see Table 3-5. CAN-FD supports data rates up to 10 Mbps. For information about supported lower data rates, contact QTI.

Table 3-40 Supported CAN-FD standards and exceptions

Applicable standard	Feature exception
ISO 11898-1: 2015, Road vehicles, Controller area network (CAN)	None
ISO 11898-4, Road vehicles, Controller area network (CAN)	None

3.8.11 Secure digital interfaces

The supported secure digital (SD) interface standards and their exceptions are listed here.

Table 3-41 Supported SD standards and exceptions

Applicable standard	Feature exceptions
Multi Media Card Host Specification version 5.1	None
Secure Digital: Physical Layer Specification version 3.0	None
SDIO Card Specification version 3.0	None

Table 3-42 SDC modes supported

Function	Bits	I/O voltage	Vector name	Frequency
eMMC	8	1.8 V	HS400	192 MHz
		1.8 V	HS200	192 MHz
		1.8 V	HSSDR (SDR25)	52 MHz
		1.8 V	HSDDR (DDR50)	52 MHz
SD	4	1.8 V	SDR50	100 MHz

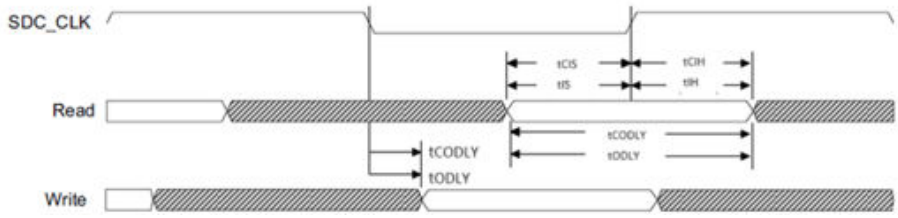


Figure 3-11 Single data rate – SDR mode

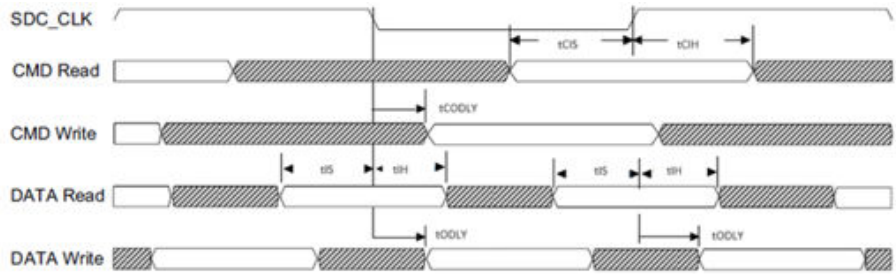


Figure 3-12 Double data rate – DDR mode

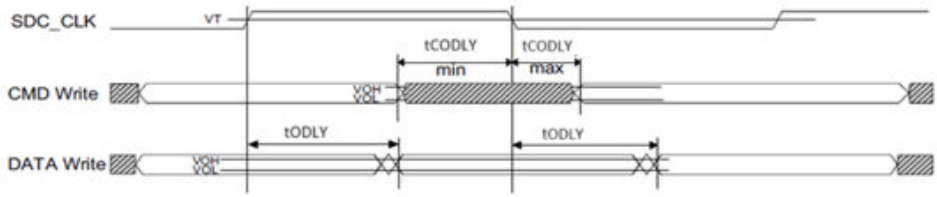


Figure 3-13 HS400 mode input timing

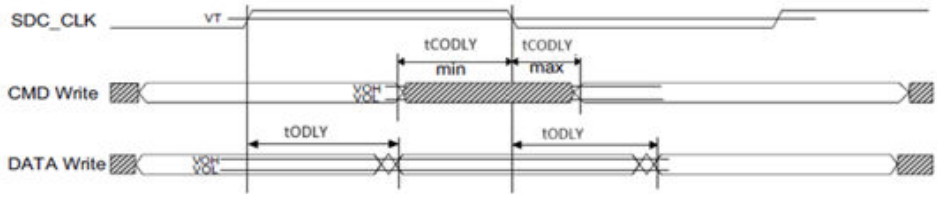


Figure 3-14 HS400 mode output timing

Table 3-43 eMMC timing specifications with respect to controller – regular mode

SDCC modes	Frequency	Parameter	Description	Minimum (ns)	Maximum (ns)
HS400 (DDR 200 MHz)	192 MHz	Tclhigh	Clock high time	2.3437	2.8646
		tRQ	DQS to DQ valid setup	0.4	–
		tRQH	DQS to DQ valid hold	0.4	–
		tCIDV	Command input valid window	2.4	–
		tCODLY	Command output delay	-1.45	0.85
		tIDV	Data input valid window	1.7	–
		tODLY	Data output delay	0.4	1.85
HS200 (SDR 200 MHz)	192 MHz	Tclhigh	Clock high time	2.3437	2.8646
		tCIDV	Command input valid window	2.4	–
		tCODLY	Command output delay	-1.45	0.85
		tIDV	Data input valid window	1.7	–
		tODLY	Data output delay	-1.45	0.85
HSDDR (DDR 50)	52 MHz	Tclhigh	Clock high time	9	11
		tCIS	Command input setup	6.3	–
		tCIH	Command input hold	1.5	–
		tCODLY	Command output delay	-8.2	3

Table 3-43 eMMC timing specifications with respect to controller – regular mode (cont.)

SDCC modes	Frequency	Parameter	Description	Minimum (ns)	Maximum (ns)
		tIS	Data input setup	2	–
		tIH	Data input hold	1.5	–
		tODLY	Data output delay	0.8	6
HSSDR (SDR25)	52 MHz	Tclhigh	Clock high time	9	11
		tCIS	Command input setup	6.3	–
		tCIH	Command input hold	1.5	–
		tCODLY	Command output delay	-8.2	3
		tIS	Data input setup	2	–
		tIH	Data input hold	1.5	–
		tODLY	Data output delay	-3.7	1.5

Table 3-44 SD card timing specifications with respect to controller – regular mode

SDCC modes	Frequency	Parameter	Description	Minimum (ns)	Maximum (ns)
SDR 50	100 MHz	Tclhigh	Clock high time	T*0.45	T*0.45
		tCIS	Command input setup	2.5	–
		tCIH	Command input hold	1.5	–
		tCODLY	Command output delay	-3.7	1.5
		tIS	Data input setup	2.5	–
		tIH	Data input hold	1.5	–
		tODLY	Data output delay	-3.7	1.5

3.9 Internal functions

Some internal functions require external interfaces to enable their operation. These include modes and resets, and JTAG functions.

3.9.1 Modes and resets

Mode and reset functions are basic digital I/Os that meet the performance specifications presented in Section 3.5.

3.9.2 JTAG

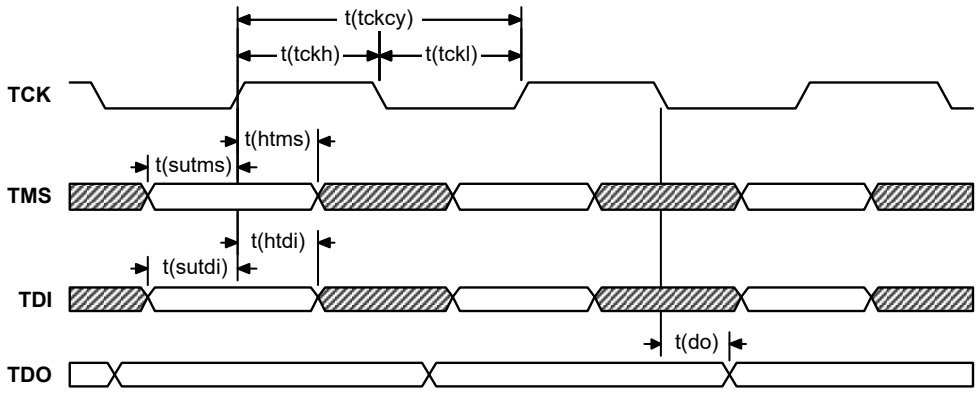


Figure 3-15 JTAG interface timing diagram

Table 3-45 JTAG interface timing characteristics

Parameter		Min	Typ	Max	Unit
t(tckcy)	TCK period	50	–	–	ns
t(tckh)	TCK pulse width high	20	–	–	ns
t(tckl)	TCK pulse width low	20	–	–	ns
t(sutms)	TMS input setup time	5	–	–	ns
t(htms)	TMS input hold time	20	–	–	ns
t(sutdi)	TDI input setup time	5	–	–	ns
t(htdi)	TDI input hold time	20	–	–	ns
t(do)	TDO data output delay	–	–	15	ns
t(tckcy)	TCK period	50	–	–	ns

4 Mechanical information

This topic provides IC mechanical information including dimensions, markings, ordering information, and thermal characteristics.

4.1 Device physical dimensions

The QCS8275 device is available in the FCBGA1326+HS, with a 25.0 mm × 25.0 mm body and a maximum height of 2.76 mm. Pin A1 is located by an indicator mark on the top of the package, and by the ball pattern when viewed from below. The following figure shows a simplified version of the package outline drawing.

Click the *QCS8275 Package Outline Drawing, FCBGA1326+HS, 25.0 × 25.0 × 2.76 mm, L1620, S716, PL1, HS1* (NT90-31845-1) to download the drawing.

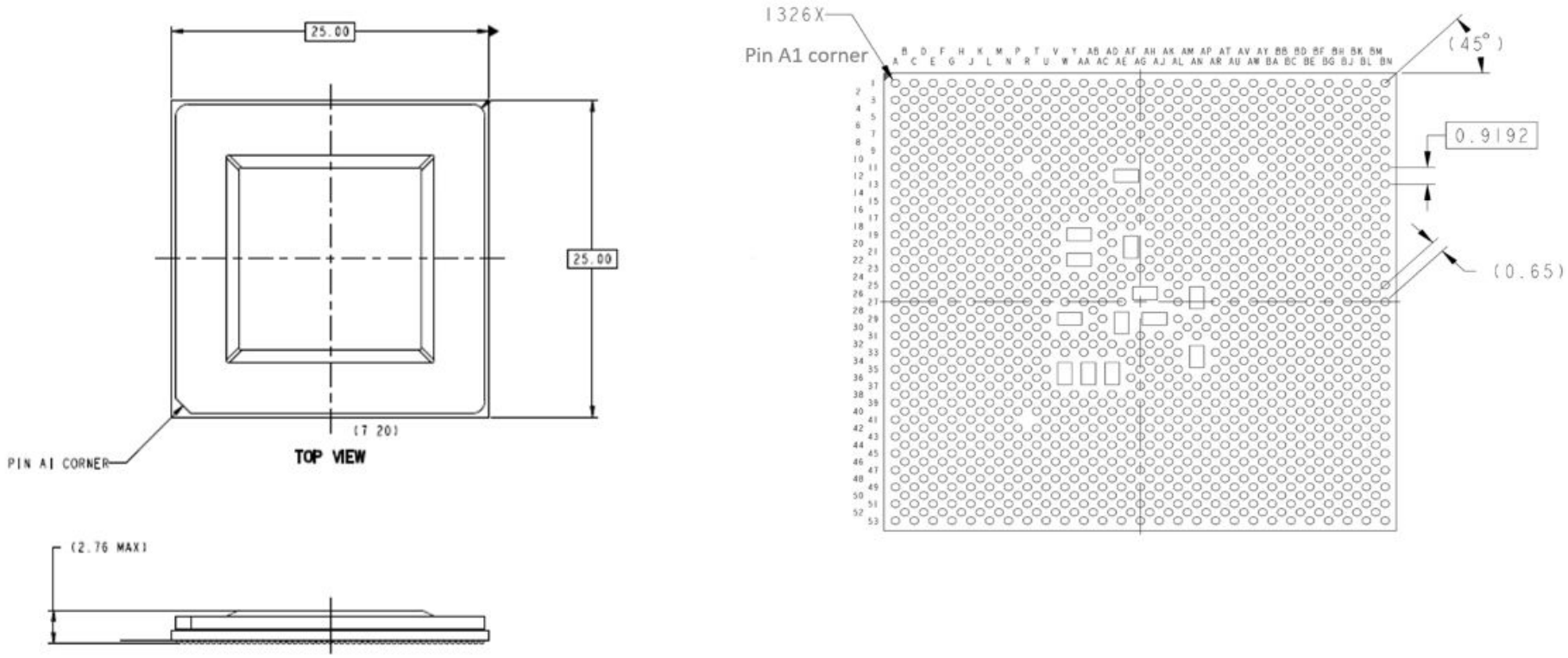


Figure 4-1 Simplified FCBGA1326+HS outline drawing

4.2 Part marking



Figure 4-2 QCS8275 device marking (top view, not to scale)

Table 4-1 QCS8275 device marking line definitions

Line	Marking	Description
C1	Qualcomm	Qualcomm name
P1	PRODUCT	Qualcomm Technologies, Inc. (QTI) product name ■ See Table 4-3 for the assigned values.
P2	VARIANT	Device variant information ■ See Table 4-3 for the assigned values.
	OPTIONAL	Optional information
		2D bar code
T1	•	Ball A1 indicator
	FAYWWXXX	F = supply source code ■ F = J (Samsung) A = assembly site code ■ A = C (Amkor, Korea) ■ A = K (SPIL, Taiwan) Y = single/last digit of year WW = two-digit work week of year specified by Y XXX = traceability number

4.3 Device ordering information

The Oracle short description is used to order QTI products, and is present on both the customer label and this document. The short description includes the product name, configuration code, package type, product revision

code, source code, and feature code/program ID of the part. This device can be ordered using the identification code shown in the following table.

Table 4-2 Example device identification code

Device ID code	AAA-AAAA	-P	-TTTTTT	NNNN	A	+FF	-EE	-RR	-S [a]	-BB or -PID [b]
Symbol definition	**Product name**	**Configuration code**	**Package type**	**Number of pins**	**Package variable**	**Additional package information**	**Shipping package**	**Product revision**	**Source code**	**Feature code**
Example	QCS-8275	-0	-FCBGA	1326		+HS	-MT	-00	-0	-AA

[a] Source code is always '0'.
[b] The feature code (BB) and the program ID (PID) are mutually exclusive. A product may have one of them or none of them, but it will never have both. If there is no feature code/program ID, this field is blank, and the Oracle short description ends after the source configuration code (S).

For example: QCS-8275-0-FCBGA1326+HS-MT-00-0-AA

Table 4-2 shows the current package-type nomenclature. For legacy parts, the Oracle short description has the position of package type and number of pins reversed.

4.4 Device identification for each sample type

Device identification details for all samples available to date are summarized in the following table.

Table 4-3 Device identification details

Device	Sample type	Variant (PRR-BB) P = product configuration code RR = product revision code BB = feature code [a]	FEATURE_ID [b]	Hardware revision number	Hardware version	Source configuration code (S) [c]	Comments
QCS8275	ES	000-AA	0x3	0x0 02E7 0E1	v1.0	0	QCS8275, Octa Kryo Gen6, Adreno GPU 623, neural signal processor, 4 × 16-bit LPDDR5, IoT Industrial, 4 × PMIC
QCS8275	EA	000-AC	0x4	0x0 02E7 0E1	v1.0	0	QCS8275, Octa Kryo Gen6, Adreno GPU 623, neural signal processor, 4 × 16-bit LPDDR5, IoT low, 2 × PMIC **Note**: Contact QTI for availability of parts.
QCS8275	CS CS date codes are as follows: ■ Amkor ≥ 527 ■ SPIL ≥ 527	000-AA	0x3	0x0 02E7 0E1	v1.0	0	QCS8275, Octa Kryo Gen6, Adreno GPU 623, neural signal processor, 4 × 16-bit LPDDR5, IoT Industrial, 4 × PMIC
QCS8275	CS CS date codes are as follows: ■ Amkor ≥ 527 ■ SPIL ≥ 527	000-AC	0x4	0x0 02E7 0E1	v1.0	0	QCS8275, Octa Kryo Gen6, Adreno GPU 623, neural signal processor, 4 × 16-bit LPDDR5, IoT low, 2 × PMIC

[a] BB is the feature code that identifies an IC's specific feature set, which distinguishes it from other versions or variants. Feature sets are detailed in the comments column.
[b] The FEATURE_ID combined with the hardware revision number (JTAG_ID) defines unique product variants. This information is shown for situations where other device identification information (such as device marking information) is not easily accessible.
[c] S is the source configuration code that identifies all of the qualified die fabrication-source/assembly site combinations available when the particular sample type was shipped.

4.5 Daisy chain devices

For daisy chain part information, contact the Qualcomm Sales team for support.

4.6 Thermal characteristics

Rather than providing thermal resistance values θ_{JC} and θ_{JA}, validated thermal package models are provided through the Qualcomm website. Designers can extract thermal resistance values by conducting their own thermal simulations.

NOTE Click the following link to download the QCS8275 thermal package model from the Qualcomm website.

QCS8275 Package Thermal Models (Icepak, FloTHERM, And 6Sigma) (HS11-73475-5HW)

4.7 Package loading during heat sink attachment

The following table specifies the maximum static load.

Table 4-4 Maximum loading specification

Component	Loading
QCS8275	56 N

NOTE Loading should be compressive and non-zero in magnitude. Tensile loads should be avoided on the SoC during and post-SMT and mechanical assembly.

The maximum static compressive load should not exceed 56 N.

The specification above has been determined through limited compressive testing with uniform loading applied perpendicular to the integrated heat spreader lid surface of the package at 125°C up to 2000 hours. It was determined that the above load recommendation is within safe working limits to ensure negligible solder ball collapse and BGA shorting risk. The end user is encouraged to validate the acceptable loading range based on intended application, motherboard, and heat sink attach configurations.

4.8 Board flexure guidance

The following table specifies the maximum strain limit of the PCB after SMT assembly.

Table 4-5 Maximum strain specification

Parameter	Value
Maximum PCB flexure	500 με

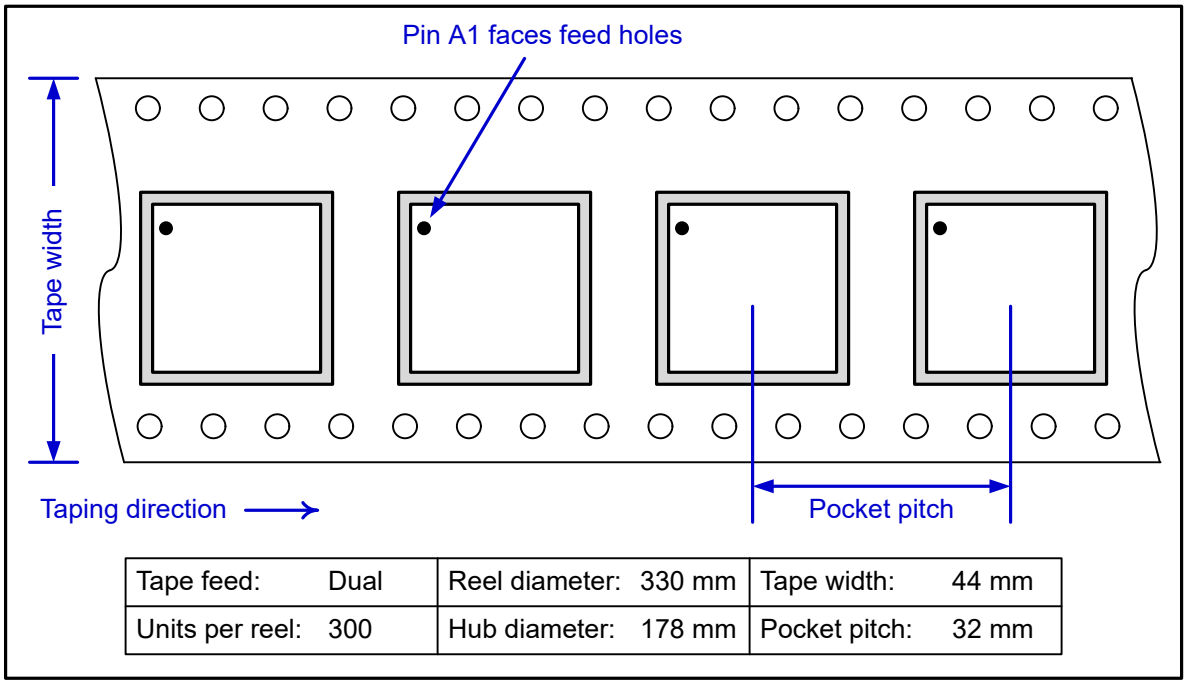
This value is based on testing according to IPC/JEDEC-9707 using a 1.6 mm thick test board, to ensure no mechanical damage to the SoC or the solder interconnects that attach the SoC to the PCB.

5 Carrier, storage, and handling information

5.1 Tape and reel information

All QTI tape carrier systems conform to EIA-481 standards.

A simplified sketch of the QCS8275 tape carrier is shown in the following figure including the proper part orientation, maximum number of devices per reel, and key dimensions.



Tape feed:	Dual	Reel diameter:	330 mm	Tape width:	44 mm
Units per reel:	300	Hub diameter:	178 mm	Pocket pitch:	32 mm

Figure 5-1 Carrier tape drawing with part orientation

Tape-handling recommendations are shown in the following figure.

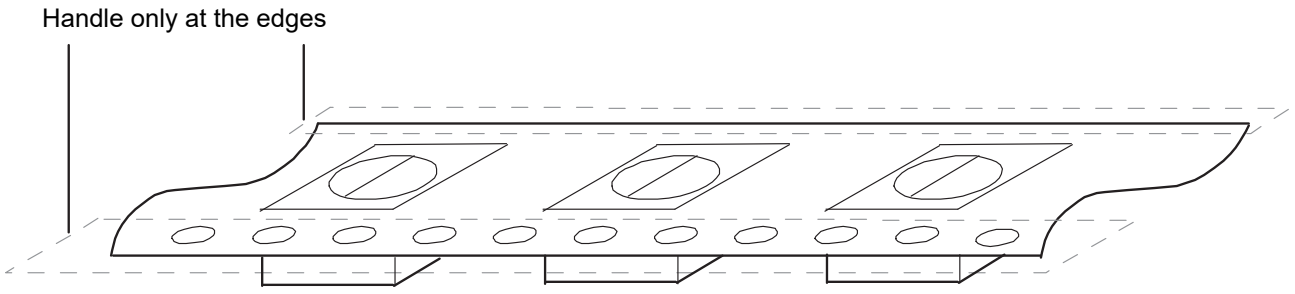


Figure 5-2 Tape handling

5.2 Matrix tray information

All QTI matrix tray carriers conform to JEDEC standards.

The device pin 1 is oriented to the chamfered corner of the matrix tray.

All matrix tray media is available for sampling only. Production is supported on tape and reel.

Table 5-1 Matrix tray approved sources of supply

Key dimensions (mm)	
Array	4 × 11 (44)
Standard bundle	5 + 1 trays (220 units)
M	26.70
M1	20.00
M2	27.50
M3	27.50

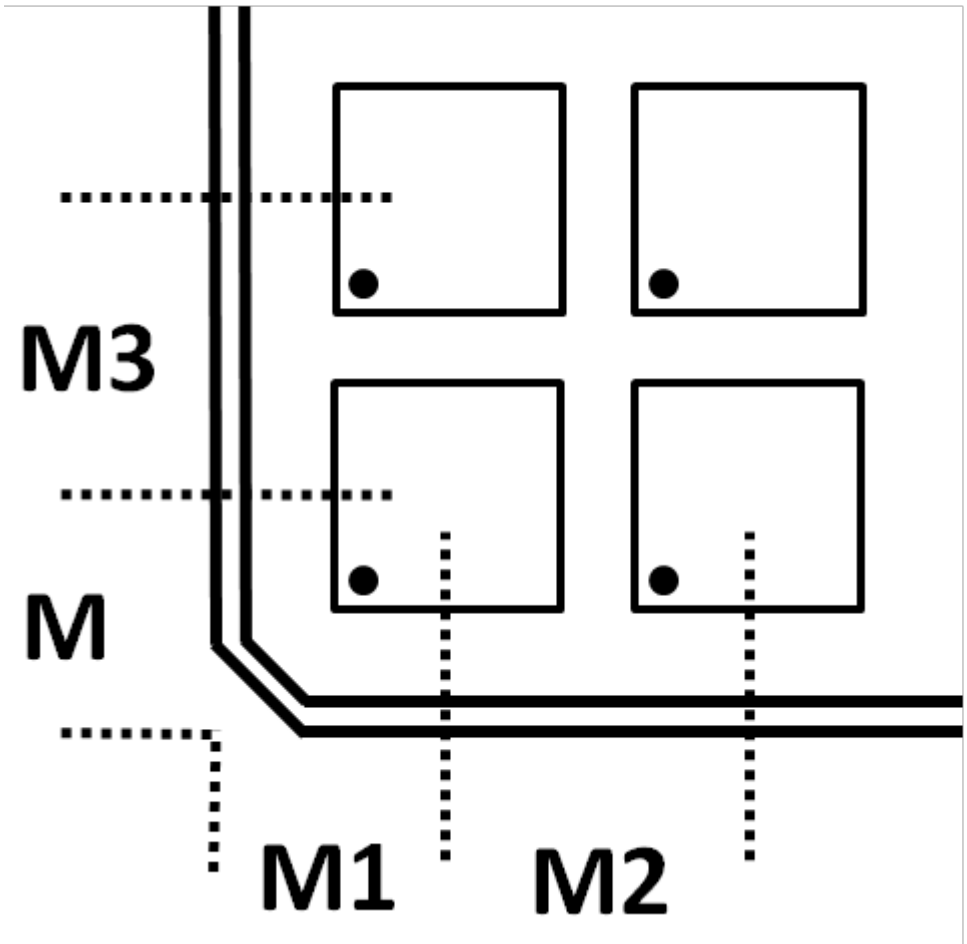


Figure 5-3 Matrix tray part orientation

5.3 Device moisture sensitivity level

Non-hermetically sealed packages are susceptible to damage induced by absorbed moisture and high temperature. A package's moisture sensitivity level (MSL) indicates its ability to withstand exposure after it is removed from its shipment bag, while it is on the factory floor awaiting PCB installation. A low MSL rating is better than a high rating; a low MSL device can be exposed on the factory floor longer than a high MSL device. All pertinent MSL ratings are summarized in the following table.

Table 5-2 MSL ratings summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	≤ 30°C/85%RH
2	1 year	≤ 30°C/60%RH
2a	4 weeks	≤ 30°C/60%RH
3	168 hours	≤ 30°C/60%RH, QCS8275 target rating
4	72 hours	≤ 30°C/60%RH
5	48 hours	≤ 30°C/60%RH
5a	24 hours	≤ 30°C/60%RH
6	Mandatory bake before use. After bake, it must be reflowed within the time limit specified on the label.	≤ 30°C/60%RH

QTI follows the latest IPC/JEDEC J-STD-020 standard revision for moisture-sensitivityqualification. ***The QCS8275 devices are targeted as MSL3; the qualification temperature was 255°C.***

5.4 Bagged storage conditions

QCS8275 devices delivered in tape and reel carriers must be stored in sealed, moisture barrier, anti-static bags. See *IC Products Packing Method* (80-VK055-1) for the expected shelf life.

5.5 Out-of-bag duration

The out-of-bag duration is the time a device can be on the factory floor before being installed onto a PCB. It is defined by the device MSL rating, as described in Device moisture sensitivity level.

5.6 Handling

Tape handling was described in Tape and reel information. Other (IC-specific) handling guidelines are presented in the following subsections.

5.6.1 Baking

It is not necessary to bake the QCS8275 if the conditions specified in Bagged storage conditions and Out-of-bag duration have not been exceeded. It is necessary to bake the QCS if any condition specified in Bagged storage conditions or Out-of-bag duration has been exceeded. The baking conditions are specified on the moisture-sensitive caution label attached to each bag; see the *IC Products Packing Method* (80-VK055-1) document for details.

CAUTION: If baking is required, the devices must be transferred into trays that can be baked to at least 125°C for 12 hours. Devices should not be baked in tape and reel carriers at any temperature.

5.6.2 Electrostatic discharge

Electrostatic discharge (ESD) occurs naturally in laboratory and factory environments. An established high-voltage potential is always at risk of discharging to a lower potential. If this discharge path is through a semiconductor device, destructive damage may result.

ESD countermeasures and handling methods must be developed and used to control the factory environment at each manufacturing site.

QTI products must be handled according to the ESD Association standard: ANSI/ESD S20.20-1999, Protection of Electrical and Electronic Parts, Assemblies, and Equipment.

5.7 Bar code label and packing for shipment

See the *IC Products Packing Method* (80-VK055-1) document for all packing-related information, including bar code label details.

6 PCB mounting guidelines

6.1 ELV and RoHS requirements

The device meets the substance restriction requirements of the EU ELV directive and the EU RoHS directive. Its SnAgCu solder balls have SAC405 composition.

6.2 SMT assembly guidelines

For recommendations on SMT process development, see the *SMT Assembly Guidelines* (SM80-P0982-1)

6.3 Daisy-chain components

Daisy-chain packages use the same processes and materials as actual products; they are recommended for SMT characterization and board-level reliability testing. The SMT process recommendations described in SMT assembly guidelines can be performed using daisy-chain components.

For Daisy-chain PCB routing recommendations, see the *Daisy Chain Interconnect, FCBGA1326+HS, 25.0 × 25.0 mm,* (DS90-31845-R1).

7 Part reliability

This topic provides the reliability data, including a definition of the qualification samples and a summary of qualification test results.

7.1 Reliability qualifications summary

Table 7-1 Silicon reliability results

Stress	ABV	Test no.	Test method	Test conditions/pre and post ATE (identify temperature, RH, and bias)	Requirements		Results	Comments
					S.S. per lot	# Lots	Fails/total S.S.	
Test group A – accelerated environment stress tests								
Preconditioning	PC	A1	JEDEC-J-STD-020, JESD22-A113	MSL 3, 3X reflow (pre and post ATE at R)	231	3	0F/693 (693)	Pass SPIL, (ATK)
Preconditioning + biased HAST	HAST	A2	JEDEC-JESD22-A110	130°C/85%RH for 96 hours (pre and post ATE at R and H)	77	3	0F/231 (231)	Pass SPIL, (ATK)
Preconditioning + unbiased HAST/temperature humidity	TH	A3	JEDEC-JESD22-A101	130°C/85%RH for 96 hours (pre and post ATE at R)	77	3	0F/231 (231)	Pass SPIL, (ATK)
Preconditioning + temperature cycle	TC	A4	JEDEC-JESD22-A104	Ta = -55°C to +125°C for 1000 cycles (pre and post ATE at R and H)	77	3	0F/231 (231)	Pass SPIL, (ATK)
High temperature storage life	HTSL	A6	JEDEC-JESD22-A103	Ta = +150°C for 500 hours (pre and post ATE at R and H)	77	3	0F/231 (231)	Pass SPIL, (ATK)
Test group B – accelerated lifetime simulation tests								
High temperature operating life	HTOL	B1	JEDEC-JESD22-A108	Tj = 125°C for 1000 hours (pre and post ATE at R, C, and H)	77	3	0F/231	–
Early life failure rate	ELFR	B2	AEC-Q100-008	Tj = 125°C for 48 hours (pre and post ATE at R and H)	800	3	0F/2400	–
NVM endurance, data retention, and operational life	EDR	B3	–	–	–	–	–	Not applicable, no NVM memory on device
Test group C – package assembly integrity tests								

Table 7-1 Silicon reliability results (cont.)

Stress	ABV	Test no.	Test method	Test conditions/pre and post ATE (identify temperature, RH, and bias)	Requirements		Results	Comments
					S.S. per lot	# Lots	Fails/total S.S.	
Wire bond shear	WBS	C1	–	–	–	–	–	Not applicable, not a wirebonded package
Wire bond pull strength	WBPS	C2	–	–	–	–	–	Not applicable, not a wirebonded package
Solderability	SD	C3	–	–	–	–	–	Not applicable, not a leadframe package
Physical dimensions	PD	C4	JESD22-B100	Package outline drawing	10	3	0F/30, (30)	CPK ≥ 1.67 SPIL, (ATK)
Solder ball shear	SBS	C5	AEC-Q100-010	5 balls each from a min. of 10 devices, ≥ 300 g/ball for pad opening 350 µm	10	3	0F/30, (30)	CPK ≥ 1.67 SPIL, (ATK)
Lead integrity	LI	C6	–	–	–	–	–	Not applicable, not a leadframe package
Test group D – die fabrication reliability tests								
Electromigration	EM	D1	JP001A	–	–	–	–	Pass
Time-dependent dielectric breakdown	TDDB	D2	JP001A	–	–	–	–	Pass
Hot carrier injection	HCI	D3	JP001A	–	–	–	–	Pass
Negative bias temperature instability	NBTI	D4	JP001A	–	–	–	–	Pass
Stress migration	SM	D5	JP001A	–	–	–	–	Pass
Test group E – electrical verification tests								
Pre- and post-stress electrical test	TEST	E1	–	–	–	–	–	Pre and post ATE
ESD – human body model	HBM	E2	AEC-Q100-002	Pass target ±1 kV (pre and post ATE at R and H)	3	1	0F/3	–
ESD – Charged Device Model	CDM	E3	AEC-Q100-011	Pass target ±150 V (Pre and Post ATE at R & H)	3	1	0F/3, (3)	Pass SPIL, (ATK)
Latch-up	LU	E4	AEC-Q100-004	Ta=105°C (Class II) for SOV (1.5X Vop Max) and I-Test (±105 mA) (Pre and Post ATE at R & H)	6	1	0F/6	–
Electrical distributions	ED	E5	–	–	–	–	–	Available for review

Table 7-1 Silicon reliability results (cont.)

Stress	ABV	Test no.	Test method	Test conditions/pre and post ATE (identify temperature, RH, and bias)	Requirements		Results	Comments
					S.S. per lot	# Lots	Fails/total S.S.	
Fault grading	FG	E6	AEC-Q100-007	–	–	–	–	Available for review
Characterization	CHAR	E7	–	–	–	–	–	Available for review
Electromagnetic compatibility	EMC	E9	–	–	–	–	–	Not performed. System level test
Short circuit characterization	SC	E10	–	–	–	–	–	Not applicable, not >12 V or a power device
Soft error rate	SER	E11	–	–	–	–	–	Leverage foundry simulation data
Test group F – defect screening tests								
Process average test	PAT	F1	–	–	–	–	–	Available for review
Statistical bin/yield analysis	SBA	F2	–	–	–	–	–	Available for review

7.2 Qualification sample description

Table 7-2 Device characteristics

Category	Definition
Device name	QCS8275
Package type	FCBGA1326+HS
Package body size	25.0 × 25.0 × 2.76 mm
Ball count	1326
Ball composition	SAC405
Fab process	4 nm
Supply sources (fab sites)	Samsung
Assembly sites	Amkor, Korea SPIL, Taiwan
Solder ball pitch	0.9 mm

8 Samples and known issues

8.1 Sample testing

See Table 4-3 for device identification details for all available samples.

8.1.1 Engineering samples (ES)

These devices undergo limited testing and sometimes have significant feature limitations. They are suitable to assist with PCB development, to conduct board-level electrical evaluation tests, and to explore manufacturing considerations. Engineering samples should not be used for product-level qualification.

8.1.2 Commercial samples (CS)

These devices undergo full production-level testing, and meet the specifications and features described in the data sheet, except as otherwise noted in this document. They have passed device level qualification. Commercial samples are suitable for performance testing, and also for product level production and qualification.

8.2 Known issues

There are no known issues for QCS8275.

9 Revision history

The following table lists the technical content changes for all revisions.

Revision	Date	Description
AK	September 2026	Added information for clarity in: ■ Figure 2-1 *QCS8275 pin assignments* ■ Figure 4-1 *Simplified FCBGA1326+HS outline drawing*
AJ	July 2026	■ Section 1.2 *QCS8275 features*: Updated VPU specifications for accuracy ■ Table 2-3 *Pin descriptions – general-purpose input/output ports*: Removed software-configurable GPIO alternate functions
Revision AI was omitted in accordance with QTI document conventions.		
AH	June 2026	■ Section *Device description*: Updated for improved clarity ■ Section *Key features*: Updated for improved clarity ■ Table 2-2 *Pin descriptions – primary pins*: Updated the functional descriptions for DDR_RESET_N, DS_EN, EBI01_CAL, EBI23_CAL, REFGEN_REXT2, REFGEN_REXT3, CXO_0, CXO_1, and RTSS_CXO ■ Table 3-3 *Operating conditions for voltage rails with non-AVS*: Added minimum and maximum values ■ Added VDD_RTSS_PX_RGMII rail in: □ Table 3-3 *Operating conditions for voltage rails with non-AVS* □ Table 3-7 *DC specifications for RGMII 1.8 V mode (VDD_RTSS_PX_RGMII)* ■ Table 3-8 *Possible drive strength*: Added this table ■ Added the timing specifications in: □ Section 3.7.1 *Camera interfaces* □ Section 3.7.2 *Display support* □ Section 3.7.3 *DisplayPort (DP)* □ Section 3.8.1 *USB interfaces* □ Section 3.8.2 *PCIe interface* □ Section 3.8.4 *SGMII interface* □ *RGMII interface* ■ Added the following content: □ Section 3.8.10 *CAN-FD-interface* □ Section 3.8.11 *Secured digital interfaces* ■ Figure 3-2 *Octa-SPI DDR timing diagram*: Updated this diagram
AG	January 2026	■ Table 2-2 *Pin descriptions – primary pins*: Updated the functional descriptions of PS_HOLD, MD_PS_HOLD, RTSS_PS_HOLD, RESIN_N, RTSS_RESIN_N, RESOUT_N, and RTSS_RESOUT_N
AF	December 2025	■ Figure 1-1 *QCS8275 functional block diagram*: Corrected the DSP information ■ □ Section 1.2 *QCS8275 features*: LPASS – corrected the number of GPDSPs to one □ HS-I²S – corrected the number of receive interfaces to three □ PCIe – corrected the 4-lane PCIe Gen4 to PCIe1 ■ Table 2-1 *I/O parameter definitions*: Removed unsupported parameters and updated the voltage to 1.2 V in the PX_11 description ■ Table 2-2 *Pin descriptions – primary pins*: □ Updated the pad type for multiple pins □ Updated the PCIe pin descriptions to Gen 4

Revision	Date	Description
		■ Table 2-6 *Pin descriptions – power supply pins*: ▫ Updated the functional description for application processor pins ▫ Updated the functional description for the pins N13, BD28, and BD30 ■ Section 3.5 *Digital logic characteristics*: Removed the DC specification information for VDD_RTSS_PX_8 ■ Figure 3-7 *SPI slave timing diagram*: Added the figure ■ Table 3-27 *SPI slave timing characteristics*: Corrected the parameters ■ Added the following topics: ▫ Section 4.7 *Package loading during heat sink attachment* ▫ Section 4.8 *Board flexure guidance*
AE	August 2025	■ Table 2-3 *Pin descriptions – general-purpose input/output ports*: Removed all instances of QUP2 ■ Added the following sections: ▫ Section 3.1 *Absolute maximum ratings* ▫ Section 3.3 *Power distribution network* ▫ Section 3.4 *Average operating current* ▫ Section 3.5 *Digital logic characteristics* ▫ Section 3.6 *Memory support* ▫ Section 3.7 *Multimedia* ▫ Section 3.7.1 *Camera interfaces* ▫ Section 3.7.2 *Display support* ▫ Section 3.7.3 *DisplayPort (DP)* ▫ Section 3.8 *Connectivity* ▫ Section 3.8.1 *USB interfaces* ▫ Section 3.8.2 *PCIe interface* ▫ Section 3.8.3 *UFS interface* ▫ Section 3.8.4 *SGMII interface* ▫ Section 3.8.5 *Octa-SPI/Quad-SPI interface* ▫ Section 3.8.6 *RGMII interface* ▫ Section 3.8.7 *Audio support* ▫ Section 3.8.7.1 *I²S interfaces* ▫ Section 3.8.7.2 *PCM/TDM interfaces* ▫ Section 3.8.8 *I²C interface* ▫ Section 3.8.9 *Serial peripheral interface* ▫ Section 3.9 *Internal functions* ▫ Section 3.9.1 *Modes and resets* ▫ Section 3.9.2 *JTAG* ▫ Section 7.1 *Reliability qualifications summary* ▫ Section 7.2 *Qualification sample description* ■ Table 4-3 *Device identification details*: Added CS samples ■ Section 5.3 *Device moisture sensitivity level*: Added the MSL rating
AD	17 January 2025	Editorial changes made; no technical updates.
AC	17 January 2025	■ Introduction: Added a note ■ Figure : QCS8275 functional block diagram: Updated this figure and added a note ■ Table : QCS8275 variant feature set comparison: Added this table ■ QCS8275 features: Added RF information ■ Physical interfaces: Updated QSPI description

Revision	Date	Description
		■ Added the following sections: □ Table : Pin descriptions – general-purpose input/output ports: Updated GPIO[10] through GPIO[16] □ Part marking □ Device ordering information □ Device identification for each sample type □ Daisy chain devices Device moisture sensitivity level □ Thermal characteristics ■ Samples and known issues: Added this chapter
AB	September 17, 2024	■ Table: Pin descriptions – general-purpose input/output ports: Updated GPIO_9 configurable function ■ Table : Pin descriptions – RTSS I/O: Updated RTSS_IO_52 functional description ■ Device physical dimensions: Updated the NT90 download link
AA	September 12, 2024	Initial release

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